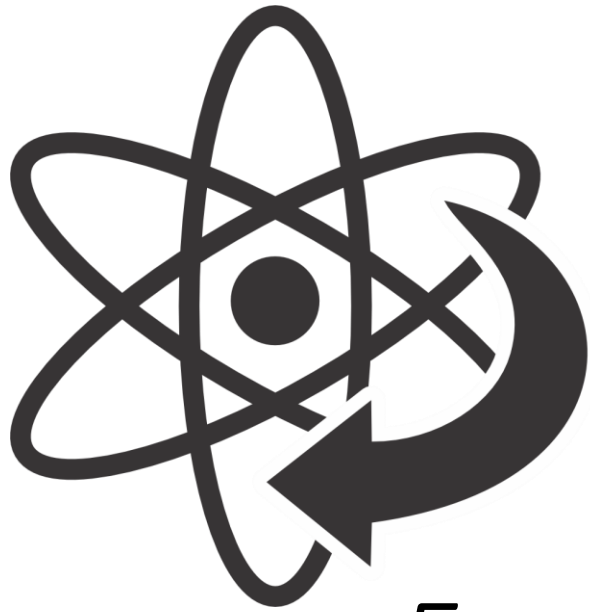




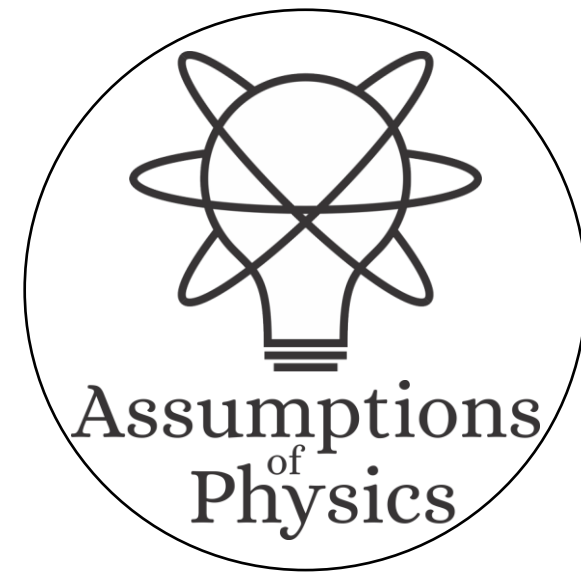
Reverse Physics

From Laws to Physical Assumptions

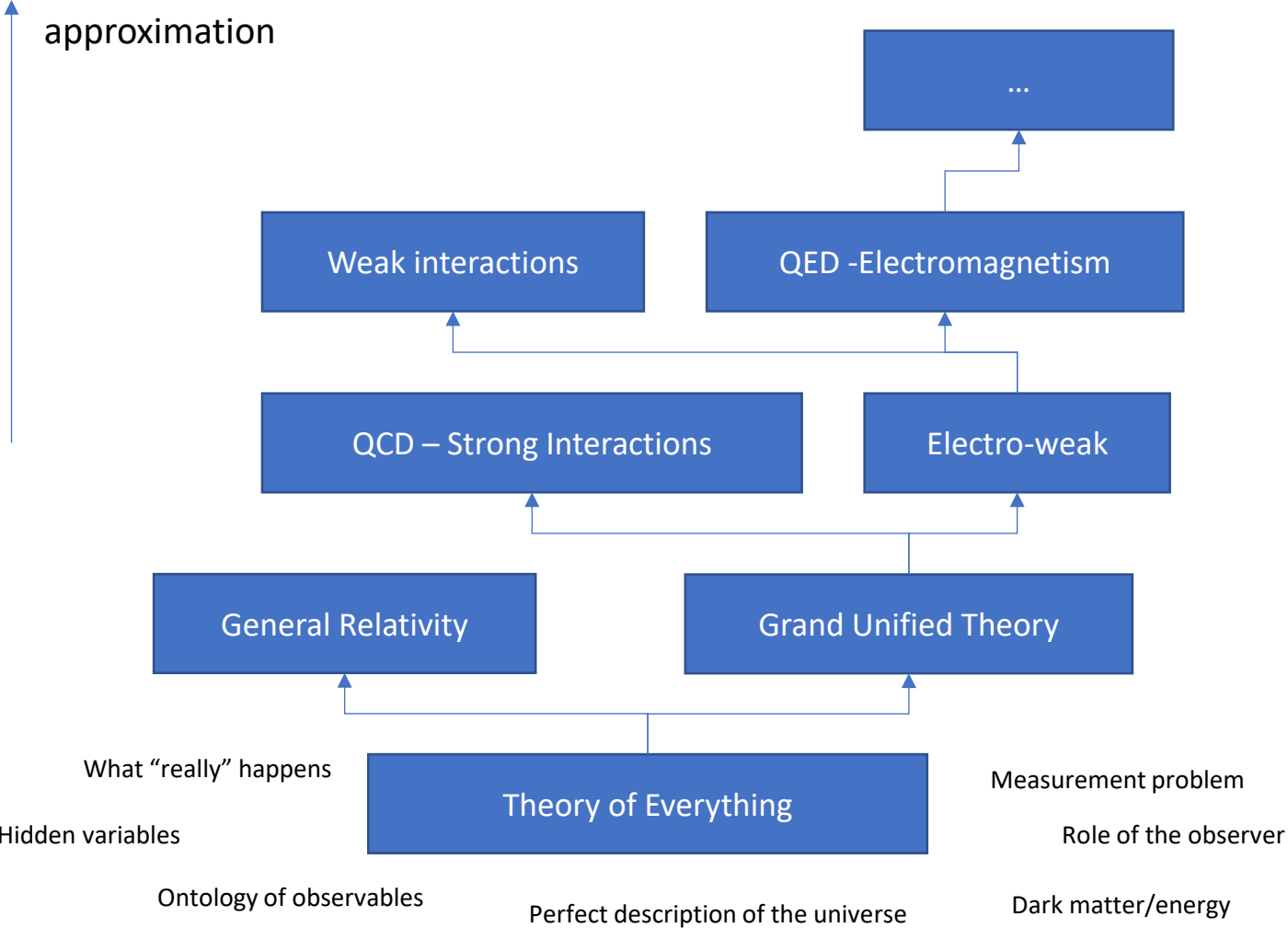


Christine A. Aidala
Physics Department
University of Michigan

Physics Department Colloquium
Duke University



Standard view of the foundations of physics

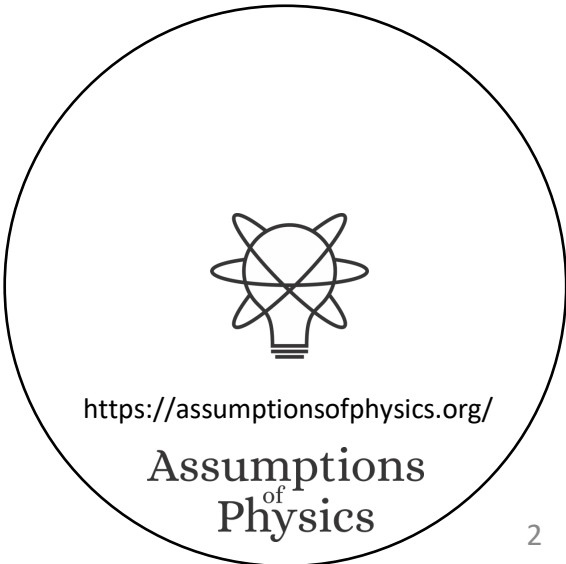


Goal of physics is to find the true laws of the universe!

The “real” physics!

The foundations of physics!

Everything else is an approximation



We found:

Physical requirements and assumptions drive most of the theoretical apparatus,
for example:

Experimental verifiability $\Rightarrow$ topologies and σ -algebras

Geometric structures $\Leftrightarrow$ Entropic structures

Hamiltonian evolution $\Leftrightarrow$ det-rev/isolation + DOF independence

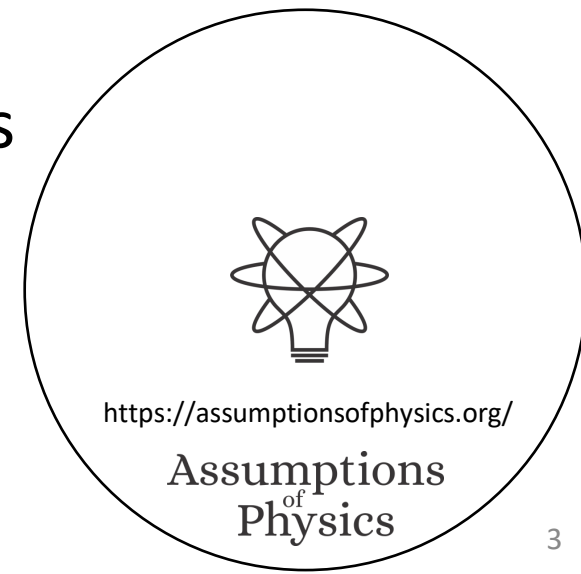
Massive particles and potential forces $\Leftrightarrow$  + Kinematic eq

~~Goal of physics is to find the
true laws of the universe!~~

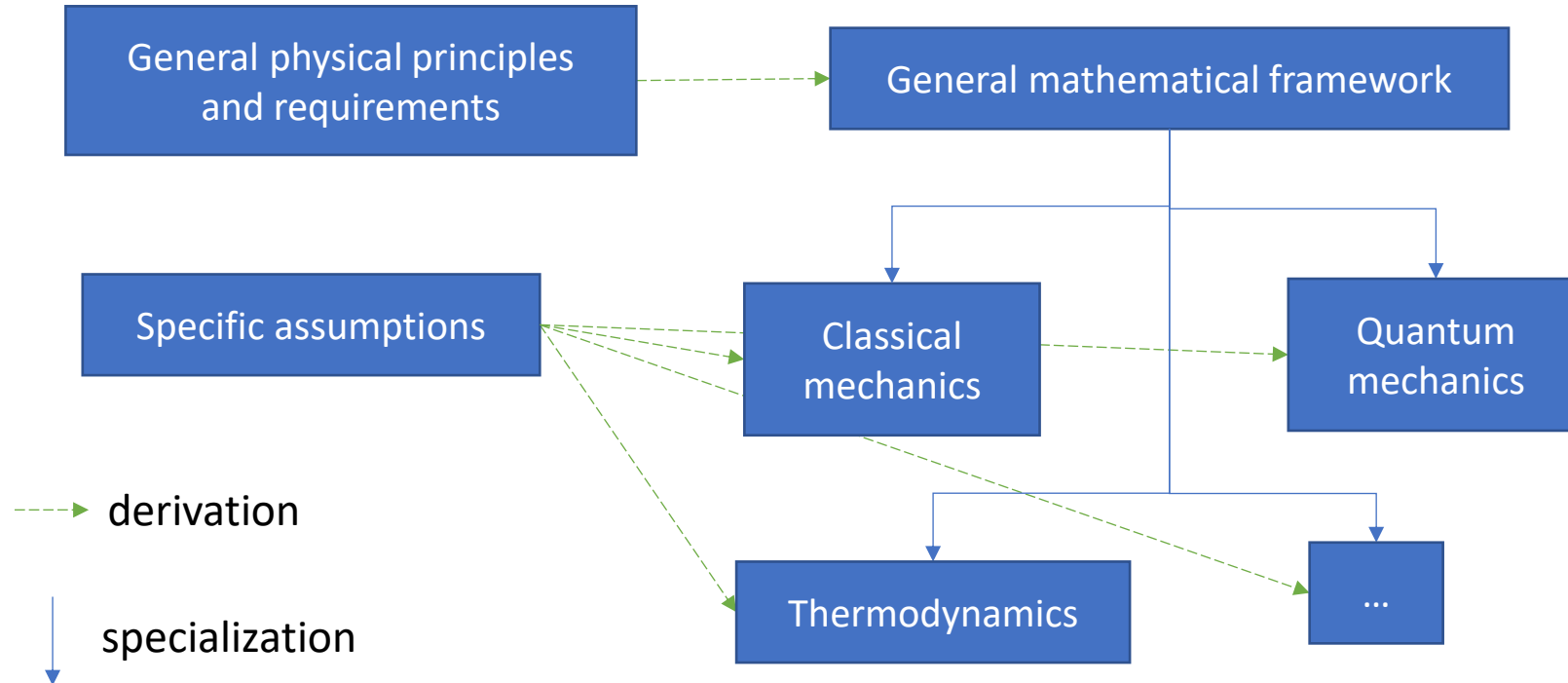
Less productive point of view

Goal of physics is to find models
that can be empirically tested

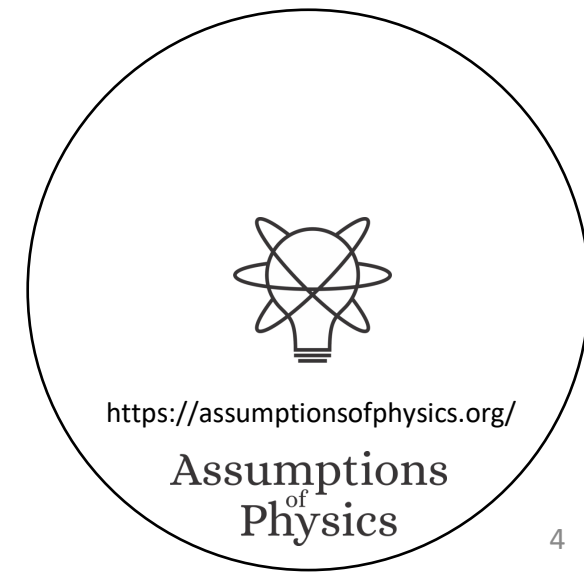
More productive point of view



Our view of the foundations of physics



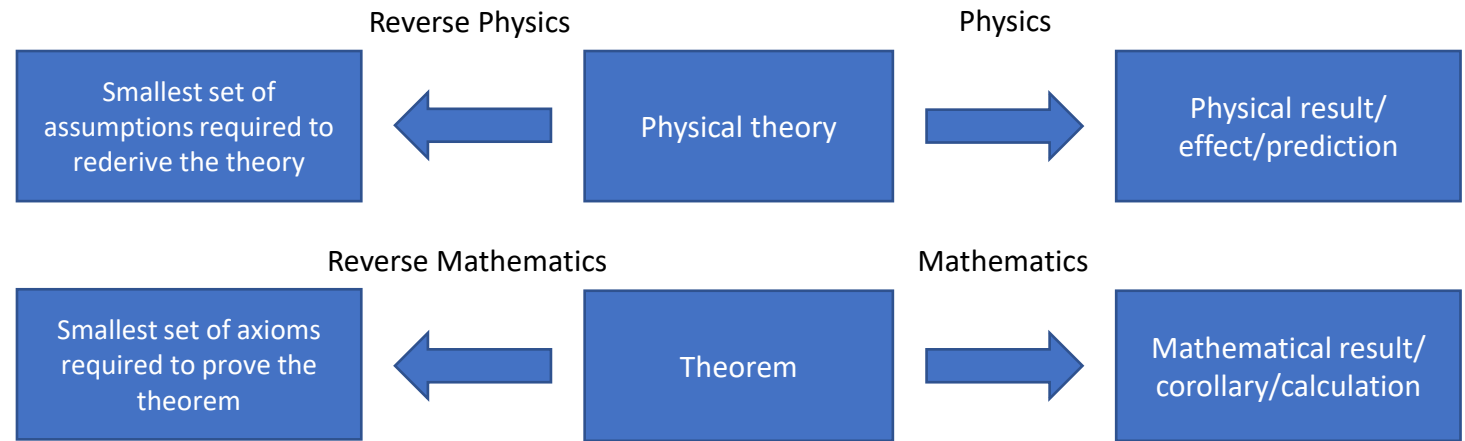
Foundations of
physics
↕
The theory of
physical theories



Reverse Physics:

Start with the equations,
reverse engineer physical
assumptions/principles

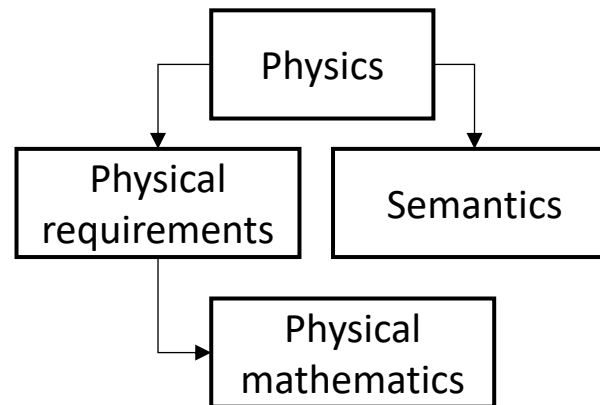
Found. Phys. **52**, 40 (2022)



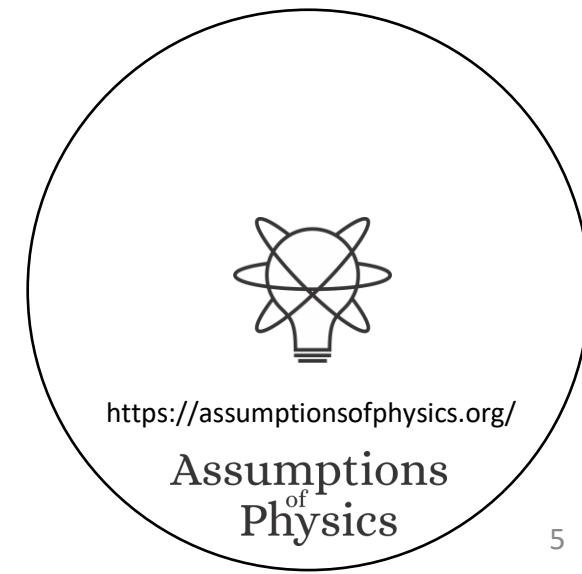
Goal: find the right overall physical concepts, “elevate” the discussion from mathematical constructs to physical principles

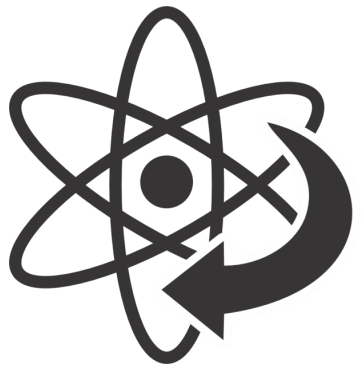
Physical Mathematics:

Start from scratch and rederive
all mathematical structures from
physical requirements



Goal: get the details right, perfect one-to-one map between mathematical and physical objects





Reverse Physics

“When the math is derived from the right physical assumptions, the physical assumptions can be derived from the math.”

Reverse Physics aims to adopt approaches that are more in line with the foundations of mathematics, to break physics into a set of physical and mathematical conditions with well-understood logical relationships.

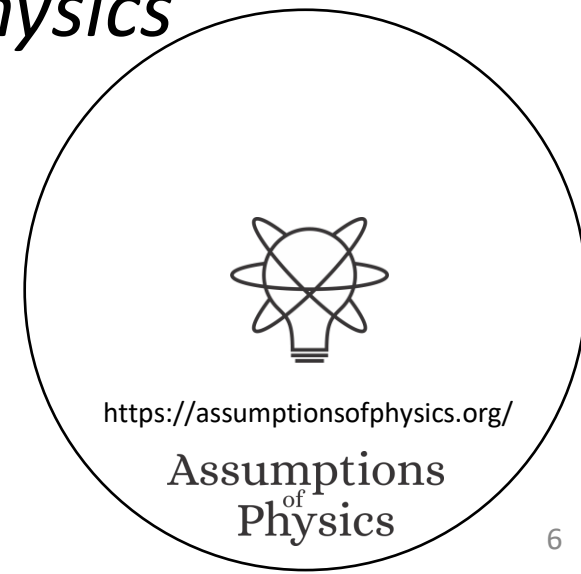
A new methodology for the foundations of physics

Assumptions of Physics, Michigan Publishing (v3 2025)
<https://doi.org/10.3998/mpub.12204707>

<https://arxiv.org/abs/2608.27543>

Found. Phys. **52** 40 (2022)
<https://link.springer.com/article/10.1007/s10701-022-00555-z>

J. Phys. Commun. **2** 045026 (2018)
<http://iopscience.iop.org/article/10.1088/2399-6528/aaba25>



Break physics into different conditions and study their logical relationships



Similar to what is done in the foundations of mathematics

Theorem 1.4. *Over Zermelo-Fraenkel set theory, the following are equivalent:*

AoC (*Axiom of Choice*) *For every non-empty set X there exists a function f that assigns to each nonempty set $S \subseteq X$ some representative element $f(S) \in S$;*

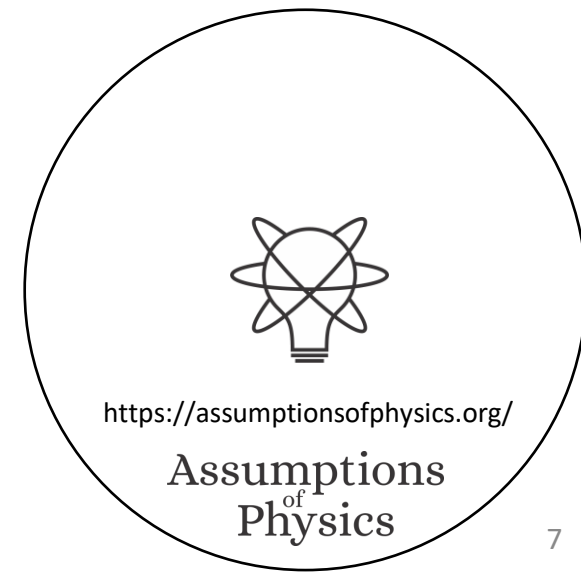
WOP (*Well-Ordering Principle*) *Every set can be well-ordered;*

ZL (*Zorn's Lemma*) *Every non-empty partially ordered set P with the property that every chain in P is bounded has a maximal element.*

“The Axiom of Choice is obviously true, the well-ordering principle obviously false, and who can tell about Zorn’s lemma?”

Jerry Bona, quoted in E. Schechter, *Handbook of Analysis and Its Foundations*, 1997

Clarifying what any given starting point commits us to is particularly valuable when intuition can be faulty!



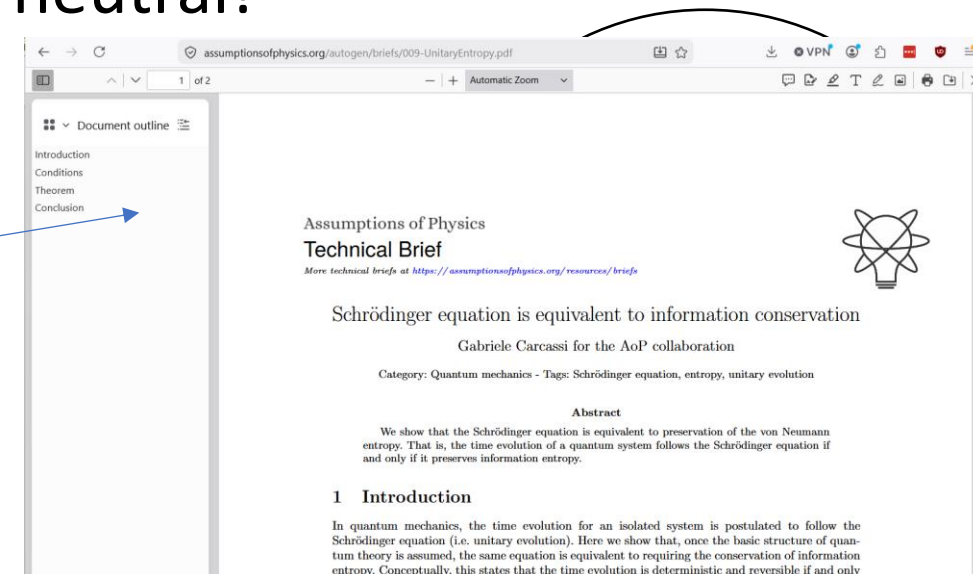
Goals of Reverse Physics

(and structure of the talk)

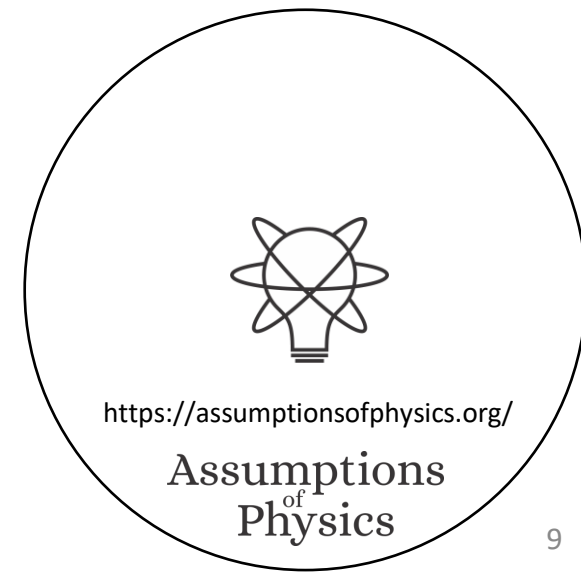
- Examine the actual content of mathematical structures used in physics
- Uncover “paradoxes” and develop standards of rigor appropriate for physics
- Clarify the logical dependencies of physical/mathematical conditions
- Identify core constitutive requirements of physical theories

Must be framework/interpretation/formulation neutral!

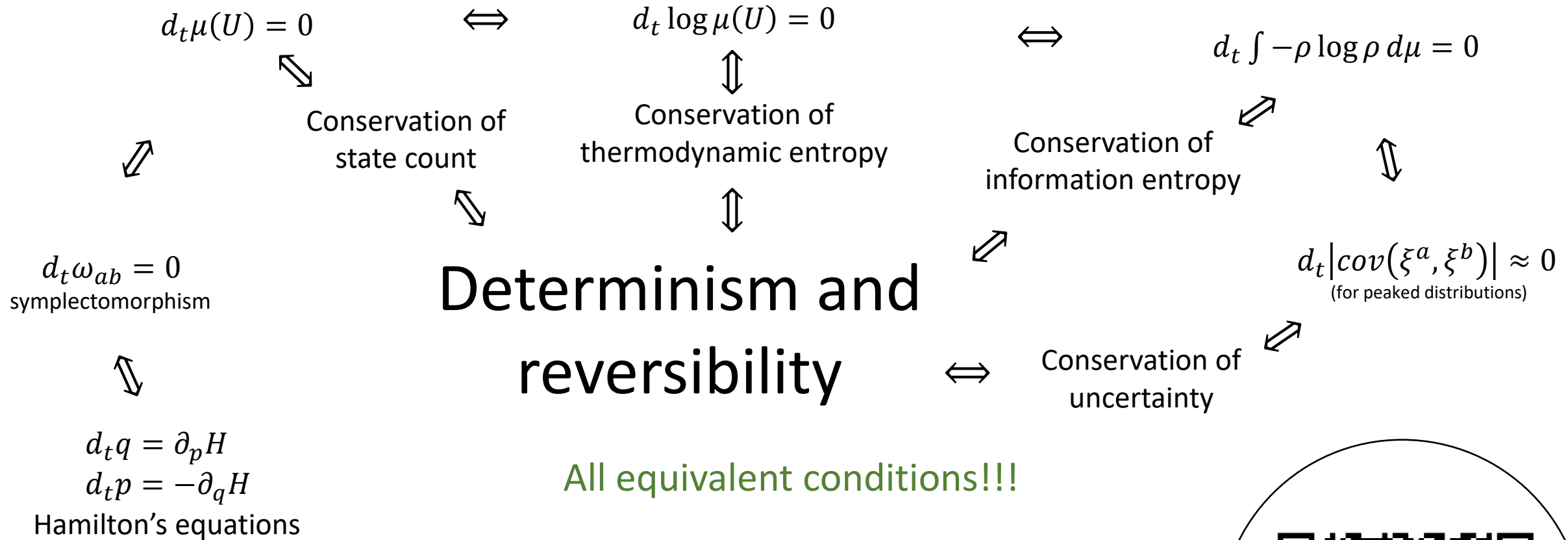
Where relevant throughout,
QR codes and links to papers or
technical briefs



Examining the actual content of mathematical structures



Reversing classical Hamiltonian mechanics in 1 DOF

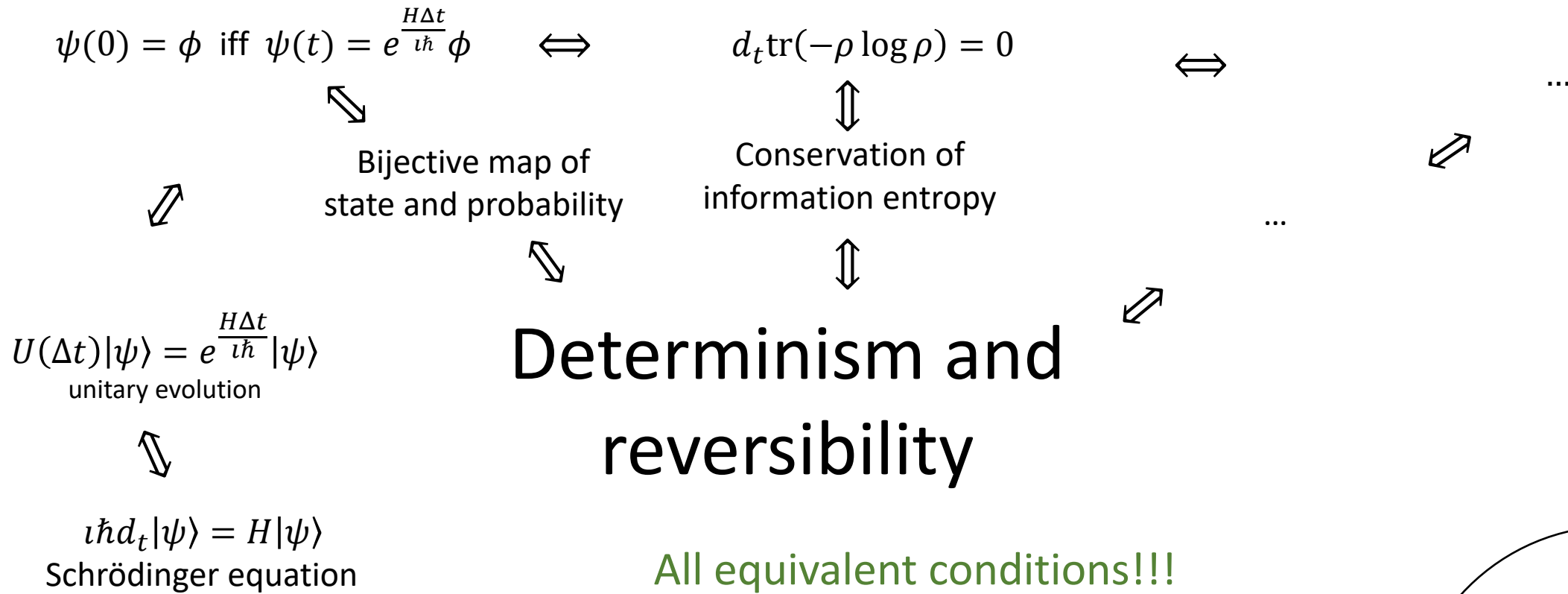


Base theory: State: $\xi^a = [q \ p] \in \mathbb{R}^2$

Count of states: $\mu(U) = \int_U dq dp = \int_U \omega$



Reversing unitary evolution

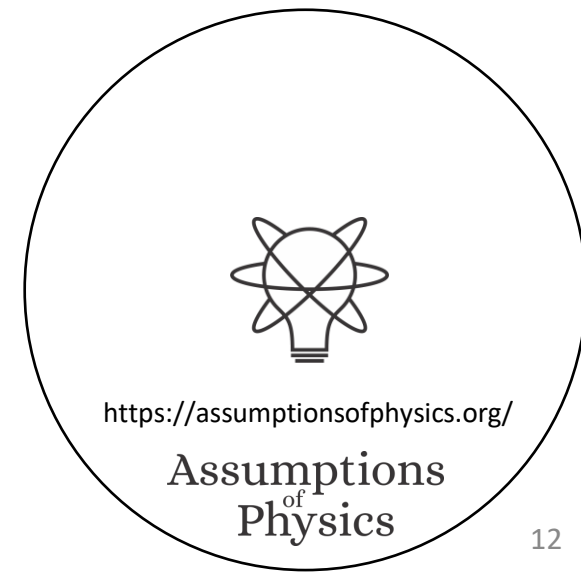


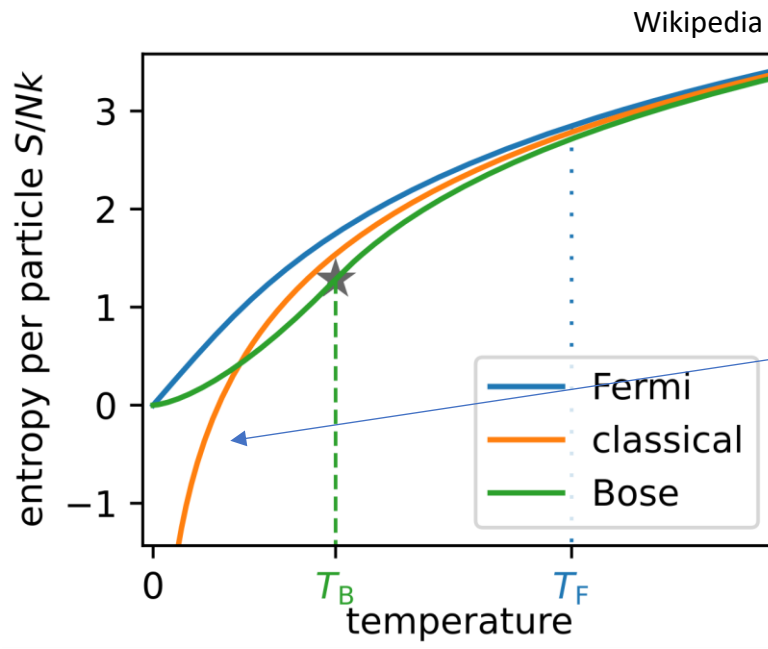
Base theory: State: $\psi \in P(\mathcal{H})$

Probability of ψ given ϕ : $p(\psi|\phi) = |\langle\psi|\phi\rangle|^2$



Why does classical mechanics fail? What is quantum mechanics fixing?



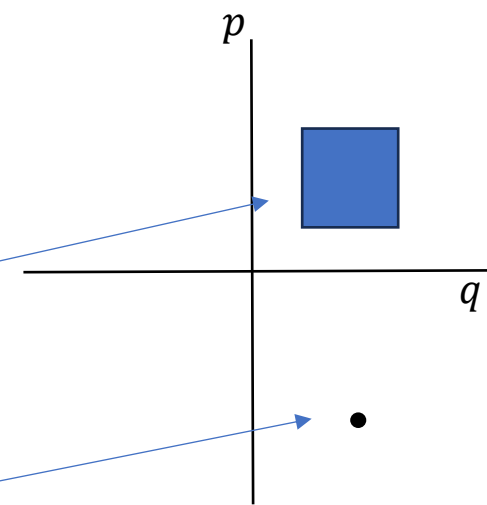


Temperature/entropy curve for ideal gases

to minus infinity!

$$S = \log \Delta q \Delta p$$

For a single microstate
 $S = \log 0 = -\infty$



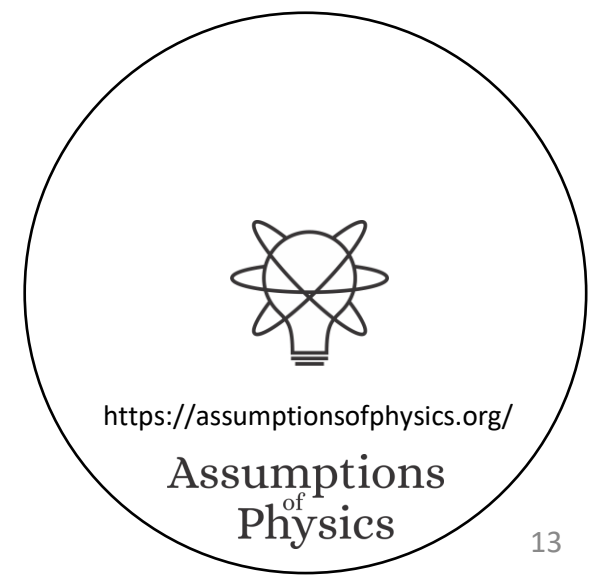
Third law of thermodynamics

Every substance has a finite positive entropy, but at the absolute zero of temperature the entropy may become zero, and does so become in the case of perfect crystalline substances.

G. N. Lewis and M. Randall, Thermodynamics and the free energy of chemical substances (McGraw-Hill, 1923)

$$S \geq 0$$

Classical physics allows arbitrarily small entropy...
... but thermodynamics doesn't!!!



Classical uncertainty principle

$$S(\rho) = -\int \rho \log h \rho \, dq \, dp$$

Fixes units.

h : Area in phase space over which a uniform distribution has zero entropy.

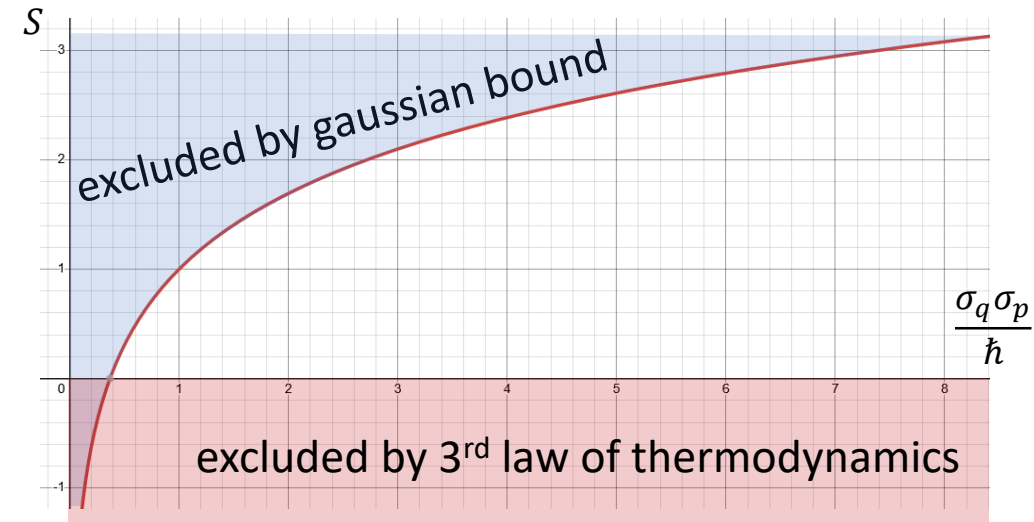
$$S(\rho) \leq \log 2\pi e \frac{\sigma_q \sigma_p}{h}$$

Gaussian maximizes entropy for a given uncertainty

$$\sigma_q \sigma_p \geq \frac{h}{2\pi e} e^{S(\rho)} = \frac{\hbar}{e} e^{S(\rho)}$$

Entropy puts a lower bound on the uncertainty

$$S \geq 0 \Rightarrow \sigma_q \sigma_p \geq \frac{\hbar}{e}$$



<https://assumptionsofphysics.org/autogen/briefs/004-ClassicalUncertaintyPrinciple.pdf>

<https://link.springer.com/article/10.1007/s10701-022-00555-z>

Comparing theories

$$\begin{array}{cc} \text{classical} & \text{quantum} \\ \sigma_q \sigma_p \geq \frac{\hbar}{e} & \sigma_q \sigma_p \geq \frac{\hbar}{2} \end{array}$$

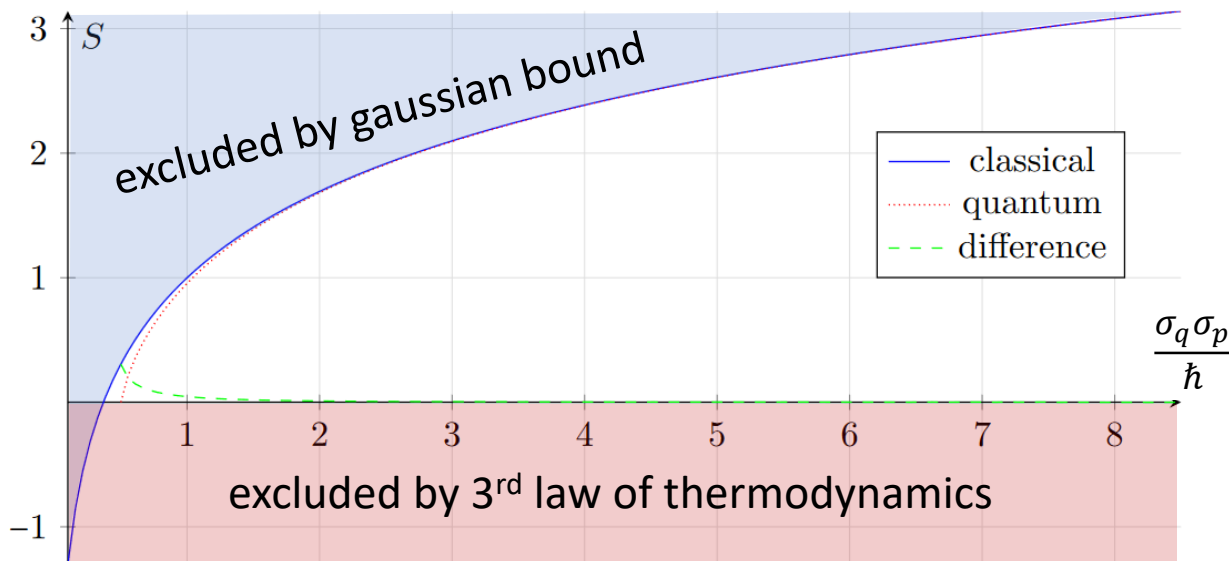
2.71828...

Entropy of quantum states is already non-negative

The gaussian bound quickly becomes very similar across theories

Quantum mechanics incorporates the third law.
Classical mechanics does not.

Is this the only difference?



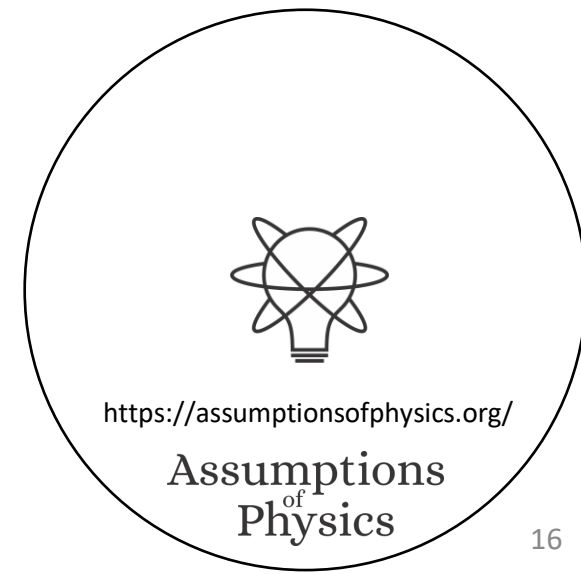
$$S_C = \ln e \sigma$$

$$S_Q = \left(\sigma + \frac{1}{2}\right) \ln \left(\sigma + \frac{1}{2}\right) - \left(\sigma - \frac{1}{2}\right) \ln \left(\sigma - \frac{1}{2}\right)$$



Suppose the lower bound on the entropy is the only difference,
then in the limit of high entropy of quantum mechanics we should
recover classical mechanics

Can we?





Classical mechanics as high entropy limit?

606 views • 10 months ago 06/01/2024

greetings



Manuele Landini <manulando@gmail.com>
To: carcassi@umich.edu

You replied to this message on 7/10/2024 10:00 AM.

Caro Gabriele,

Mi chiamo Manuele Landini e lavoro a Innsbruck (Austria) come senior scientist in un gruppo di fisica atomica sperimentale. Puoi vedere di cosa ci occupiamo sul nostro sito: <https://quantummatter.at>.

Ho visto un po' dei tuoi video su youtube. Mi sembra un progetto molto ambizioso, ma promettente. Mi farebbe piacere riuscire a spiegare agli studenti in futuro in termini piu' fisici concetti come le sovrapposizioni o il teorema spin-statistica.

Per la storia della metrica, da quel che ho capito hai bisogno di una metrica che non sia basata sull'entropia, visto che vuoi definire una distanza a entropia costante. Ci sono varie opzioni, ma la trace distance [Trace distance - Wikipedia](#) funziona perche' ha una proprieta' fondamentale che puoi usare. Chiamala: $T(\rho, \sigma)$

Se parti da stati puri, si riduce a $(1 - |\langle \psi | \phi \rangle|)^2$. Quindi per massimizzarla, scegli due stati ortogonali (non importa quali). Il massimo e' $T_0 = 1$. Una volta che hai questi stati, che hanno entropia 0, li puoi trasformare in stati con entropia finita (in particolare quelli con massima distanza) tramite una trace preserving map M .

Siccome T si contrae, hai che $T(M(\rho), M(\sigma)) \leq T(\rho, \sigma)$. L'uguale vale se la mappa e' unitaria. Così definisci un serie di step in cui la distanza massima decresce $T_{n+1} < T_n$, fino ad arrivare a 0 per stati fully mixed.

arXiv > quant-ph > arXiv:2411.00972

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Quantum Physics

[Submitted on 1 Nov 2024 (v1), last revised 3 Dec 2024 (this version, v2)]

Classical mechanics as the high-entropy limit of quantum mechanics

Gabriele Carcassi, Manuele Landini, Christine A. Aidala

We show that classical mechanics can be recovered as the high-entropy limit of quantum mechanics. That is, the high entropy masks quantum effects, and mixed states of high enough entropy can be approximated with classical distributions. The mathematical limit $\hbar \rightarrow 0$ can be reinterpreted as setting the zero entropy of pure states to $-\infty$, in the same way that non-relativistic mechanics can be recovered mathematically with $c \rightarrow \infty$. Physically, these limits are more appropriately defined as $S \gg 0$ and $v \ll c$. Both limits can then be understood as approximations independently of what circumstances allow those approximations to be valid. Consequently, the limit presented is independent of possible underlying mechanisms and of what interpretation is chosen for both quantum states and entropy.

Comments: 14 pages, 3 figures

Subjects: Quantum Physics (quant-ph)

Cite as: arXiv:2411.00972 [quant-ph]

(or arXiv:2411.00972v2 [quant-ph] for this version)

<https://doi.org/10.48550/arXiv.2411.00972>

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[v1] Fri, 1 Nov 2024 18:48:04 UTC (19 KB)

[v2] Tue, 3 Dec 2024 13:52:45 UTC (20 KB)

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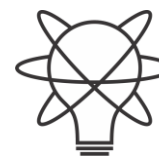
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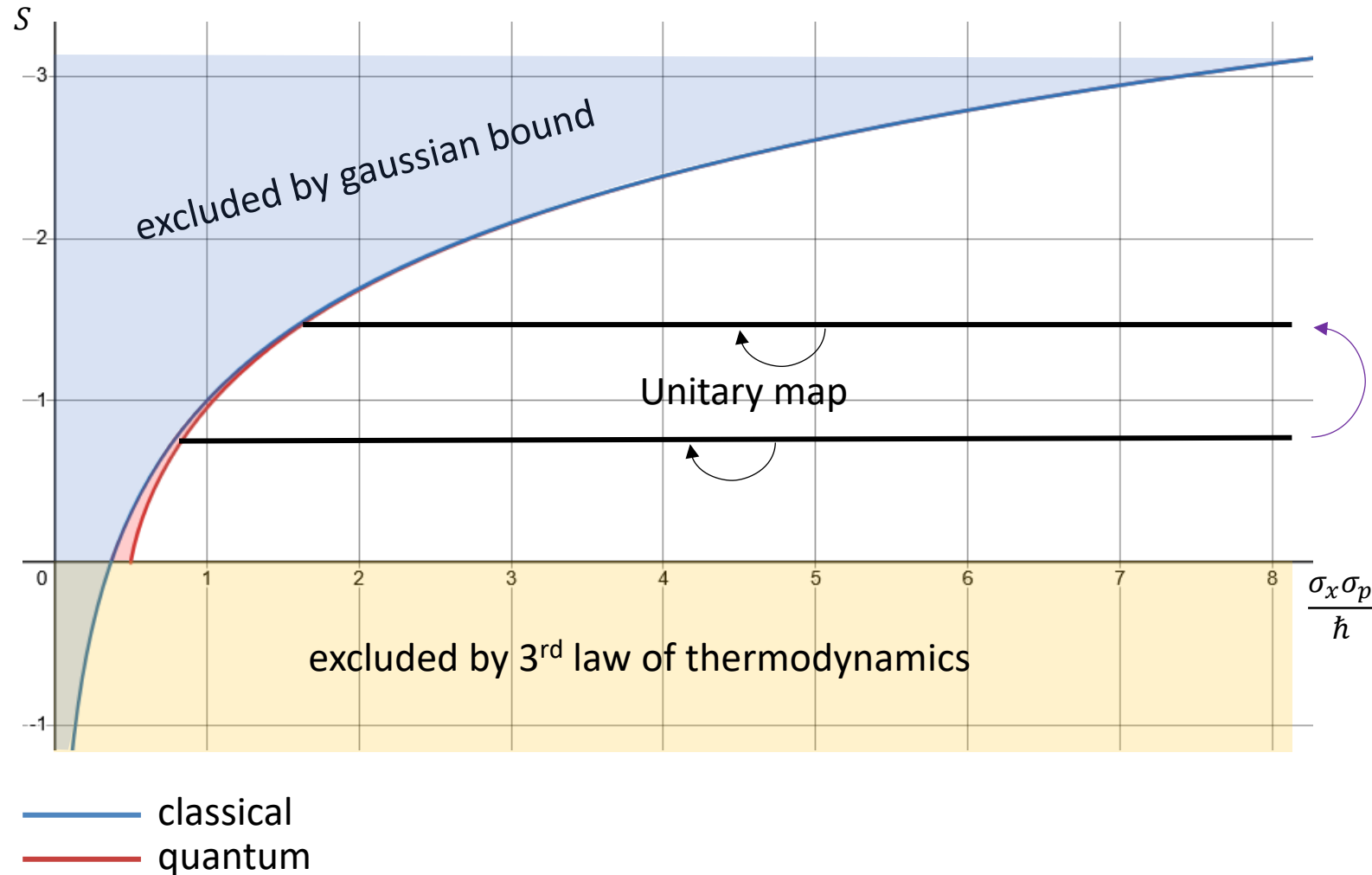


<https://assumptionsofphysics.org/>

Assumptions
of
Physics

Looking for a map $R(\rho)$ that increases entropy of all mixed states, such that every level set of entropy maps to another level set

In the limit!



⇒ Unitary maps must be mapped to unitary maps



<https://arxiv.org/pdf/2411.00972>

<https://iopscience.iop.org/article/10.1088/1402-4896/ae3a20>

In classical mechanics – stretch phase space

$$S(R(\rho)) = S(\rho) + \log \lambda \iff \{R(x), R(p)\} = \lambda \{x, p\}$$

Jacobian is a constant:
all volumes rescaled
by the same factor

In quantum mechanics

Stretching map $[R(X), R(P)] = \lambda [X, P]$

Pure stretching map $T(X) = \sqrt{\lambda} X \quad T(P) = \sqrt{\lambda} P$

Need to take care of
operator ordering!!!

Infinitesimal pure
stretching map
must satisfy:

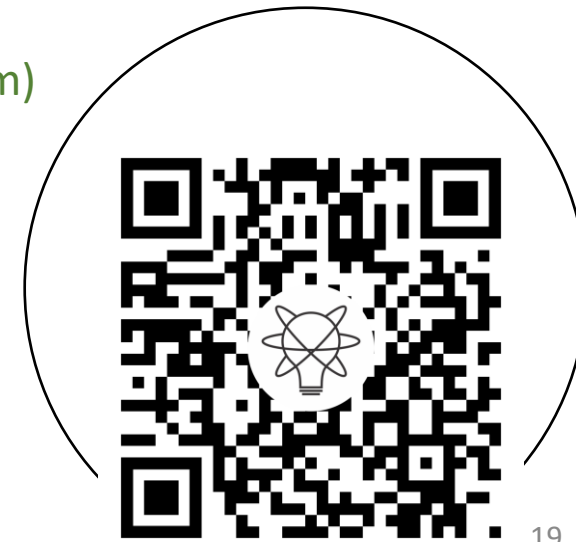
$$\frac{dX}{dt} = \frac{i}{\hbar} [H, X] + \gamma \left(L^\dagger X L - \frac{1}{2} \{L^\dagger L, X\} \right)$$

Lindblad equation
(open quantum system)

with $H = 0$, $\gamma = \lambda$, and $L = a^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(X + \frac{i}{m\omega} P \right)$

No unitary/conservative part:
purely entropy-increasing

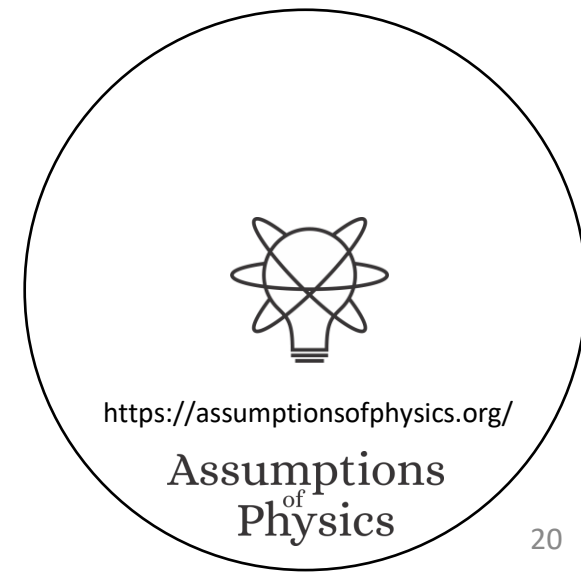
Need anti-normal ordering ($aa^\dagger$), associated with
Husimi Q quasiprobability distribution in phase space



Effects of stretching map on phase-space representations

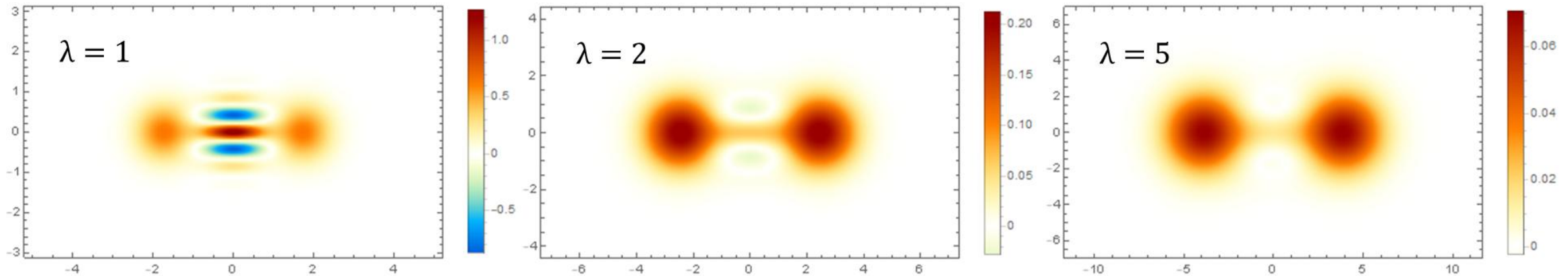
Husimi Q distribution is simply stretched

Wigner function is stretched and convolved with a Gaussian,
and coincides with the Husimi Q in the limit $\lambda \rightarrow \infty$

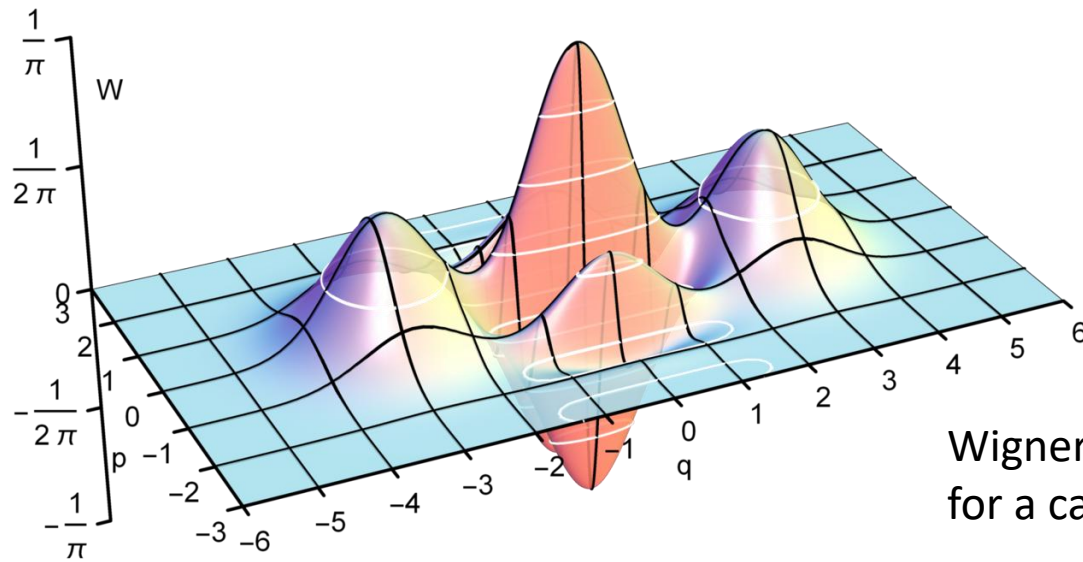


Stretching a cat state

- a quantum superposition of two macroscopically distinct states



Physica Scripta 101, 065105 (2026)



Wigner quasiprobability distribution
for a cat state

Wikipedia



Another perspective: move the pure states to minus infinite entropy

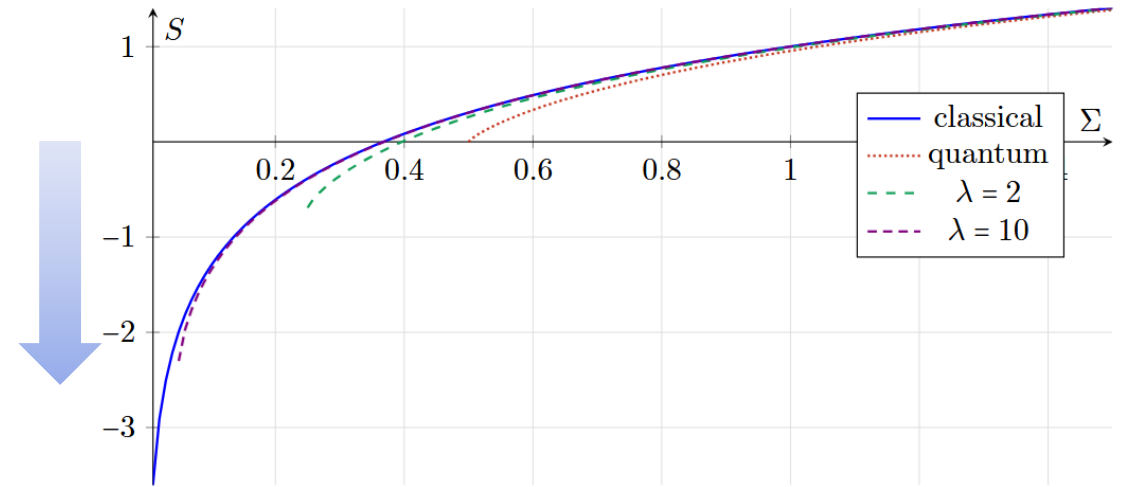
Instead of

$$[X, P] = i\hbar \quad [T(X), T(P)] = \lambda i\hbar$$

Redefine original space such that

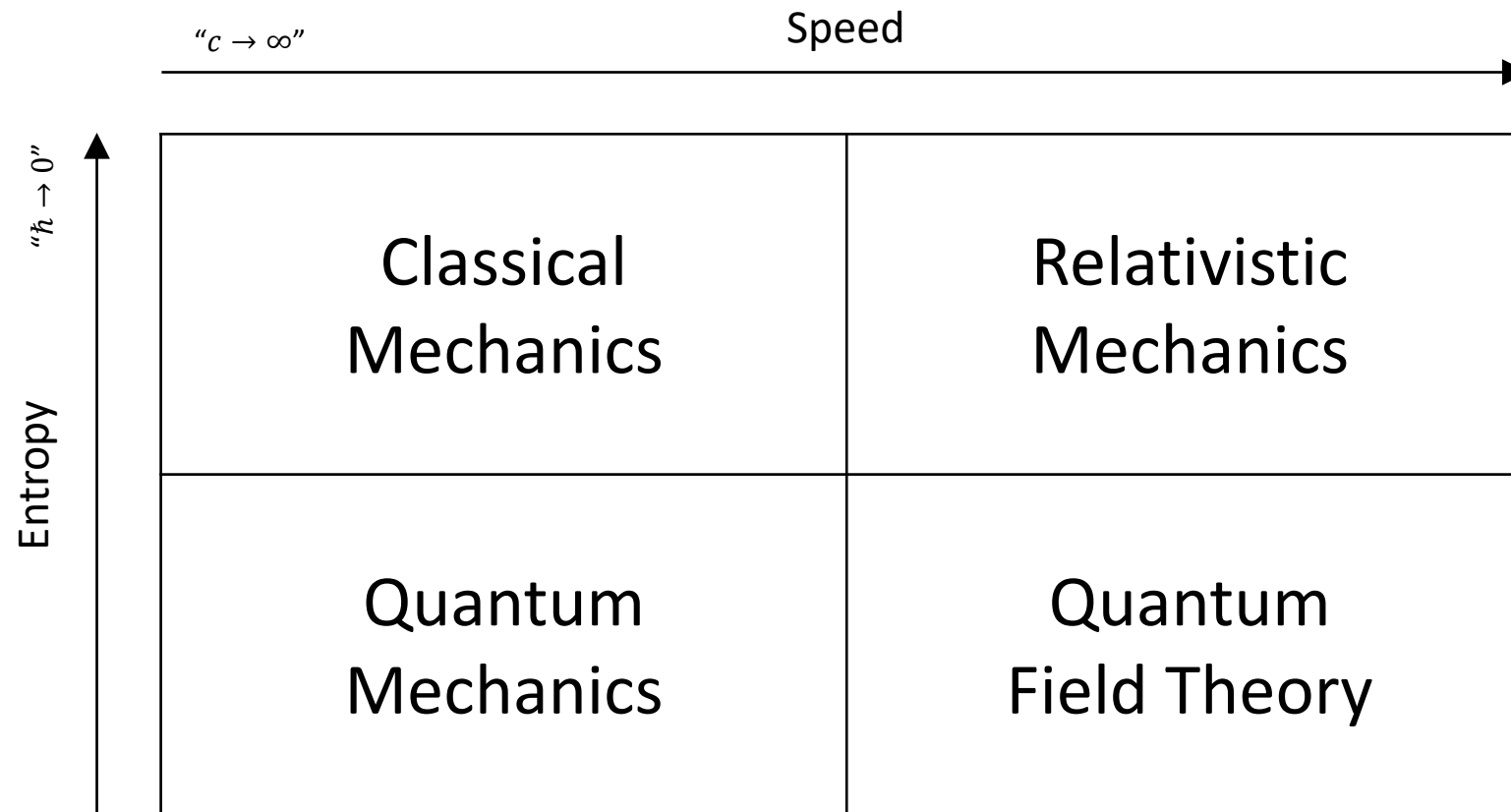
$$[\hat{X}, \hat{P}] = \frac{i\hbar}{\lambda} \quad [T(\hat{X}), T(\hat{P})] = i\hbar$$

$$\lambda \rightarrow \infty \Rightarrow \frac{\hbar}{\lambda} \rightarrow 0$$



Mathematically equivalent to lowering the entropy of a pure state to $-\infty$, or $\hbar \rightarrow 0$ (group contraction)





No-mechanism limit
(same as non-relativistic limit)

Physica Scripta 101, 065105 (2026)

Christine Aidala - University of Michigan



$$\{A, B\} \longrightarrow \frac{[A, B]}{i\hbar}$$

Dirac's correspondence principle: give me a theory with an entropic lower bound that recovers the classical one at high entropy

Only one way to do it

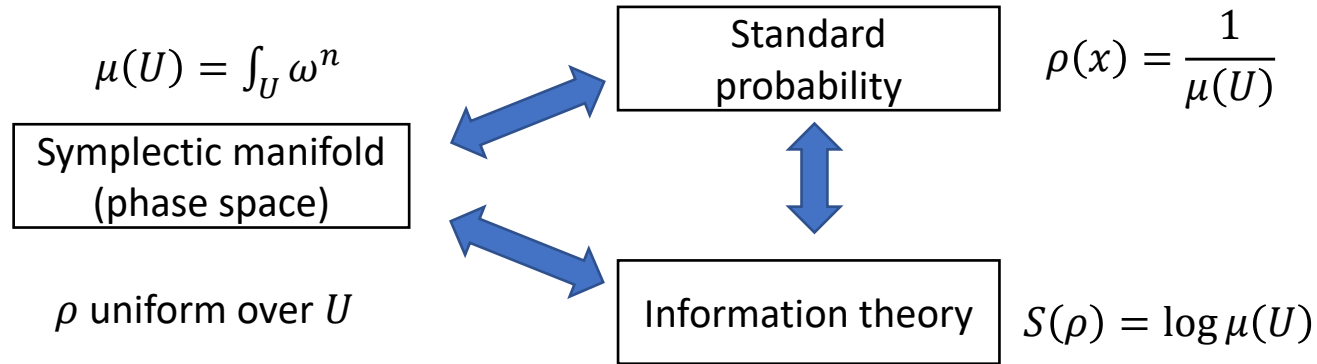
Moyal bracket is the unique one-parameter Lie-algebraic deformation of the Poisson bracket

Quantizing a classical theory
means putting a lower bound
on the entropy



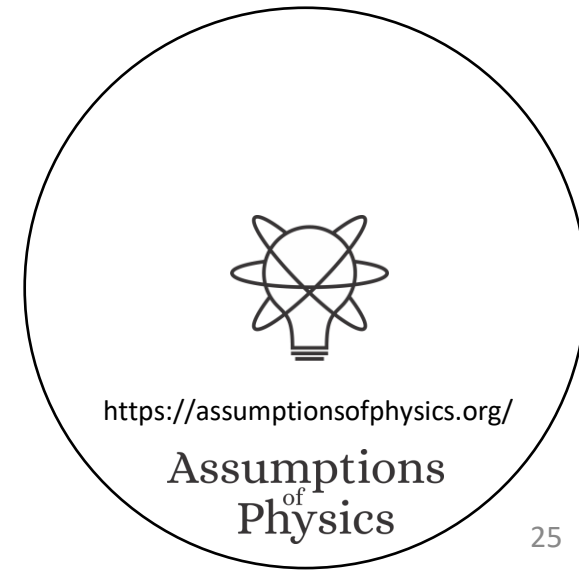
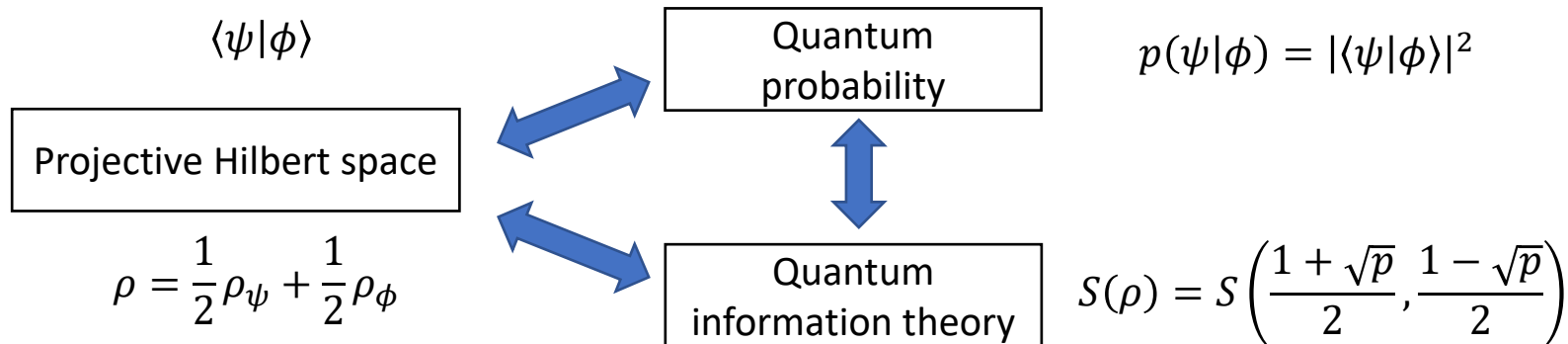
Geometry is entropy!

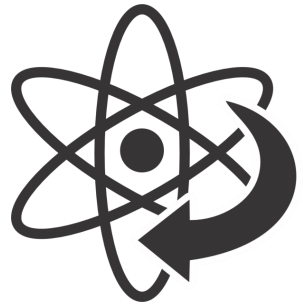
The geometric structures of both classical and quantum mechanics are equivalent to the entropic structures



Thermodynamics/Statistical mechanics are not built on top of mechanics

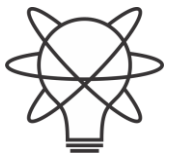
Mechanics is the ideal case of thermodynamics/statistical mechanics





**Reverse
Physics**

Illuminating paradoxes and developing standards of rigor



<https://assumptionsofphysics.org/>

**Assumptions
of
Physics**

Poor dimensional checking in statistical mechanics

Example 1: Definition of Gibbs entropy

$$-\int_X \rho \log \rho \, dq^i \, dp_j \Rightarrow -\int_X \rho \log(h^n \rho) \, dq^i \, dp_j$$

Density in phase space
is not a pure number!

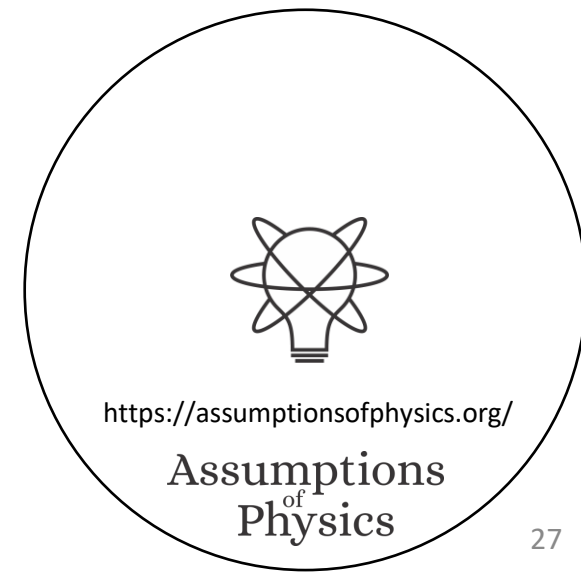
Need to make a dimensionless argument by multiplying by the
volume in phase space corresponding to zero entropy for a uniform
distribution

Example 2: When deriving Sackur–Tetrode equation for the entropy of a monatomic ideal gas

$$\Omega_N = \frac{1}{N!} \frac{V^N}{h^{3N}} \frac{2\pi^{3N/2}}{(\frac{3N}{2} - 1)!} (\sqrt{2mU})^{3N-1} \approx \frac{1}{N!} \frac{V^N}{h^{3N}} \frac{\pi^{3N/2}}{(3N/2)!} (\sqrt{2mU})^{3N}.$$

You can't “approximate physical dimensions”!!!

$$\text{length}^N \neq \text{length}^{N-1}$$



Inconsistent expectations in Hilbert spaces

Quantum Stud.: Math. Found. **12**, 13 (2025)

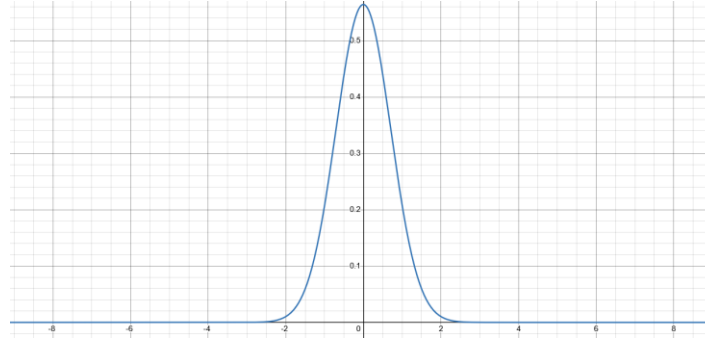
$$\psi(x) = \sqrt{\frac{e^{-x^2}}{\sqrt{\pi}}}$$

$$\int |\psi|^2 dx = 1$$

$$\rho_\psi(x) = \frac{e^{-x^2}}{\sqrt{\pi}}$$

$$\langle X^2 \rangle_\psi = \frac{1}{2}$$

Expectation of position² finite in terms of x ...



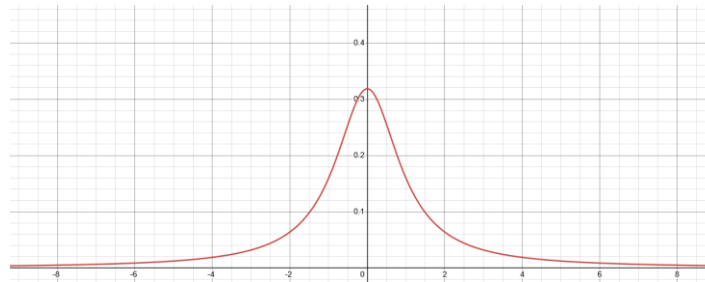
$$\phi(y) = \sqrt{\frac{1}{\pi(y^2 + 1)}}$$

$$\int |\phi|^2 dy = 1$$

$$\rho_\phi(y) = \frac{1}{\pi(y^2 + 1)}$$

$$\langle Y^2 \rangle_\phi \rightarrow \infty$$

...but infinite in terms of y !



Re-expressing the same state in a different coordinate system can turn a finite expectation value into an infinite one—unphysical!!!

$$y = \tan\left(\frac{\pi}{2} \operatorname{erf}(x)\right)$$

$$\psi(y) = \psi(x) \sqrt{\frac{dx}{dy}}$$

Expectations can also oscillate between finite and infinite in finite time—unphysical!!!

$$x(x_0, t) = x_0 \cos^2 \frac{\pi t}{2} + \tan\left(\frac{\pi}{2} \operatorname{erf}(x_0)\right) \sin^2 \frac{\pi t}{2}$$

Every continuous linear operator defined on the whole Hilbert space is bounded $\Rightarrow$ position/momentum/energy/number of particles are not defined on the whole Hilbert space!!!

<https://arxiv.org/pdf/2308.06669>

<https://link.springer.com/article/10.1007/s40509-024-00357-0>



Inconsistent convergence in Hilbert spaces

Consider the sequence of quantum harmonic oscillator states given by:

$$|\psi_i\rangle = \sqrt{\frac{i-1}{i}} |0\rangle + \sqrt{\frac{1}{i}} |i\rangle \text{ for } i = 1, \dots, \infty$$

In the limit that $i \rightarrow \infty$, the state converges to the ground state:

$$|\psi_i\rangle \rightarrow |0\rangle$$

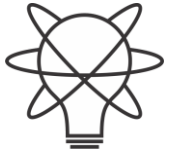
But the energy level converges to the first level—inconsistent!!

$$\begin{aligned} \langle \psi_i | \hat{N} | \psi_i \rangle &= \frac{i-1}{i} \langle 0 | \hat{N} | 0 \rangle + \frac{1}{i} \langle i | \hat{N} | i \rangle \\ &= \frac{i-1}{i} (0) + \frac{1}{i} (i) = 1 \rightarrow 1 \end{aligned}$$

<https://assumptionsofphysics.org/autogen/briefs/006-InconsistentConvergence.pdf>



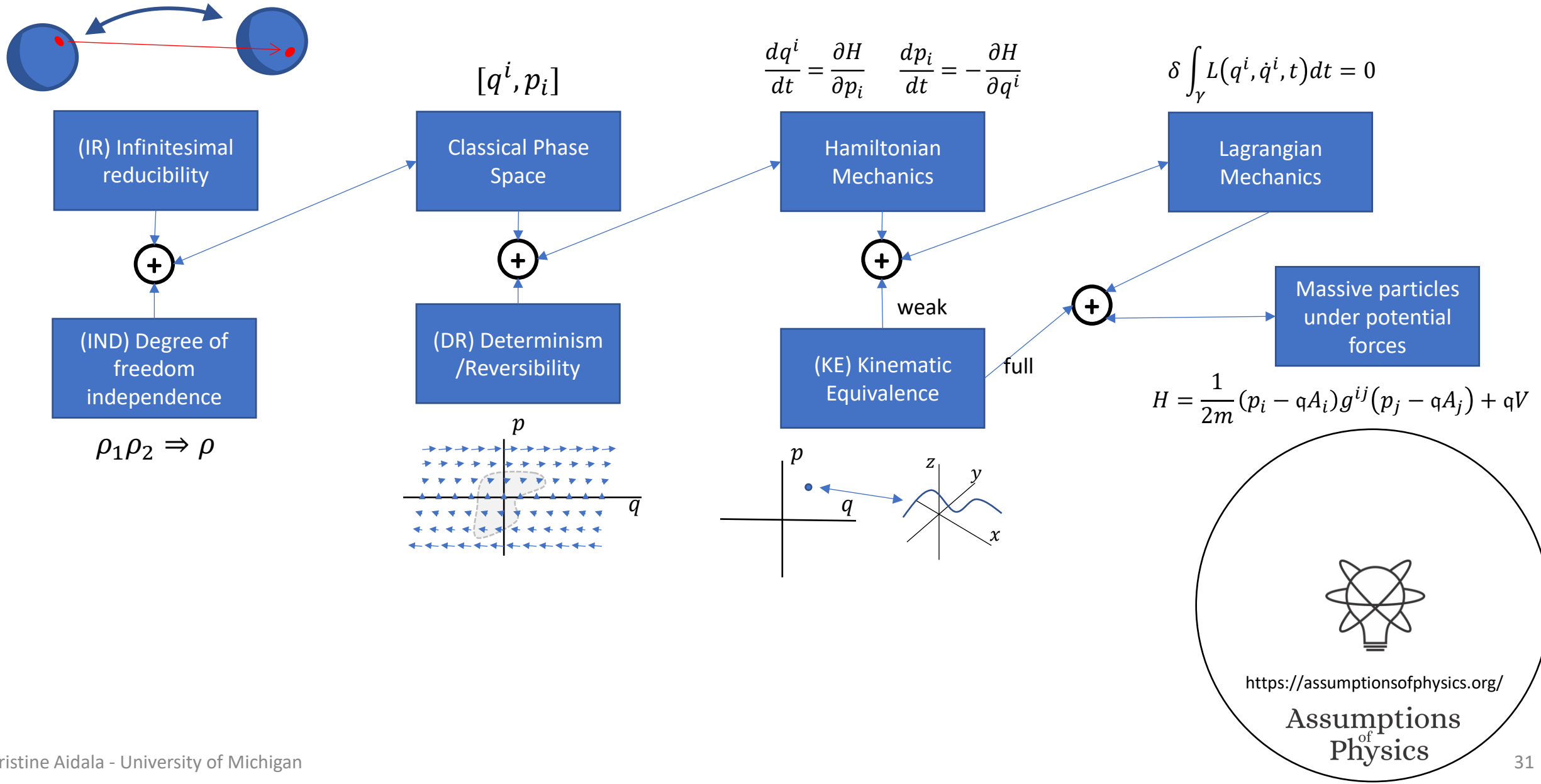
Identifying logical dependencies and clarifying base conditions



<https://assumptionsofphysics.org/>

**Assumptions
of
Physics**

Assumptions of classical mechanics



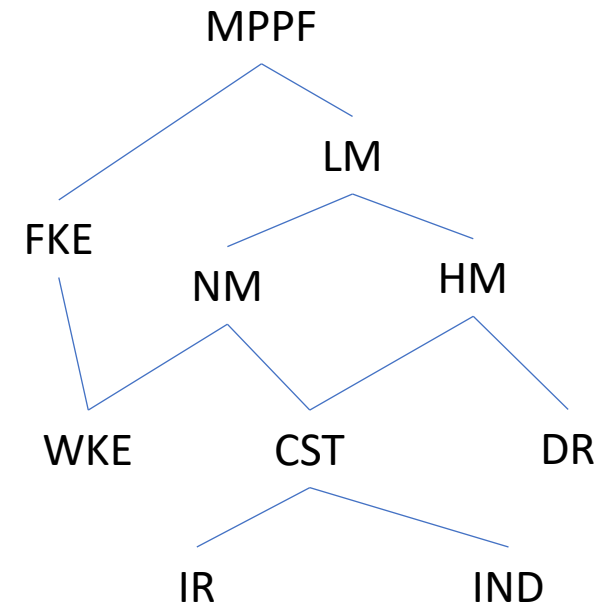
Classical conditions and their relationships

Core assumptions

IR: Infinitesimal Reducibility
 IND: INDependent DOF
 W/FKE: Weak/Full Kinematic Equivalence
 DR: Determinism+Reversibility

Physical theories

CST: Classical States
 NM: Newtonian Mechanics
 HM: Hamiltonian Mechanics
 LM: Lagrangian Mechanics
 MPPF: Massive Particles under Potential Forces



Each assumption is an equivalence class of multiple conditions

There is a linear relationship between conjugate momentum and velocity

(FKE-LIN)

At each position, there exists a local inertial frame

(FKE-INER)

The position fully defines the units of all state variables, and therefore an invertible transformation between momentum and velocity

(FKE-UNIT)

Density expressed in velocity at the same position is proportional to the density over states

(FKE-PROP)

Uniform distributions along momentum correspond to uniform distributions along velocity

(FKE-UNIF)

The system under study is a massive particle under scalar and vector potential forces

(FKE-POT)

The Jacobian of the transformation between state variables and kinematic variables is a non-singular function of position only.

(FKE-NSIN)

Densities over phase space can be expressed in terms of position and velocity by rescaling the value at each point: $\rho(x^i, v^j) |J(x^i)| = \rho(q^i, p_j)$.

(FKE-DEN)

Areas and volumes in phase space can be expressed in kinematic variables, and the transformation depends on position only: $dx^1 \dots dx^n dv^1 \dots dv^n = |J(x^i)| dq^1 \dots dq^n dp_1 \dots dp_n$.

(FKE-VOL)

The symplectic form ω_{ab} can be expressed in kinematic variables, and its components are a linear function of velocity.

(FKE-SYMP)

<https://assumptionsofphysics.org/book/AssumptionsOfPhysicsV3.0.pdf#section.1.1.4>



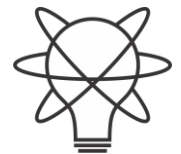
Currently applying Reverse Physics methodology to quantum mechanics

Bunch of things bundled together

Condition QSTO:H (Quantum states and observables). *The state space of a quantum system is represented by the projective space $\mathcal{P}(\mathcal{H})$ of a separable complex Hilbert space $\mathcal{H}$. The ensemble space is represented by the density operators $\mathcal{D}(\mathcal{H})$. Observables are represented by self-adjoint operators. The conditional probability is given by the Born rule $p(\psi|\phi) = \frac{\langle\phi|\psi\rangle\langle\psi|\phi\rangle}{\langle\psi|\psi\rangle\langle\phi|\phi\rangle}$. The entropy is given by the von Neumann entropy $S(\rho) = -\text{tr}(\rho \log \rho)$.*

How do we analyze this properly?

<https://arxiv.org/pdf/2608.27543>



<https://assumptionsofphysics.org/>

Assumptions
of
Physics

Reverse Physics for quantum mechanics: Some results so far

QP-RAY – Pure states are rays of a complex Hilbert space Can be classical

+

Not sufficient to distinguish QM from CM, e.g. state space of a classical magnetic dipole is a sphere, which is the complex projective line.

QE-DENS – Ensembles are density matrices

Cannot be classical

+

In CM, the ensemble space is a simplex. In QM, it is not.

BR-BORN – Orthogonal states are mutually exclusive

Must be quantum

=

The fact that orthogonal vectors represent mutually exclusive ensembles is an additional assumption.

QST:H – Quantum states on a Hilbert space

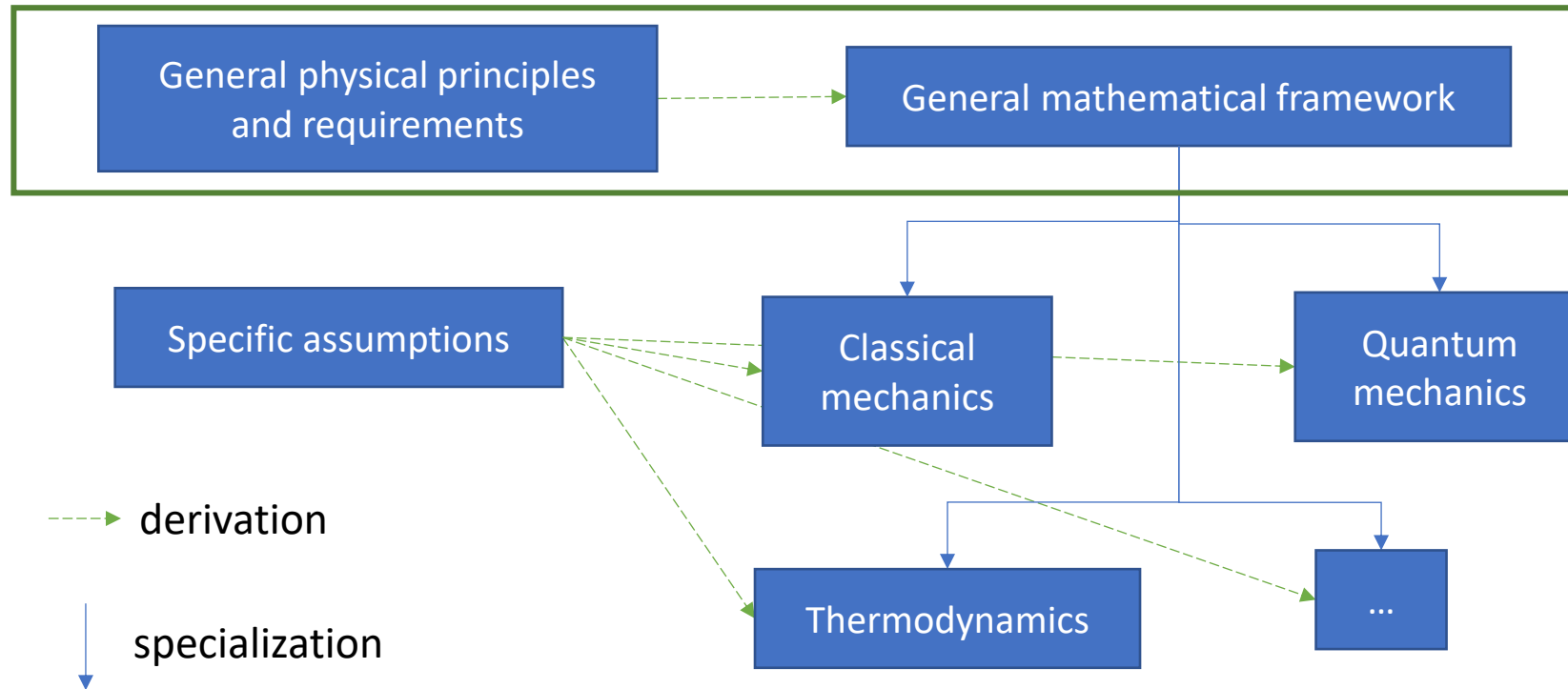


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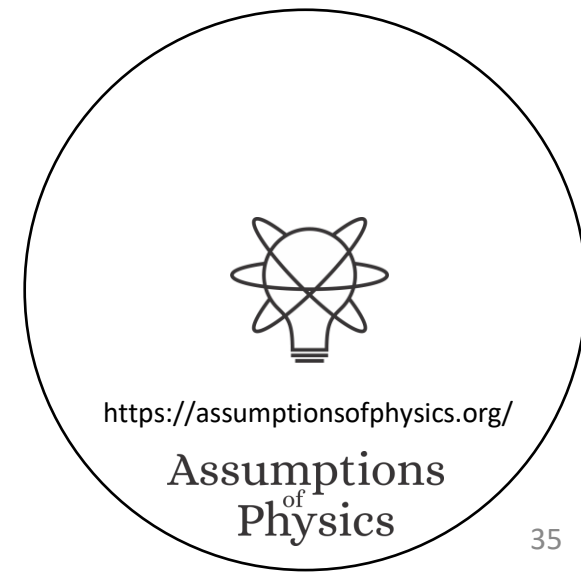
Assumptions
of
Physics

Developing a base theory for **all** of physics

Collection of conditions that **MUST** be true in all physical theories



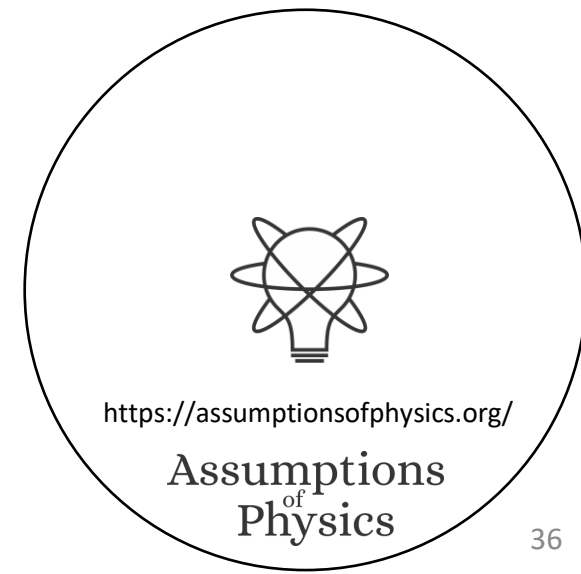
Base theory $\neq$ Theory of everything



Foundational pillars

Core constitutive requirements of physical theories:

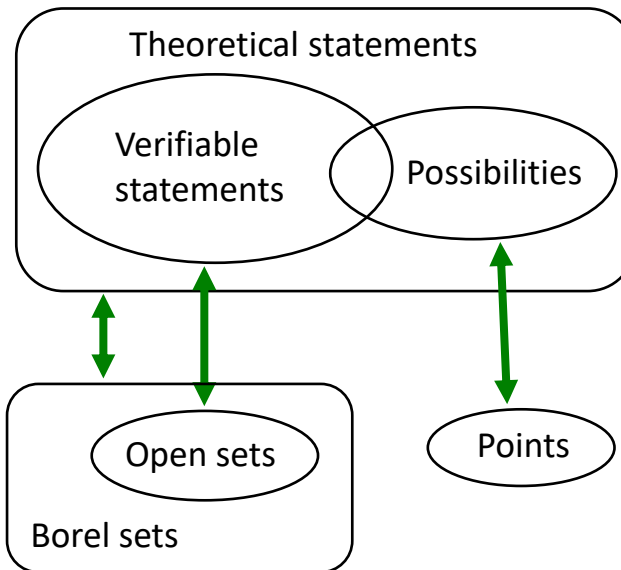
- Experimental verifiability
- Reproducibility and ensembles
- Ensemble variability
- Equilibria



Experimental verifiability

Science is evidence based $\Rightarrow$ Statements are associated to experimental tests

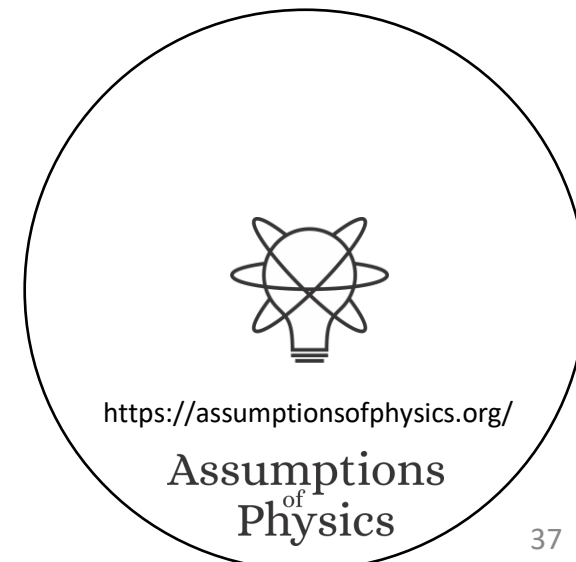
Verifiable statement	Test Result
T	SUCCESS (in finite time)
F	UNDEFINED FAILURE (in finite time)



s	Test	
T	S	$int(A)$ corresponds to the verifiable part of a statement
	U	∂A corresponds to the undecidable part of a statement
F	U	
	F	$ext(A)$ corresponds to the falsifiable part of a statement

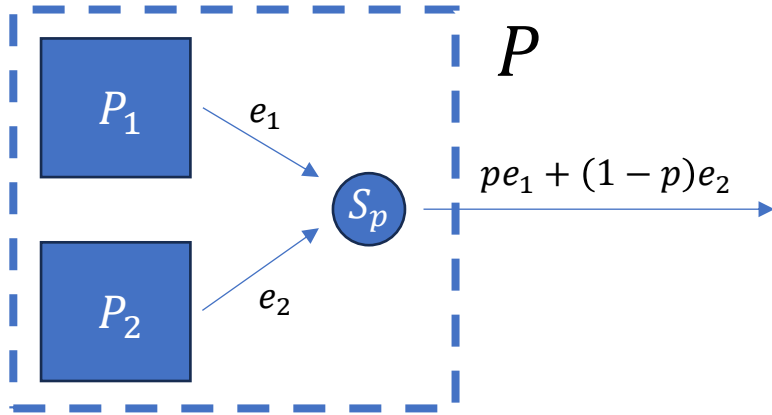
Experimental verifiability
 $\Rightarrow$ **topology and σ -algebra**

Experimental relationships must be
topologically continuous



Reproducibility and ensembles

Scientific laws describe reproducible relationships $\Rightarrow$ Ensembles

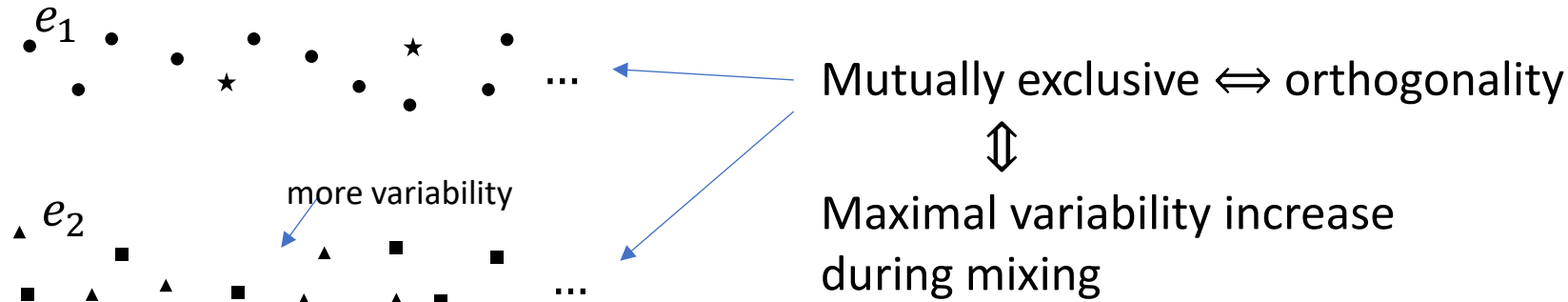


Ensembles are outputs of repeatable processes
Statistical mixing $\Rightarrow$ Linear structures in physics

Note: mixing coefficients are parameters of the preparation,
NOT probabilities in the ordinary sense

Ensemble variability

Variability of the elements within an ensemble $\Rightarrow$ Entropy



Variability $\Rightarrow$ Probabilities, inner product and geometry

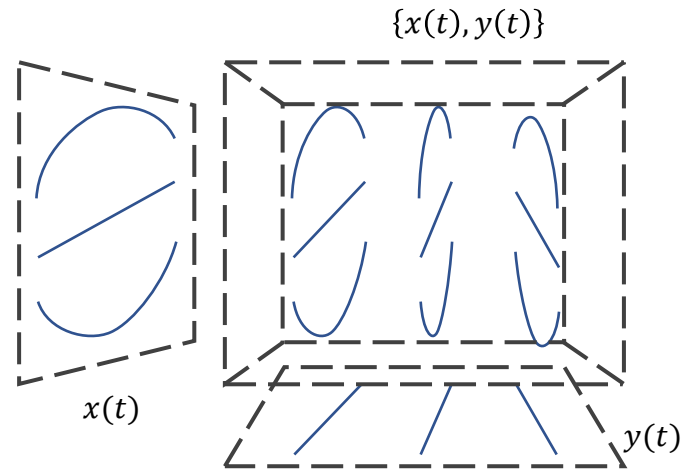
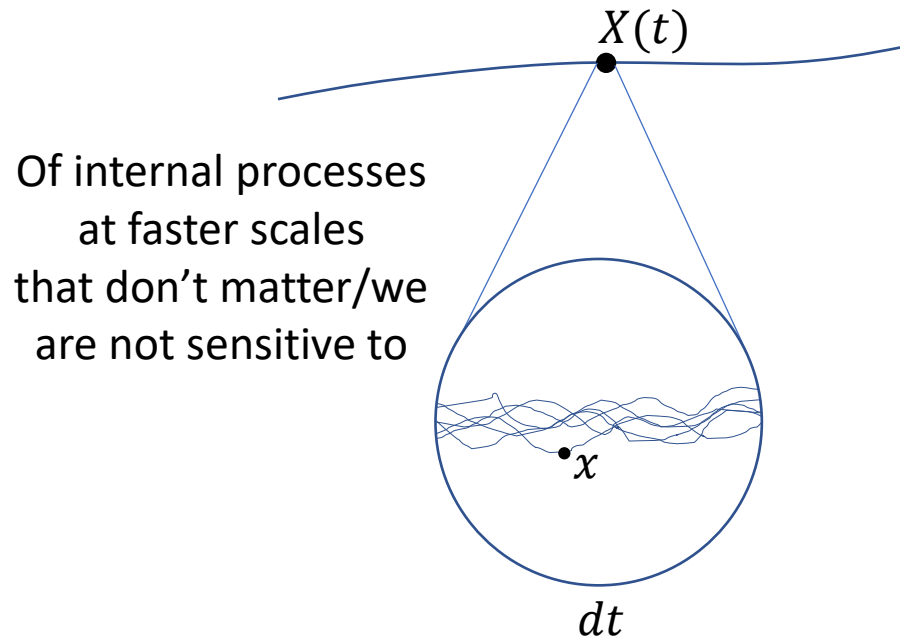
Probability $\Leftrightarrow$ Mixtures of mutually exclusive ensembles

Classical mechanics/probability $\Leftrightarrow$ All pure states are mutually exclusive

Pure states $\Leftrightarrow$ Best preparations $\nRightarrow$ No variability

Equilibria

States must exist for at least a short time $\Rightarrow$ Equilibria

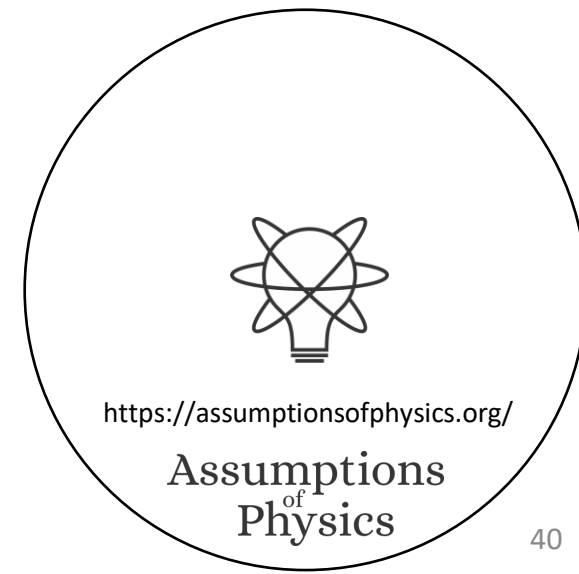


Of system/environment
interactions
that don't matter/we
are not sensitive to

Least developed part

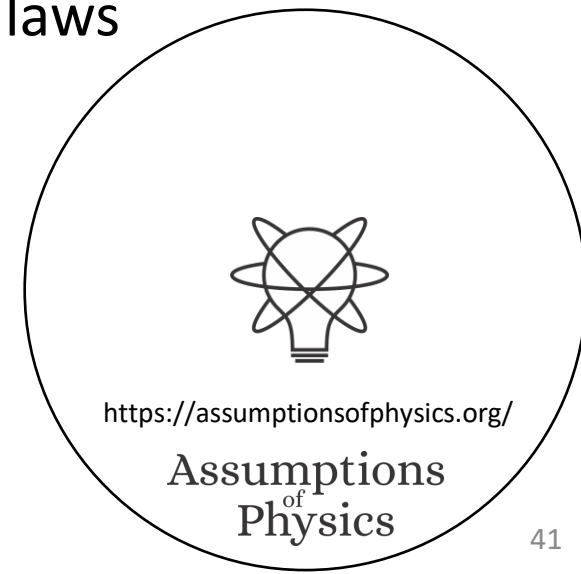
This requirement allows us to properly talk about different physical objects

Equilibria $\Rightarrow$ Symmetries, differentiability, Lie algebras



Wrapping it up

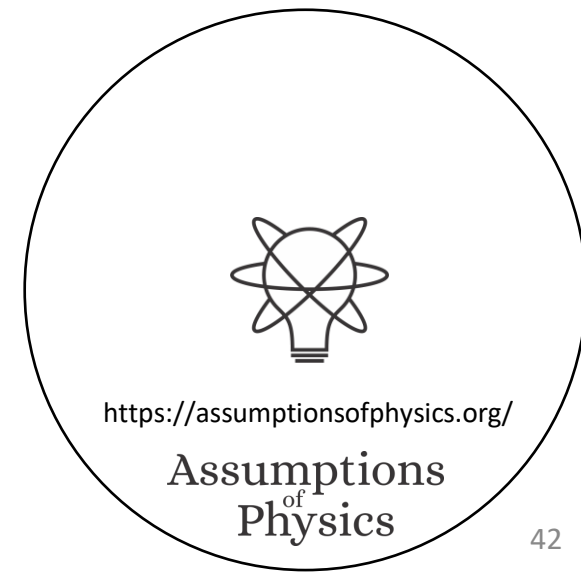
- Assumptions of Physics: different approach to the foundations of physics
 - No interpretations, no theories of everything: physically meaningful starting points from which we can rederive the laws and the mathematical frameworks they need
 - Physical theories are models that can be empirically tested
 - Need to clarify exactly what the realm of applicability of each model is
- Reverse Physics: reverse engineer principles from the known laws
 - Find physical principles and assumptions from which to rederive the laws
 - These starting points must have a clear intuitive meaning for the majority of physicists and engineers
 - Investigate physics as a whole (no theory is “more fundamental”)



Reverse Physics tells us there are not many “independent concepts”
in a physical theory

Extracting principles/assumptions behind the laws gives us solid
intuition that cuts across fields and leads to new insights,
potentially opening new avenues of research

Offers a powerful new methodology for the
foundations of physics, more in line with
foundational approaches of other disciplines



Assumptions of Physics is an open research program

Our main output is an open access book: <https://assumptionsofphysics.org/book/>

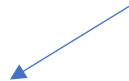
All our material is developed on GitHub: <https://github.com/assumptionsofphysics>

One YouTube channel dedicated to publicize results: <https://www.youtube.com/user/gcarcassi>

Another YouTube channel dedicated to research: <https://www.youtube.com/@AssumptionsofPhysicsResearch>

Activities coordinated through a Discord server (contact info@assumptionsofphysics.org for an invite)

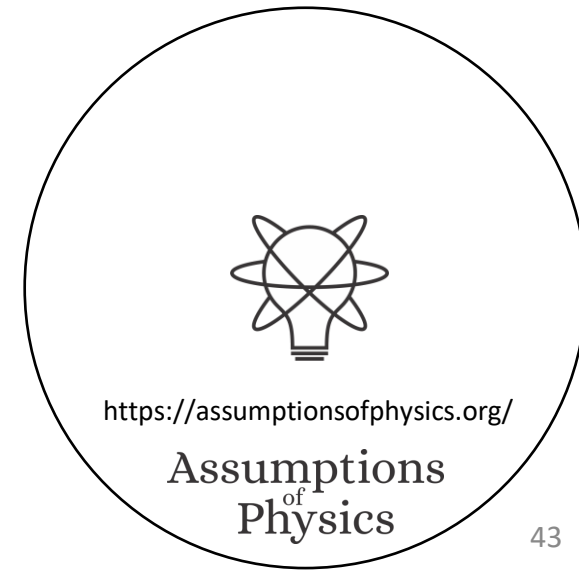
Livestream
discussions



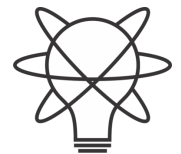
Always looking for experts to gain insights and/or help

Always looking for collaborations

Always looking for editors/journals/conferences that are sympathetic to the mission



Extra



<https://assumptionsofphysics.org/>

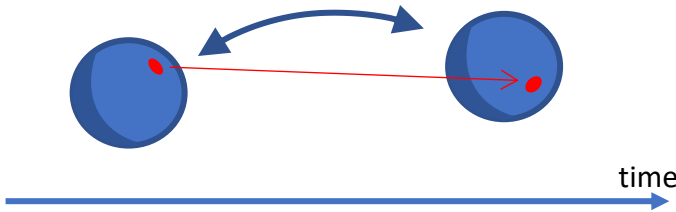
**Assumptions
of
Physics**

Main goal of the Assumptions of Physics program

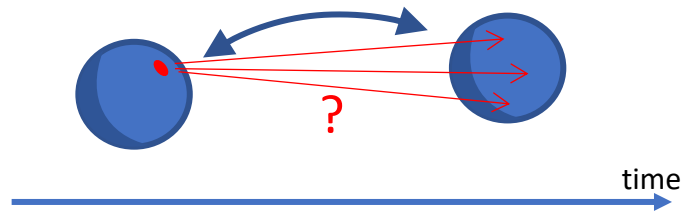
Identify a handful of physical starting points from which the basic laws can be rigorously derived

For example:

Infinitesimal reducibility $\Rightarrow$ Classical state



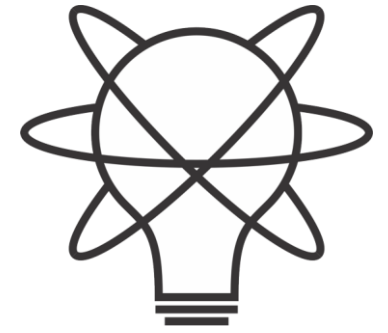
Irreducibility $\Rightarrow$ Quantum state



This also requires rederiving all mathematical structures from physical requirements

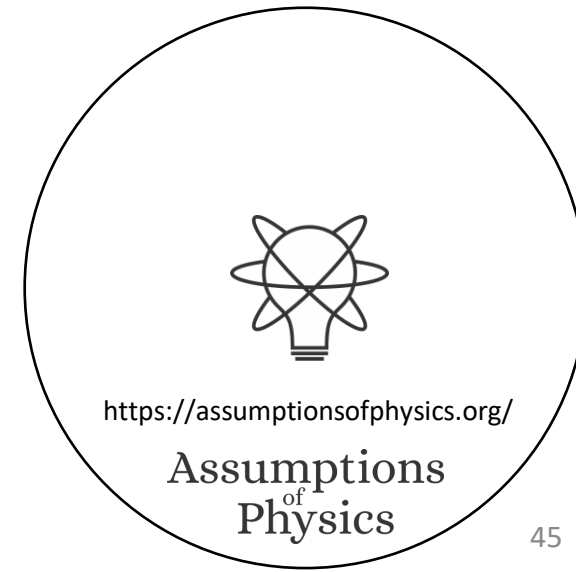
For example:

Science is evidence based $\Rightarrow$ scientific theory must be characterized by experimentally verifiable statements $\Rightarrow$ topologies and σ -algebras



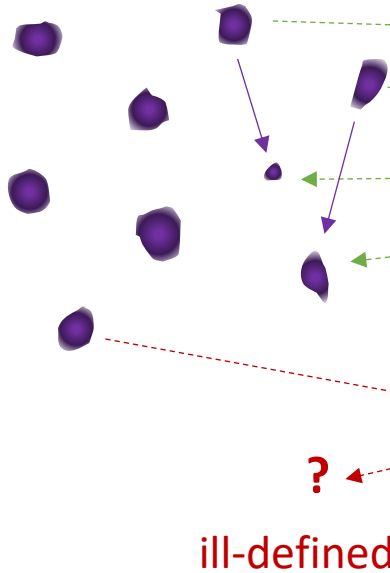
Assumptions
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Physics

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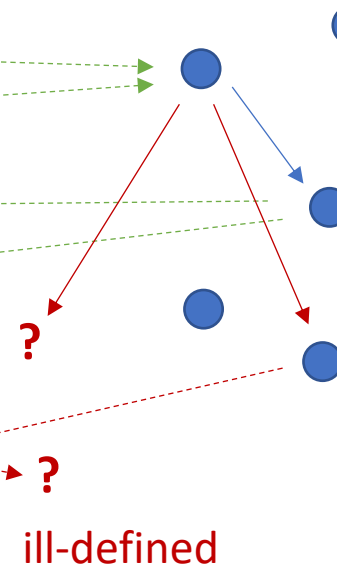
Physical world (informal system)

well-defined
physical
objects

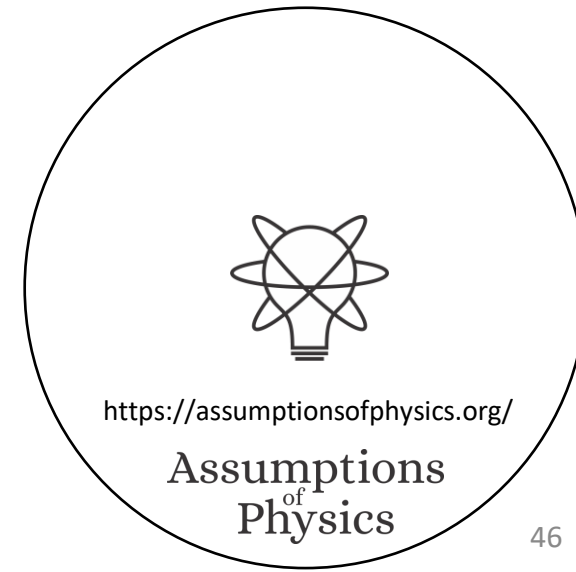


Mathematical representation (formal system)

well-defined
mathematical
objects

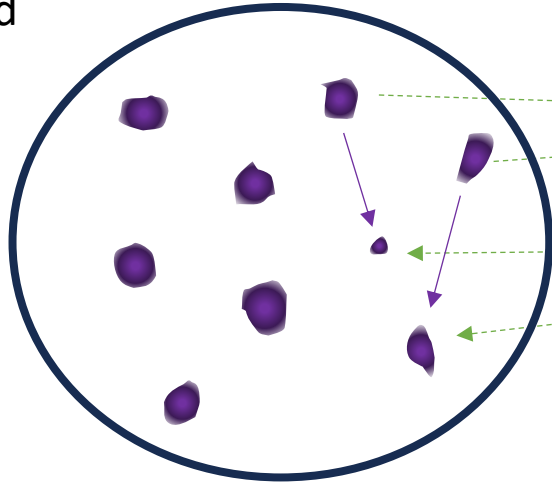


Current state of the art in theoretical physics



Physical world (informal system)

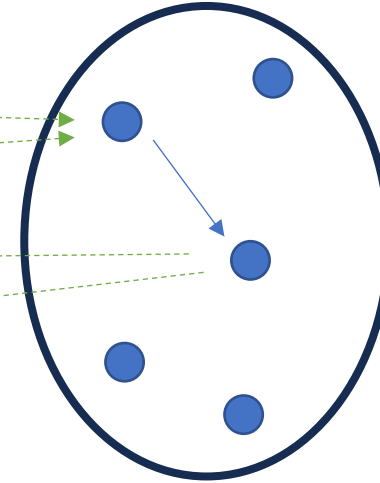
well-defined
physical
objects



Physical specifications

Mathematical representation (formal system)

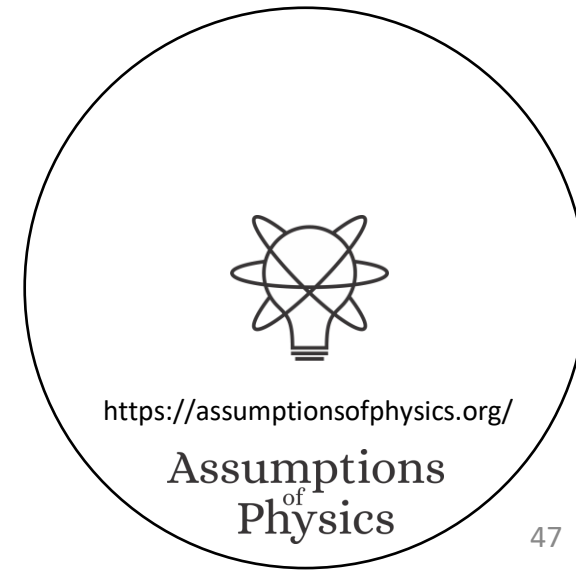
well-defined
mathematical
objects



Mathematical definition



A mathematical definition is **physical** if it captures and only captures an aspect of the physical system



Superposition is multi-decomposition

Property of the **vector space representation** of states in quantum mechanics

$$\text{Superposition: } |\phi\rangle = c_1|\psi_1\rangle + c_2|\psi_2\rangle$$



Impossible in classical probability
(i.e. Choquet simplex)

$$\text{Multi-decomposition: } \rho = p_\phi \rho_\phi + p_{\hat{\phi}} \rho_{\hat{\phi}} = p_1 \rho_{\psi_1} + p_2 \rho_{\psi_2}$$

Linear transformations



Output of mixtures = mixture of outputs

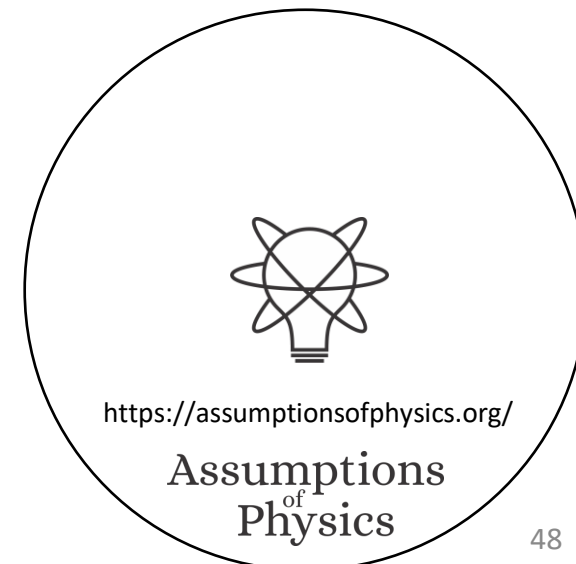
Property of **distinguishability of preparations**

Indistinguishability of different mixtures



No physical process can depend on them

<https://assumptionsofphysics.org/autogen/briefs/002-SuperpositionAndMultipleMixtures.pdf>



Non-linear representation in QM

This map introduces a total phase based on all coefficients

$$m(c_+|z^+\rangle + c_-|z^-\rangle) = e^{i\theta \frac{|c_+|^2}{|c_+|^2 + |c_-|^2}} (c_+|z^+\rangle + c_-|z^-\rangle)$$

const. $\theta \in (0, 2\pi)$ Non-linear map...

... that preserves the physics
(i.e. Born rule)

$$\frac{\langle m(\psi) | m(\phi) \rangle \langle m(\phi) | m(\psi) \rangle}{\langle m(\psi) | m(\psi) \rangle \langle m(\phi) | m(\phi) \rangle} = \frac{\langle \psi | \phi \rangle \langle \phi | \psi \rangle}{\langle \psi | \psi \rangle \langle \phi | \phi \rangle}$$

Linearity is a mathematical convenience,
not a physical necessity



Geometry is entropy

Classical mechanics

$$S(\rho_U) = \log \mu(U) = \log \int_U \omega$$

Symplectic form determines and is determined by the entropy of all uniform joint and marginal (DOF) distributions

$$\omega_{ab} = \begin{bmatrix} -mG_{\alpha\beta\gamma}u^\gamma - qF^{\alpha\beta} & mg_{\alpha\beta} \\ -mg_{\alpha\beta} & 0 \end{bmatrix}$$

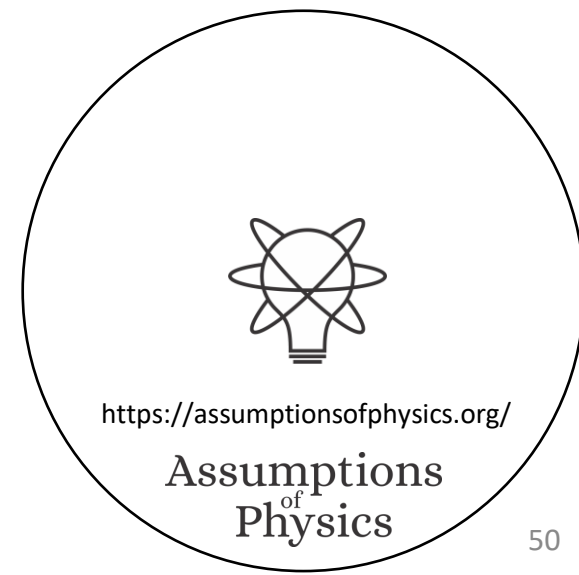
$\partial_\alpha g_{\beta\gamma} - \partial_\beta g_{\alpha\gamma}$

If you write the symplectic form in terms of position and velocity rather than position and conjugate momentum, metric tensor appears

Quantum mechanics

$$S\left(\frac{1}{2}\rho_\psi + \frac{1}{2}\rho_\phi\right) = I\left(\frac{1+|\langle\psi|\phi\rangle|}{2}, \frac{1-|\langle\psi|\phi\rangle|}{2}\right) \\ = -\frac{1+|\langle\psi|\phi\rangle|}{2}\log\frac{1+|\langle\psi|\phi\rangle|}{2} - \frac{1-|\langle\psi|\phi\rangle|}{2}\log\frac{1-|\langle\psi|\phi\rangle|}{2}$$

Born rule determines and is determined by the entropy of all equal mixtures of pairs of pure states



Uncertainty and entropy

- In both classical and quantum mechanics, Gaussian states maximize entropy for a single independent DOF at fixed uncertainty, or equivalently, minimize uncertainty at fixed entropy
- If we fix the entropy S , will have $\Delta x \Delta p \geq \Sigma(S)$
- Specific value of the uncertainty Σ depends on S and on whether we're using classical or quantum mechanics, but relationship always saturated by Gaussian states with no correlation between x and p
- Classical mechanics: $S_C(\Sigma) = \ln\left(2\pi e \frac{\Sigma}{h}\right) = \ln\left(\frac{\Sigma}{h}\right) + 1$
- Quantum mechanics: $S_Q(\Sigma) = \left(\frac{\Sigma}{h} + \frac{1}{2}\right) \ln\left(\frac{\Sigma}{h} + \frac{1}{2}\right) - \left(\frac{\Sigma}{h} - \frac{1}{2}\right) \ln\left(\frac{\Sigma}{h} - \frac{1}{2}\right)$

See e.g. Pathria + Beale, *Statistical Mechanics*, Academic Press, 2022 or Cover + Thomas, *Elements of Information Theory*, Wiley, 2006.

See e.g. Weedbrook et al., *Reviews of Modern Physics* 84:621, 2012, <http://www.reviewsinphysics.org/>



Reinterpreting traditional approaches: Blackbody radiation

- Ultraviolet catastrophe – a crisis for classical physics
- Failure of classical mechanics was a failure of classical mechanics with respect to *statistical* mechanics

Classical: Rayleigh-Jeans

$$\frac{2\nu^2 k_B T}{c^2} = \frac{2\nu^2}{c^2 \beta}$$

Quantum: Planck

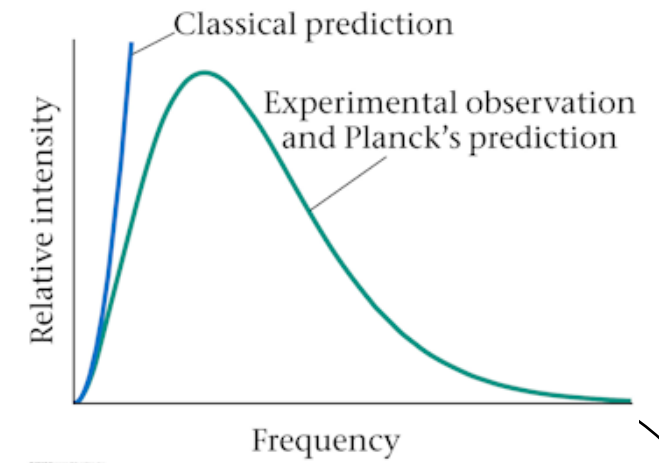
$$\frac{2h\nu^3}{c^2} \frac{1}{\exp\left(\frac{h\nu}{k_B T}\right) - 1} = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\beta\nu} - 1}$$

Traditional limit:
Low frequency

$$\frac{2h\nu^3}{c^2} \frac{1}{e^{h\beta\nu} - 1} = \frac{2h\nu^3}{c^2} \frac{1}{1 + h\beta\nu + O(\nu^2) - 1} \simeq \frac{2\nu^2}{c^2 \beta}$$

Alternative limit:
High temperature (low β)

$$\frac{2h\nu^3}{c^2} \frac{1}{e^{h\beta\nu} - 1} = \frac{2h\nu^3}{c^2} \frac{1}{1 + h\beta\nu + O(\beta^2) - 1} \simeq \frac{2\nu^2}{c^2 \beta}$$



Low $\beta \leftrightarrow$ high temperature/high entropy

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Assumptions of Physics

Reinterpreting traditional approaches: Wigner's quantum correction for thermodynamic equilibrium

Phys. Rev. 40:749–759, 1932

On the Quantum Correction For Thermodynamic Equilibrium

By E. WIGNER

Department of Physics, Princeton University

(Received March 14, 1932)

- Considered Wigner distribution of a system in thermal equilibrium at inverse temperature β :
$$W(x, p) = \int dy e^{i(x+y)p/\hbar} \langle x+y | e^{-\beta \hat{H}} | x-y \rangle e^{-i(x-y)p/\hbar}$$
- With $\hat{\rho} = e^{-\beta \hat{H}}$ the mixed state that maximizes entropy at given average energy (Boltzmann dist.)
- Transformed Hamiltonian to express it as the sum of classical and quantum terms and then expanded the Wigner distribution in powers of $\hbar$
 - Quantum corrections only come in at second order in $\hbar$, recovering classical mechanics in the limit as $\hbar \rightarrow 0$

If we are dealing with a system, the behavior of which in statistical equilibrium is nearly correctly given by the classical theory, we can expand (18) into a power of $\hbar$ and keep the first few terms only. The term with the zero



Assumptions
of
Physics

Reinterpreting traditional approaches: Wigner's quantum correction for thermodynamic equilibrium

$$\tilde{H} = e^{ixp/\hbar} \hat{H} e^{-ixp/\hbar} = \frac{(p + i\hbar\partial/\partial x)^2}{2m} + V(x) = \underbrace{\epsilon(x, p)}_{\text{classical Hamiltonian}} + \underbrace{i\frac{\hbar p}{m} \frac{\partial}{\partial x} - \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2}}_{\text{quantum corrections}} \equiv \hat{Q} \quad \text{Transformed Hamiltonian}$$

$$W(x, p) = \int dy \langle x + y | e^{-\beta \tilde{H}} | x - y \rangle \quad \text{Wigner distribution in terms of } \tilde{H}$$

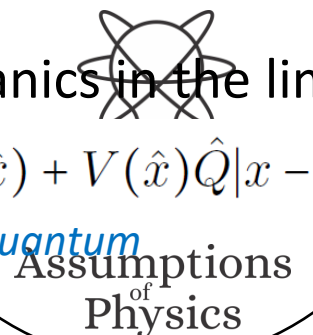
- Can expand this in powers of β for $\beta \ll 1$, rather than powers of $\hbar$
- First order: $e^{-\beta \tilde{H}} \simeq 1 - \beta(\epsilon + \hat{Q}) \Rightarrow W(x, p) \simeq 1 - \underbrace{\beta\epsilon(x, p)}_{\text{classical}} - \underbrace{\beta \int dy \langle x + y | \hat{Q} | x - y \rangle}_{\text{first-order quantum}}$
- Show first-order quantum correction in β is zero by inserting an identity in the momentum eigenbase:

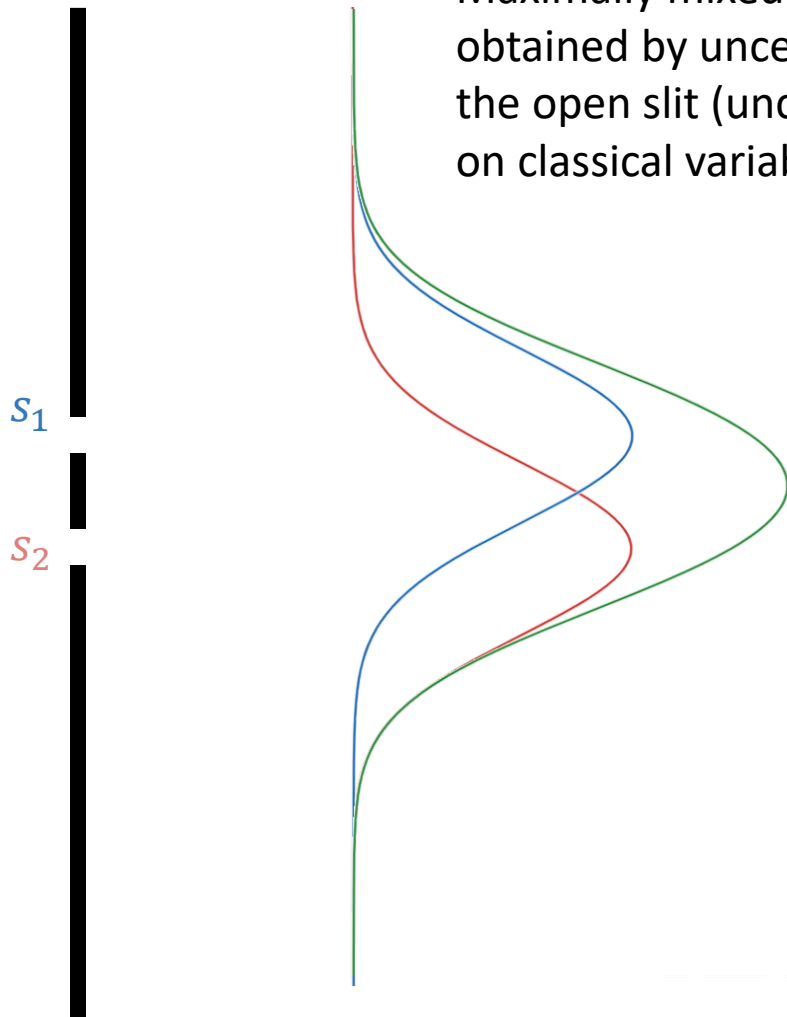
$$\int dy \langle x + y | \hat{Q} | x - y \rangle = \int \int dk dy e^{i2ky} \left(-\frac{\hbar p}{m} k + \frac{\hbar^2}{2m} k^2 \right) = \int dk \delta(2k) \left(-\frac{\hbar p}{m} k + \frac{\hbar^2}{2m} k^2 \right) = 0$$

- Thus, *quantum corrections only enter at second order in β* , recovering classical mechanics in the limit as $\beta \rightarrow 0$

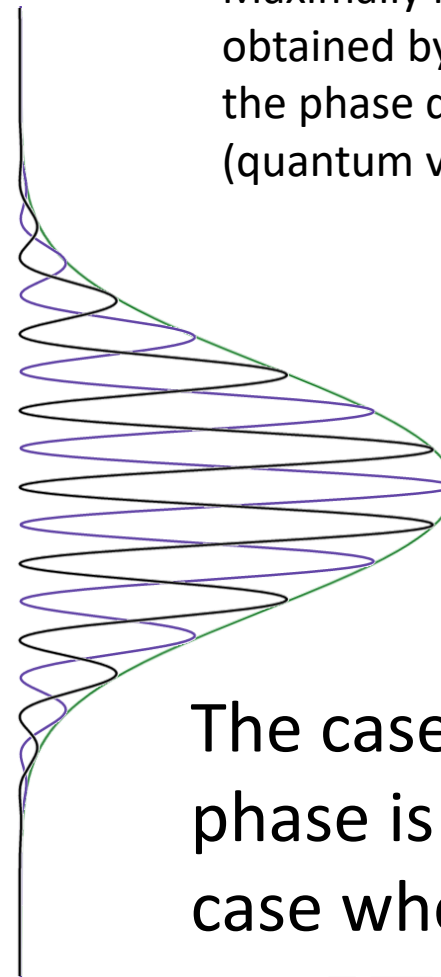
$$W(x, p) \simeq 1 - \underbrace{\beta\epsilon(x, p)}_{\text{classical}} - \underbrace{\beta^2 \epsilon^2(x, p)}_{\text{second-order quantum}} - \beta^2 \int dy \langle x + y | \hat{Q} V(\hat{x}) + V(\hat{x}) \hat{Q} | x - y \rangle$$

When entropy $S \rightarrow \infty$, $\beta \rightarrow 0$

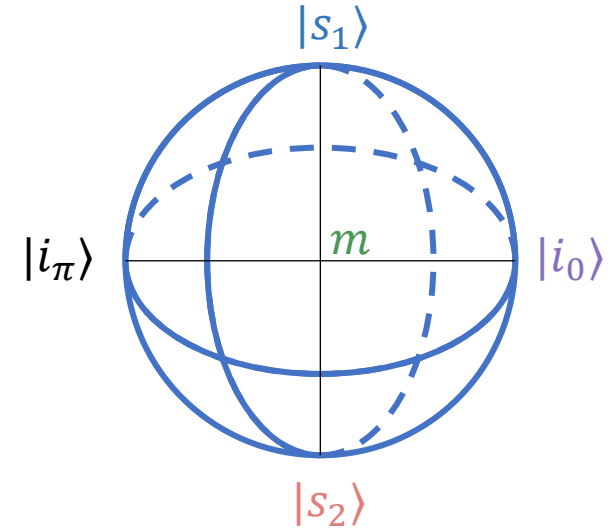




Maximally mixed state
obtained by uncertainty on
the open slit (uncertainty
on classical variable)



Maximally mixed state
obtained by uncertainty on
the phase difference
(quantum variable)



The case where we randomize the
phase is indistinguishable from the
case where we randomize the path

Entropic aliasing: high-entropy states allow (approximate)
decomposition over incompatible observables

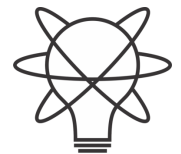
Lindblad master equation for open quantum systems

Infinitesimal
stretching map
must obey

$$\frac{dX}{dt} = \frac{i}{\hbar} [H, X] + \gamma \left(L^\dagger X L - \frac{1}{2} \{L^\dagger L, X\} \right)$$

Unitary/conservative part

Non-unitary/non-conservative part



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Assumptions
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Physics

Quantum pure stretching map

$$T(X) = \sqrt{\lambda} X \quad T(P) = \sqrt{\lambda} P$$

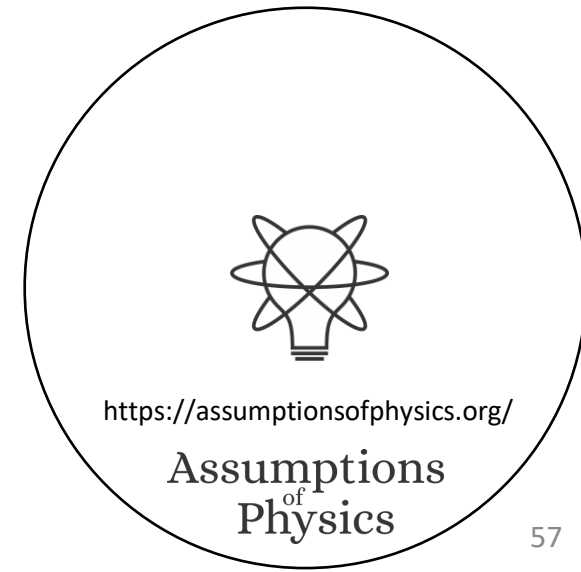
What about products?

Only unitary maps preserve products!

1. Suppose $T(\rho_1 \rho_2) = T(\rho_1)T(\rho_2)$. Since pure states are exactly the density matrices for which $\rho^2 = \rho$, $T(\rho) = T(\rho^2) = T(\rho)T(\rho) = T(\rho)^2$ and therefore T maps pure states to pure states. Therefore, it is a unitary map!
2. Suppose $T(AB) = T(A)T(B)$. Then $T([A, B]) = T(AB) - T(BA) = [T(A), T(B)]$

Which product ordering recovers

$$\langle T(x^n p^m) \rangle = (\sqrt{\lambda})^{n+m} \langle x^n p^m \rangle?$$



Symmetrized operator averages

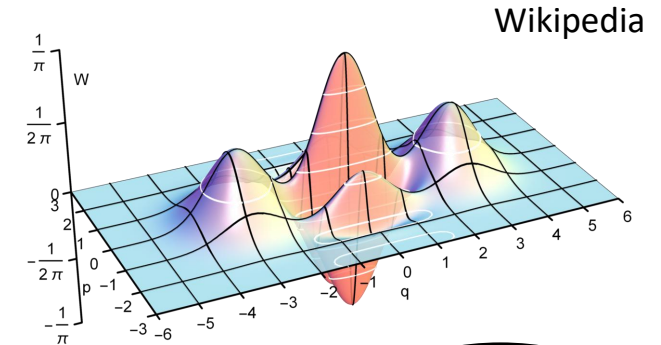
$$\langle T_W(\underbrace{\Pi(X, \dots, X)}_{n \text{ times}}, \underbrace{P, \dots, P}_{m \text{ times}}) \rangle = (\sqrt{\lambda})^{(n+m)} \langle \underbrace{\Pi(X, \dots, X)}_{n \text{ times}}, \underbrace{P, \dots, P}_{m \text{ times}} \rangle$$

$$\Pi(A_1, A_2, \dots, A_n) = \frac{1}{n!} \sum_{\pi} A_{\pi(1)} A_{\pi(2)} \cdots A_{\pi(n)}$$

We take the average of all permutations

$$\langle \underbrace{\Pi(X, \dots, X)}_{n \text{ times}}, \underbrace{P, \dots, P}_{m \text{ times}} \rangle = \int_M x^n p^m W(x, p) dx dp$$

Also given by the expectations of the Wigner function



Wikipedia

A stretching map that rescales symmetrized operator averages is a stretching map in phase space for the Wigner function...

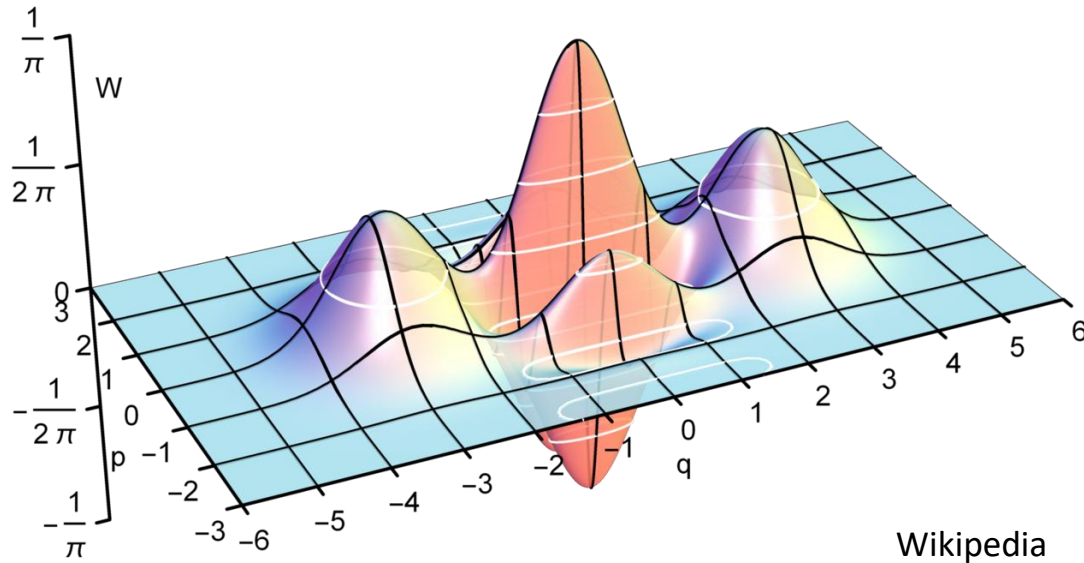
... and it can't work!!!



<https://assumptionsofphysics.org/>

Assumptions
of
Physics

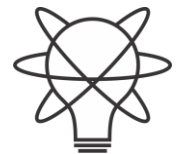
Symmetrized operator averages



Wigner functions can have negative values, but they must be restricted to “small” regions

A stretching map would make those regions arbitrarily large, so it can't map valid quantum states to valid quantum states

Symmetrized operator averages are ruled out!



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Assumptions
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Physics

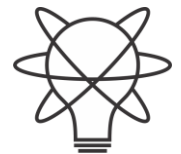
Normal ordering

$$\langle T_P((a^\dagger)^n a^m) \rangle = (\sqrt{\lambda})^{(n+m)} \langle (a^\dagger)^n a^m \rangle$$

$$a = \sqrt{\frac{m\omega}{2\hbar}} \left(X + \frac{i}{m\omega} P \right) \quad a^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(X - \frac{i}{m\omega} P \right)$$

For the vacuum state we have $a|0\rangle = 0$. The mean for all expectations for the vacuum state is zero. Therefore, after the map, all expectations are zero. Which means the vacuum is mapped to the vacuum. Entropy is not increased for all states!

Normal ordering is ruled out!



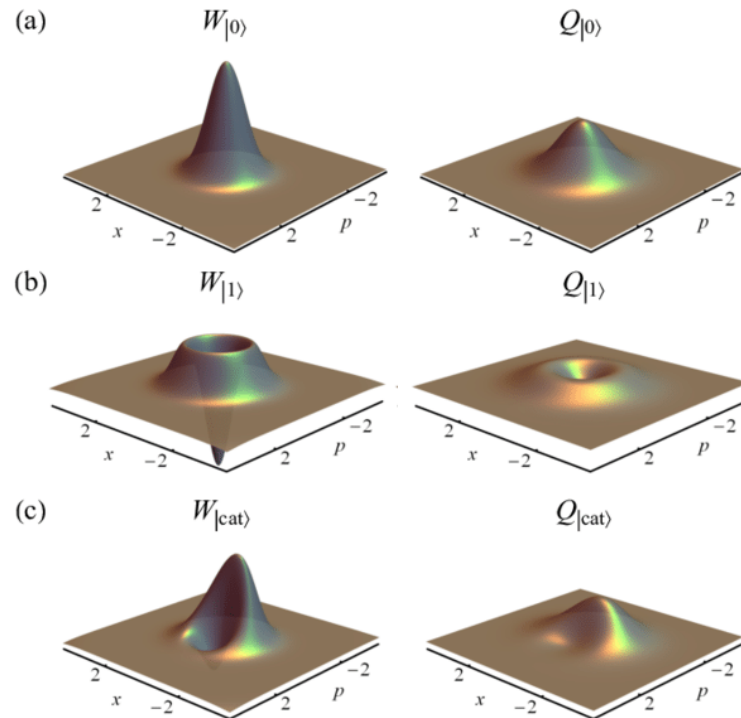
<https://assumptionsofphysics.org/>

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Physics

Anti-normal ordering

$$\langle T_Q(a^n (a^\dagger)^m) \rangle = (\sqrt{\lambda})^{(n+m)} \langle a^n (a^\dagger)^m \rangle$$

$$a = \sqrt{\frac{m\omega}{2\hbar}} \left(X + \frac{i}{m\omega} P \right) \quad a^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(X - \frac{i}{m\omega} P \right)$$



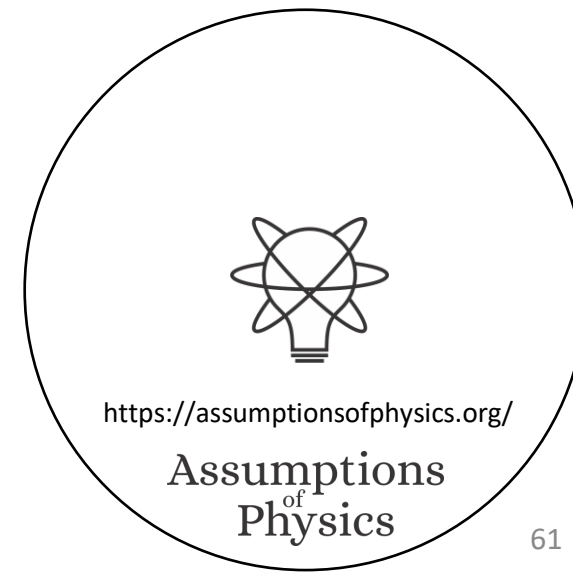
$$\langle a^n (a^\dagger)^m \rangle = \int \alpha^n (\alpha^*)^m Q_1(\alpha) d^2 \alpha$$

Also given by the expectations
of the Husimi Q function

T_Q stretches the Husimi Q...

... which is non-negative!!!

Maurice de Gosson



So, we want: $\langle T_Q(a^n(a^\dagger)^m) \rangle = (\sqrt{\lambda})^{(n+m)} \langle a^n(a^\dagger)^m \rangle$

Lindblad
master
equation

$$\frac{dX}{dt} = \frac{i}{\hbar} [H, X] + \gamma \left(L^\dagger X L - \frac{1}{2} \{L^\dagger L, X\} \right)$$

We set $H = 0$ $L = a^\dagger$

Recall $[a, a^\dagger] = 1$

$$[a, (a^\dagger)^m] = m(a^\dagger)^{m-1}$$

$$[a^n, a^\dagger] = n a^{n-1}$$

$$\frac{d}{dt} a^n (a^\dagger)^m = \gamma \left(a a^n (a^\dagger)^m a^\dagger - \frac{1}{2} a a^\dagger a^n (a^\dagger)^m - \frac{1}{2} a^n (a^\dagger)^m a a^\dagger \right)$$

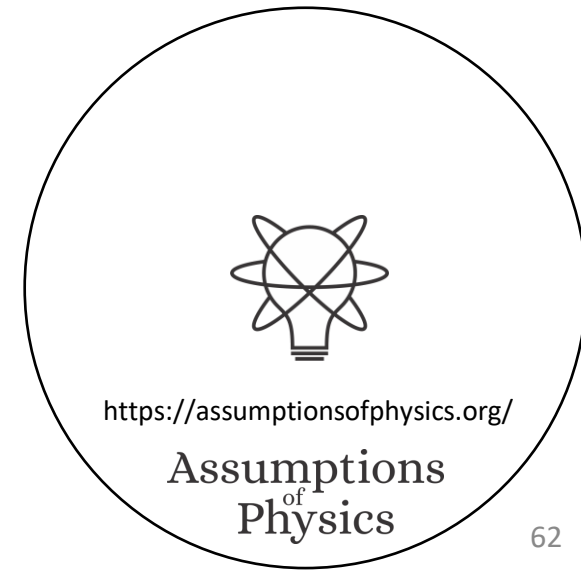
$$= \gamma \left(a a^n (a^\dagger)^m a^\dagger - \frac{1}{2} a a^n (a^\dagger)^m a^\dagger - \frac{1}{2} a [a^\dagger, a^n] (a^\dagger)^m - \frac{1}{2} a a^n (a^\dagger)^m a^\dagger - \frac{1}{2} a^n [(a^\dagger)^m, a] a^\dagger \right)$$

$$= \gamma \left(-\frac{1}{2} a (-n a^{n-1}) (a^\dagger)^m - \frac{1}{2} a^n (-m (a^\dagger)^{m-1}) a^\dagger \right)$$

$$= \frac{\gamma}{2} (n+m) a^n (a^\dagger)^m$$

first-order linear ODE

$$\Rightarrow \left(a^n (a^\dagger)^m \right) (t) = e^{\frac{\gamma}{2} t (n+m)} a^n (a^\dagger)^m$$



Effects of stretching map on phase space representations

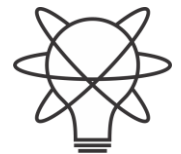
Husimi Q is simply stretched

Wigner function is stretched and convolved with a gaussian,
and coincides with the Husimi Q in the limit $\lambda \rightarrow \infty$

It is interesting to consider what happens to the negative regions of W under the stretching map. We know that W can have negative regions, but their size is limited by the uncertainty principle. In fact, convolving W with a 2D Gaussian with unitary spread, as in the definition of Q , returns a function that is never negative. In the limit $\lambda \gg 1$, the formula for W_λ reduces to

$$W_\lambda(\beta) \rightarrow_{\lambda \gg 1} \frac{2}{\pi \lambda^2} \int W_1 \left(\frac{\alpha}{\sqrt{\lambda}} \right) e^{-\frac{2}{\lambda} |\alpha - \beta|^2} d^2 \alpha = Q_\lambda(\beta). \quad (53)$$

Therefore, while negative regions can be in principle found at any finite λ , W tends to a positive function in the limit. As usual for the W distribution, the phase space size of negative regions is limited to $\hbar$ by the uncertainty principle. The weight of the function in such regions is limited between $\pm 2/\hbar$ for pure states. The effect of the stretching map is to reduce this bound to $\pm 2/(\lambda \hbar)$ for large values of λ . A clear interpretation can be made by working directly on the Fourier transform of W ; see Eq. (51). The function $F(W_\lambda)$ is a scaled version of $F(W_1)$ with a Gaussian filter applied to it. Quantum information in W_1 is carried by spectral weights with k-vectors larger than 1. The bandwidth of the Gaussian filter is given by $\lambda/(\lambda - 1)$. This cutoff approaches 1 in the limit, filtering away the interference terms.



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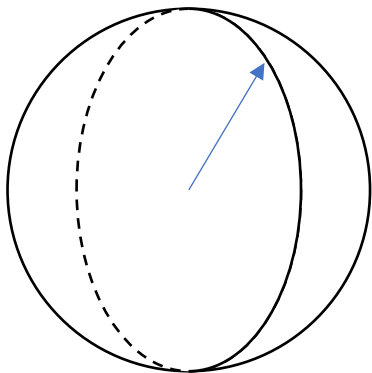
Quantum pure states are not sufficient to distinguish quantum and classical mechanics

Condition QP-RAY (Quantum state space only). *The state space of a quantum system is represented by the projective space $\mathcal{P}(V)$ of a complex topological vector space V .*

Condition CP (Classical state space only). *The state space of a classical system is represented by a symplectic manifold (X, ω) .*

Condition CQE (Classical-Quantum exclusivity). *A system cannot be both classical and quantum.*

Theorem 5. *Condition **QP-RAY**+**CP** is consistent with $\neg$ **CQE**.*



State space of a classical magnetic dipole is a sphere, which is the complex projective line

In general, every finite-dimensional complex projective space is a symplectic manifold, and therefore allows classical Hamiltonian mechanics

<https://assumptionsofphysics.org/autogen/briefs/005-PureStatesNotEnough.pdf>

<https://arxiv.org/pdf/2608.27543>



Quantum ensembles rule out classical mechanics

Condition QE-DENS (Quantum ensemble space). *The state space of a quantum system is represented by the projective space $\mathcal{P}(\mathcal{H})$ of a separable complex Hilbert space $\mathcal{H}$. The ensemble space is represented by the density operators $\mathcal{D}(\mathcal{H})$.*

Condition CE (Classical ensemble space). *The state space of a classical system is represented by a symplectic manifold (X, ω) . The ensemble space is represented by the set of probability distributions $\rho(q^i, p_j)$ (i.e. Radon–Nikodym derivatives of probability measures that are absolutely continuous with respect to the Liouville measure).*

Theorem 6. *Condition QE-DENS+CE implies CQE for every non-trivial system (i.e. the state space has more than one state).*

In classical mechanics, every ensemble is identified by a UNIQUE probability measure on pure states (i.e. ensemble space is a simplex)

In quantum mechanics, there is NO unique probability measure

Though we lack a definition of simplex that works in the infinite-dimensional case



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Assumptions
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Physics

The ensemble space is not sufficient to characterize QM

Condition [B]PMIX (Probabilities of statistical mixtures). *The conditional probability of a statistical mixture is the statistical mixture of the conditional probabilities. That is, $p(\psi \mid \sum_i \lambda_i \rho_i) = \sum_i \lambda_i p(\psi \mid \rho_i)$.*

Condition [B]MUTEX (Mutual exclusivity conditions). *Let $\psi_i \in \mathcal{P}$ be a countable sequence of states. Then the following are equivalent:*

- ψ_i are mutually exclusive
- $p(\psi_i \mid \psi_j) = \delta_{ij}$
- the entropy obeys $S(\sum_i p_i \psi_i) = -\sum_i p_i \log p_i$ for all p_i such that $\sum_i p_i = 1$.

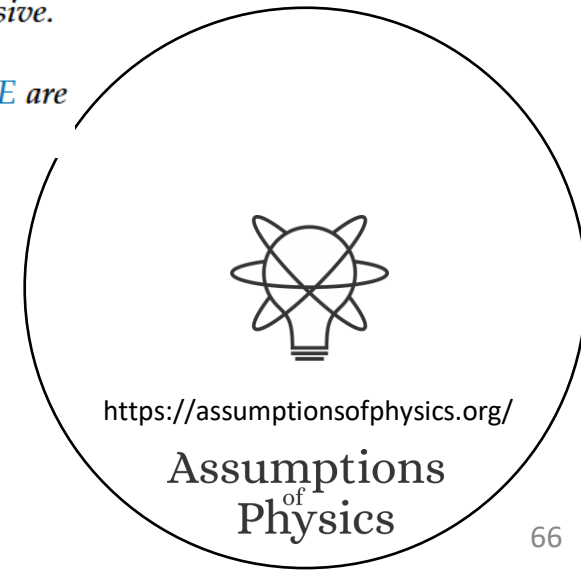
Condition BR-BORN (Conditional expectation from Born rule). *The conditional probability $p : \mathcal{P} \times \mathcal{P} \rightarrow [0, 1]$ is given by the Born rule. That is, $p(\psi \mid \phi) = \frac{\langle \phi \mid \psi \rangle \langle \psi \mid \phi \rangle}{\langle \psi \mid \psi \rangle \langle \phi \mid \phi \rangle}$.*

Condition BR-ENT (Von Neumann entropy). *The entropy function $S : \mathcal{D} \rightarrow [0, +\infty]$ is given by the von Neumann entropy $S(\rho) = -\text{tr}(\rho \log \rho)$.*

Condition BR-ORME (Orthogonality is mutual exclusivity). *Orthogonal states are mutually exclusive.*

Theorem 7. *Over $\text{QE-DENS} + [\text{B}] \text{MUTEX} + [\text{B}] \text{PMIX}$, conditions BR-BORN , BR-ENT and BR-ORME are equivalent. Moreover, BR-BORN is independent of $\text{QE-DENS} + [\text{B}] \text{MUTEX} + [\text{B}] \text{PMIX}$.*

The fact that orthogonal vectors
represent mutually exclusive ensembles
is an additional assumption

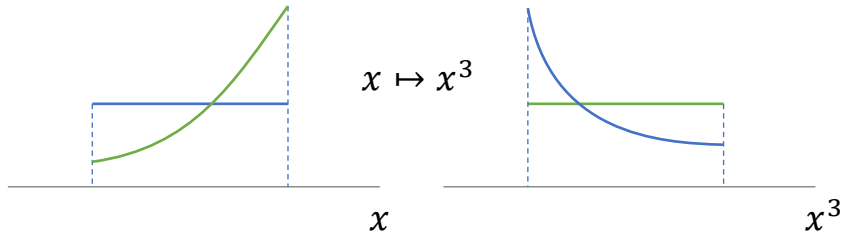


Erkenntnis 90, 443 (2025)

Stud Hist Phil Sci B 71, 082020, 60-71 (2020)

Reversing phase space

Each unit variable (i.e. coordinate) paired with a conjugate of inverse units: number of states $\Delta q \Delta k$ is invariant



$$\Delta k = 1 \text{ m}^{-1}$$



$$\Delta q = 1 \text{ m}$$

$$\hat{q} = 100 \text{ cm/m } q$$



$$\Delta \hat{q} = 100 \text{ cm}$$

$$\Delta \hat{k} = 0.01 \text{ cm}^{-1}$$

Density, entropy, uniform distributions
NOT in general coordinate invariant

Phase-space (symplectic) structure is the only one that supports
coordinate-invariant density, entropy, state count

Phase space volumes are invariant under equal-time changes of coordinates q^i

(PSV-VOL)

The Jacobian for the transformation induced by equal-time changes of coordinates q^i is unitary

(PSV-JAC)

Densities over phase space are invariant under equal-time changes of coordinates q^i

(PSV-DEN)

Thermodynamic entropy is invariant under equal-time changes of coordinates q^i

(PSV-THER)

Information entropy is invariant under equal-time changes of coordinates q^i

(PSV-INFO)

Uncertainty of peaked distributions is invariant under equal-time changes of coordinates q^i

(PSV-UNC)

The system allows statistically independent distributions over each DOF under any choice of coordinates q^i

(PSI-DEN)

The system allows informationally independent distributions over each DOF under any choice of coordinates q^i

(PSI-INFO)

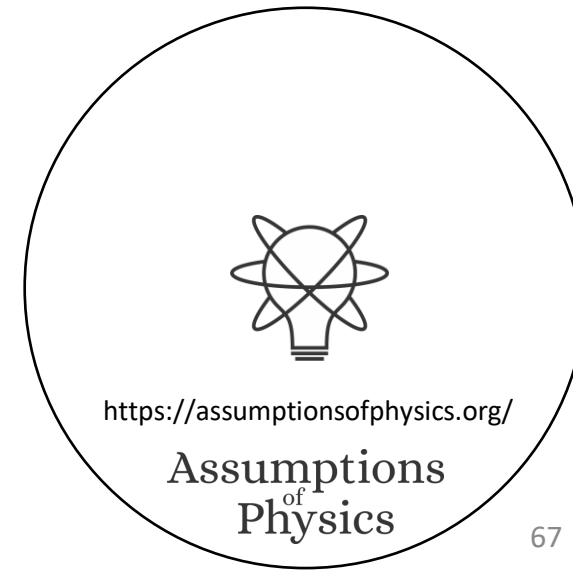
The system allows peaked distributions where the uncertainty is the product of the uncertainty on each DOF under any choice of coordinates q^i

(PSI-UNC)

Invariance of entropy/state count + DOF independence
 $\Leftrightarrow$ structure of phase space (i.e. conjugate pairs)

https://philsci-archive.pitt.edu/22202/1/Hamiltonian_Privilege_preprint.pdf

<https://link.springer.com/article/10.1007/s10670-023-00708-0>



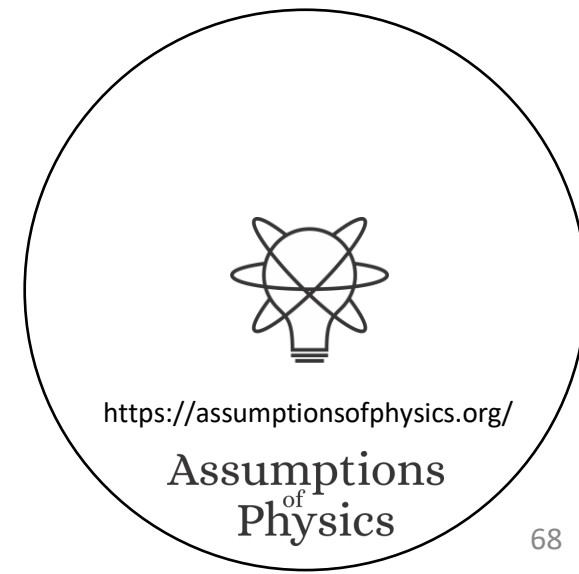
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Assumptions
of
Physics

Principle of least action - one DOF

<https://arxiv.org/pdf/2208.06428>

<https://www.nature.com/articles/s41598-023-39145-y>



Required background

- Hamiltonian mechanics

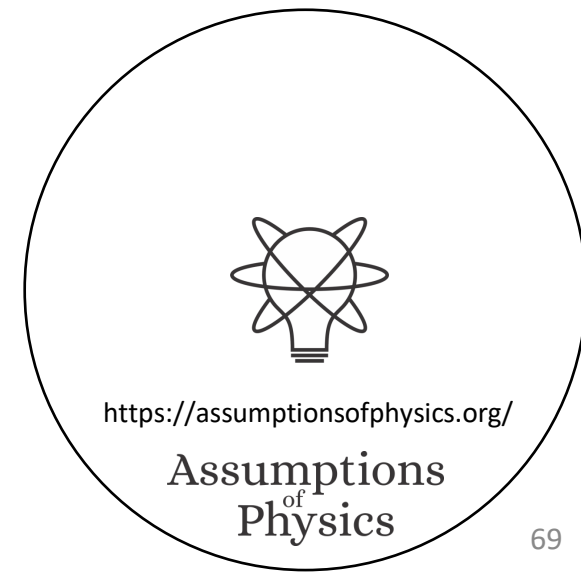
- Hamilton's equations: $\frac{dq}{dt} = \frac{\partial H}{\partial p} \quad \frac{dp}{dt} = -\frac{\partial H}{\partial q}$

- Lagrangian mechanics

- Action is the integral of the Lagrangian over a given path: $\mathcal{A}[\gamma] = \int_{\gamma} L(q(t), \dot{q}(t), t) dt$
 - Actual evolutions are the paths that make the action stationary: $\delta \mathcal{A} = 0$
 - Lagrangian is the Legendre transform of the Hamiltonian: $L = p\dot{q} - H$

- Vector calculus

- Given a vector field $\vec{v}$, a field line is a line always tangent to the field
 - The divergence $\vec{\nabla} \cdot \vec{v}$ is the flow of the field through an infinitesimal closed region
 - A divergence-free field $\vec{\nabla} \cdot \vec{B} = 0$ admits a vector potential $\vec{A}$ such that $\vec{B} = \vec{\nabla} \times \vec{A}$



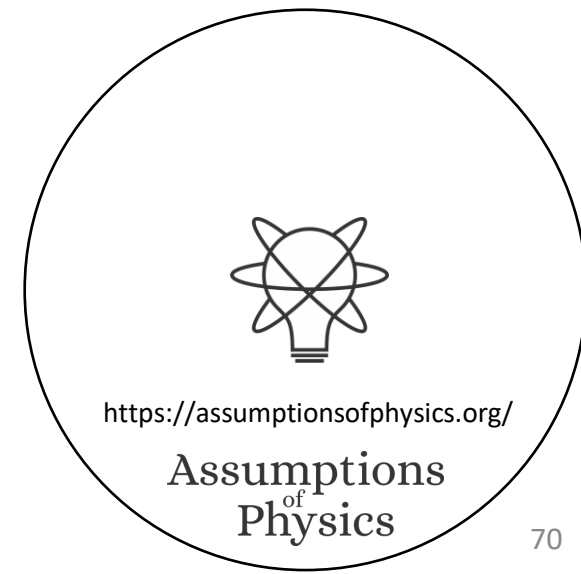
$$\text{Action: } \mathcal{A}[\gamma] = \int_{\gamma} L(q(t), \dot{q}(t), t) dt$$

$$\text{Lagrangian: } L = T - U \quad \text{Does not always work!}$$

Lagrangian for particle with charge q :

$$L = \frac{1}{2} m v^2 + q \vec{v} \cdot \vec{A} - q V$$

Kinetic
???
Potential



Displacement
field

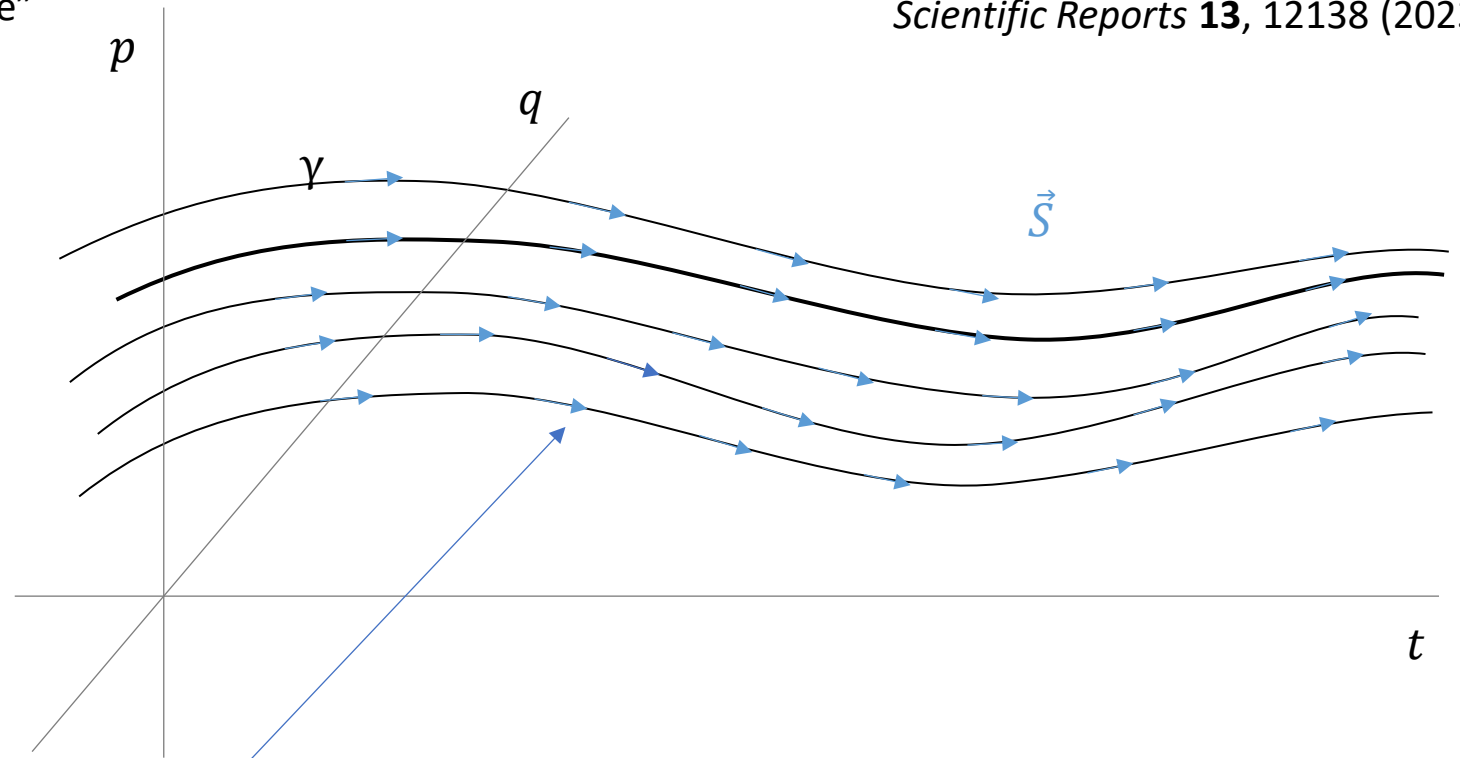
Tells us how states “move”

$$\vec{S} = \left[\frac{dq}{dt}, \frac{dp}{dt}, \frac{dt}{dt} \right]$$

Evolution

$$\gamma = [q(t), p(t), t]$$

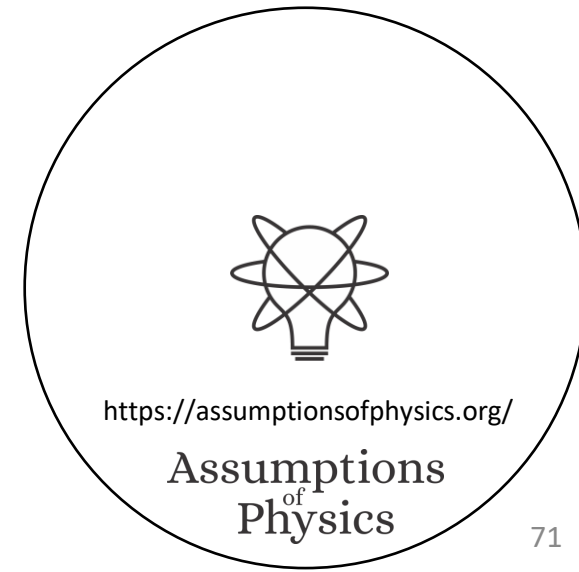
Time is both a parameter
and a variable



Scientific Reports **13**, 12138 (2023)

Evolution is field lines
of the displacement field

Always tangent
to the field



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Assumptions
of
Physics

(DR) Determinism and Reversibility

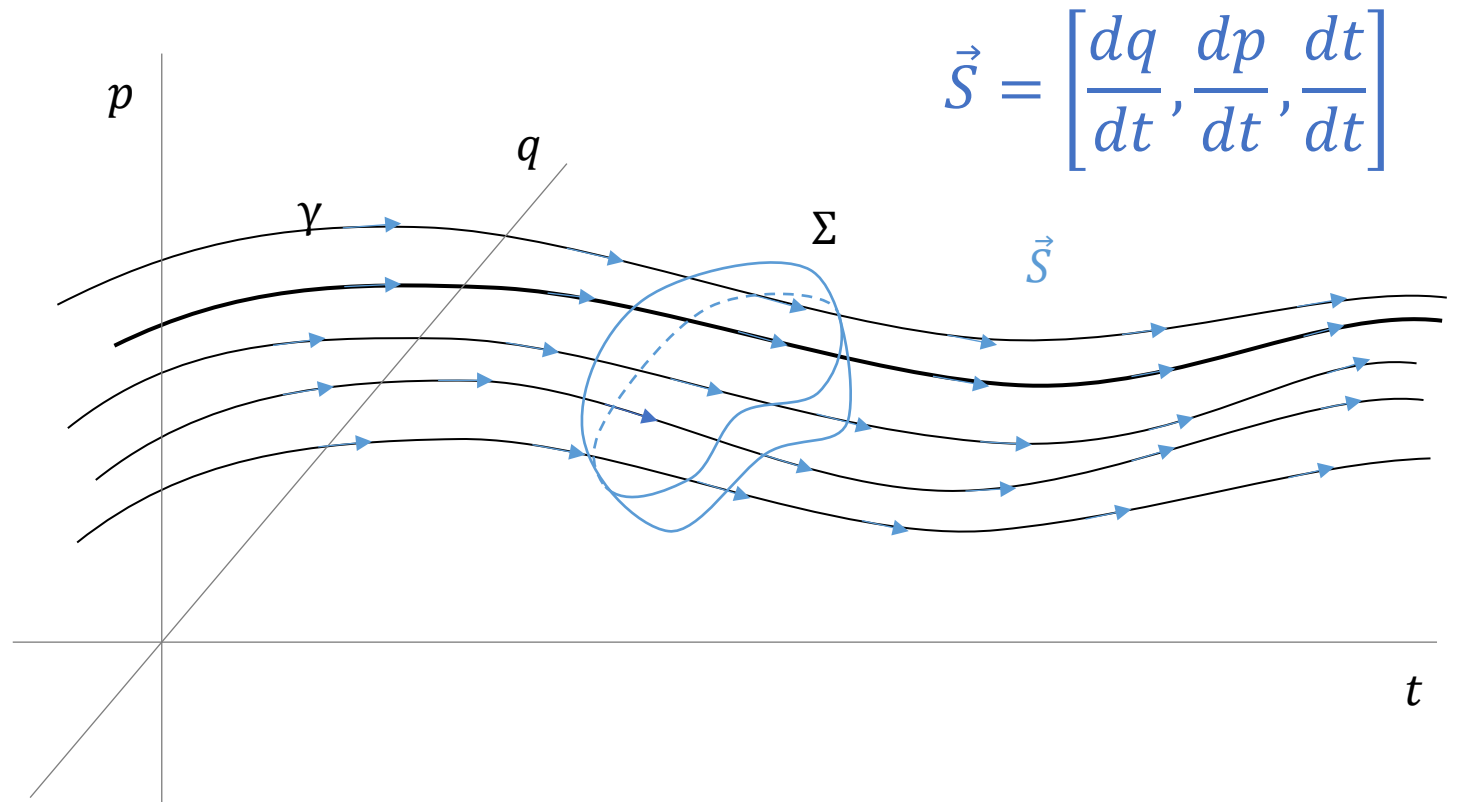
No states are lost or created
(count of states is preserved in time)

$$\oiint_{\Sigma} \vec{S} \cdot d\vec{\Sigma} = 0$$

⇒ Divergence-free displacement

$$\vec{\nabla} \cdot \vec{S} = 0$$

Displacement admits
a vector potential

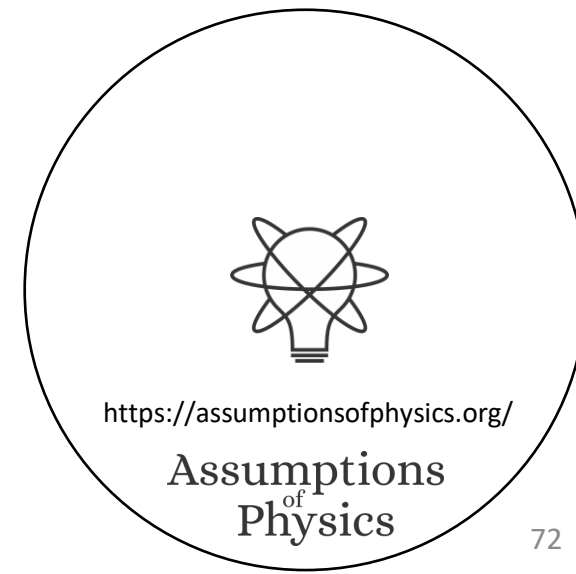


$$\vec{S} = \left[\frac{dq}{dt}, \frac{dp}{dt}, \frac{dt}{dt} \right]$$

Vector potential

$$\vec{S} = -\vec{\nabla} \times \vec{\theta}$$

Minus sign added to match conventions



$$\vec{\theta} = [\theta_q, \theta_p, \theta_t]$$

Fix gauge: $\theta_a \rightarrow \theta_a - \partial_a \int_0^p \theta_p dp$

$$\vec{\theta} = [\theta_q, 0, \theta_t]$$

$$S_t = \partial_p \theta_q - \partial_q \theta_p = \frac{dt}{dt} = 1$$

$$\vec{\theta} = [p, 0, \theta_t]$$

Rename θ_t

$$\vec{\theta} = [p, 0, -H]$$

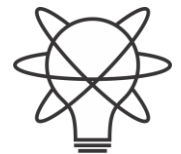
$$\vec{S} = \left[\frac{dq}{dt}, \frac{dp}{dt}, \frac{dt}{dt} \right]$$

Without loss of generality

$$\vec{S} = -\vec{\nabla} \times \vec{\theta}$$

$$\vec{\theta} = [p, 0, -H]$$

similar to four-momentum



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Assumptions
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Physics

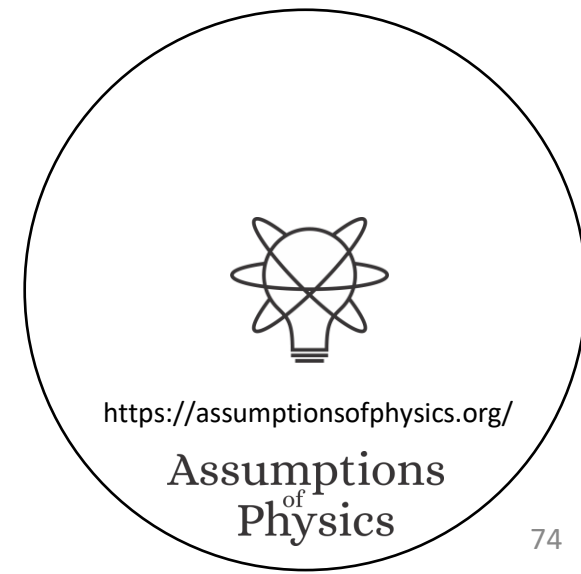
$$\vec{S} = \left[\frac{dq}{dt}, \frac{dp}{dt}, \frac{dt}{dt} \right]$$

$$\vec{S} = -\vec{\nabla} \times \vec{\theta} \quad \vec{\theta} = [p, 0, -H(q, p)]$$

$$S^q = \frac{dq}{dt} = - \left(\frac{\partial}{\partial p} \theta_t - \frac{\partial}{\partial t} \theta_p \right) = \frac{\partial H}{\partial p} \quad \leftarrow \text{Hamilton's equations}$$

$$S^p = \frac{dp}{dt} = - \left(\frac{\partial}{\partial t} \theta_q - \frac{\partial}{\partial q} \theta_t \right) = - \frac{\partial H}{\partial q}$$

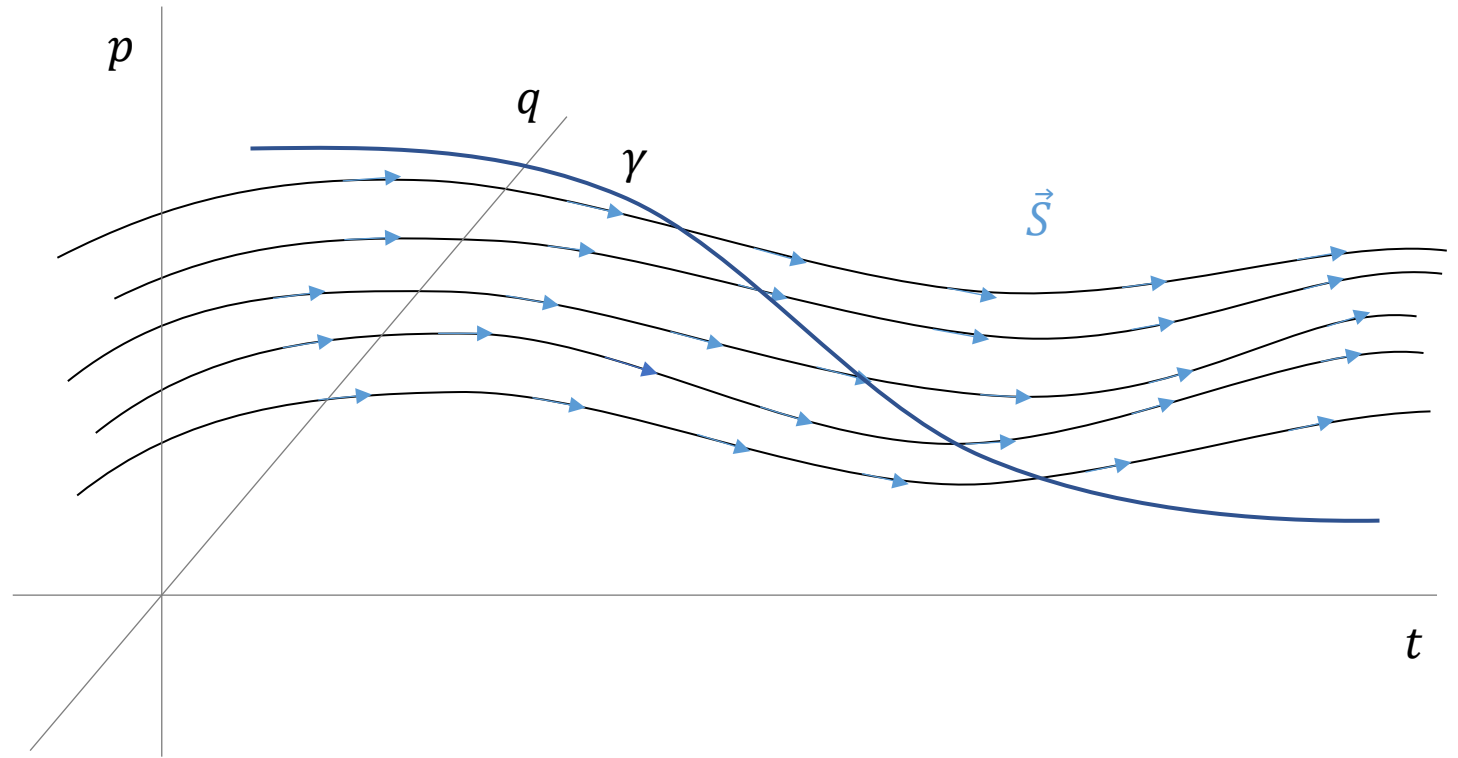
$$S^t = \frac{dt}{dt} = - \left(\frac{\partial}{\partial q} \theta_p - \frac{\partial}{\partial p} \theta_q \right) = 1$$



Line integral of $\vec{\theta}$

$$\begin{aligned} & \int_{\gamma} \vec{\theta} \cdot d\vec{\gamma} \quad \text{Express in components} \\ &= \int_{\gamma} [p, 0, -H] \\ & \quad \cdot \left[\frac{dq}{dt}, \frac{dp}{dt}, \frac{dt}{dt} \right] dt \\ &= \int_{\gamma} \left(p \frac{dq}{dt} - H \right) dt \\ &= \int_{\gamma} L(q, \dot{q}, t) dt \\ &= \mathcal{A}[\gamma] \end{aligned}$$

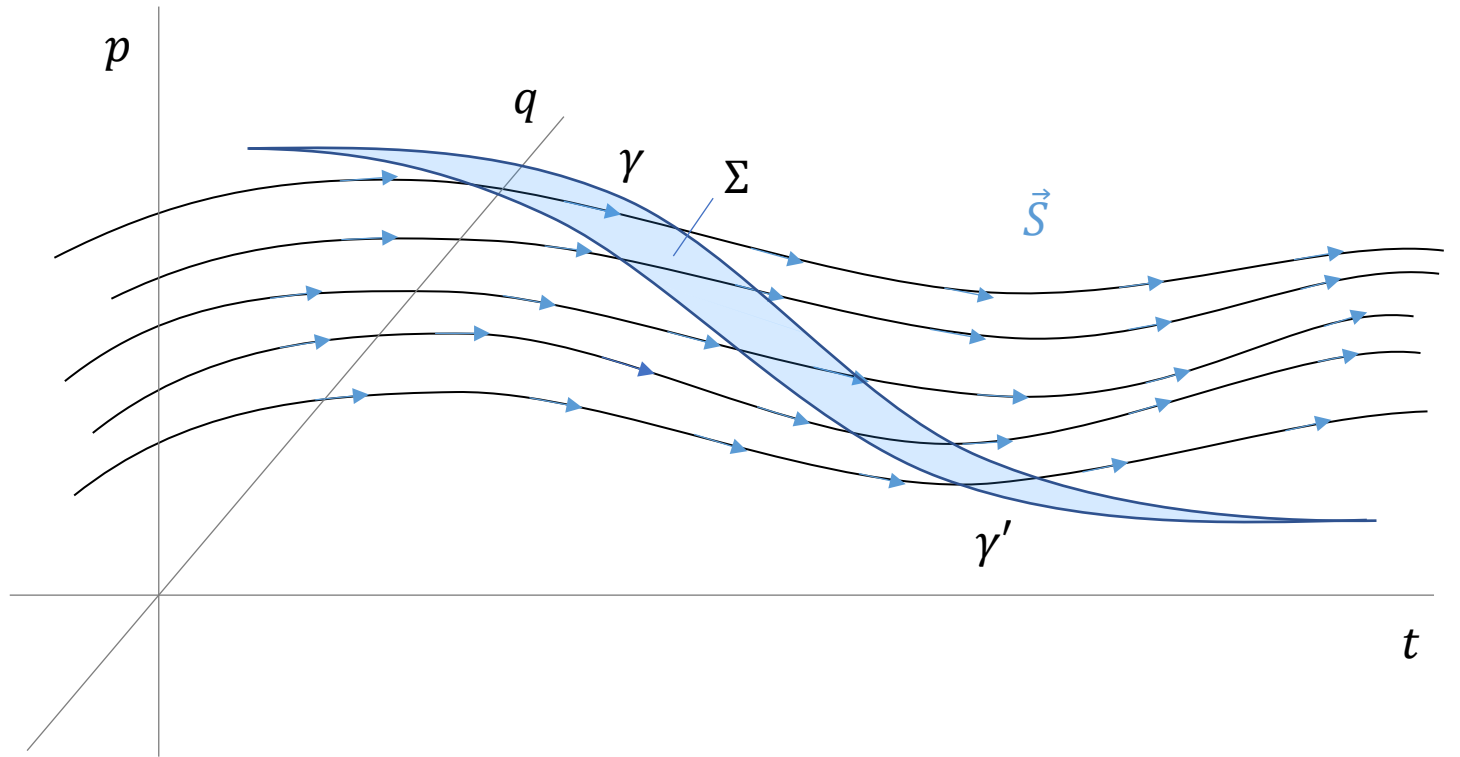
Assuming (KE) Kinematic Equivalence:
 $p = p(q, \dot{q}, t)$



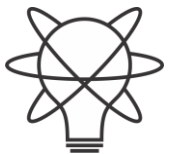
The action is the line integral
of the vector potential of the flow of states

Flow between a path γ
and its variation γ'

$$\begin{aligned}
 \Phi[\Sigma] &= - \iint_{\Sigma} \vec{S} \cdot d\vec{\Sigma} \\
 &= \iint_{\Sigma} \vec{\nabla} \times \vec{\theta} \cdot d\vec{\Sigma} && \text{Hamilton equations} \\
 &= \oint_{\partial\Sigma} \vec{\theta} \cdot d\vec{\gamma} && \text{Stokes' theorem} \\
 &= \int_{\gamma} \vec{\theta} \cdot d\vec{\gamma} - \int_{\gamma'} \vec{\theta} \cdot d\vec{\gamma} \\
 &= \delta \int_{\gamma} \vec{\theta} \cdot d\vec{\gamma} \\
 &= \delta \int_{\gamma} L dt \\
 &= \delta \mathcal{A}[\gamma]
 \end{aligned}$$



The variation of the action
is the flow between the
path and its variation



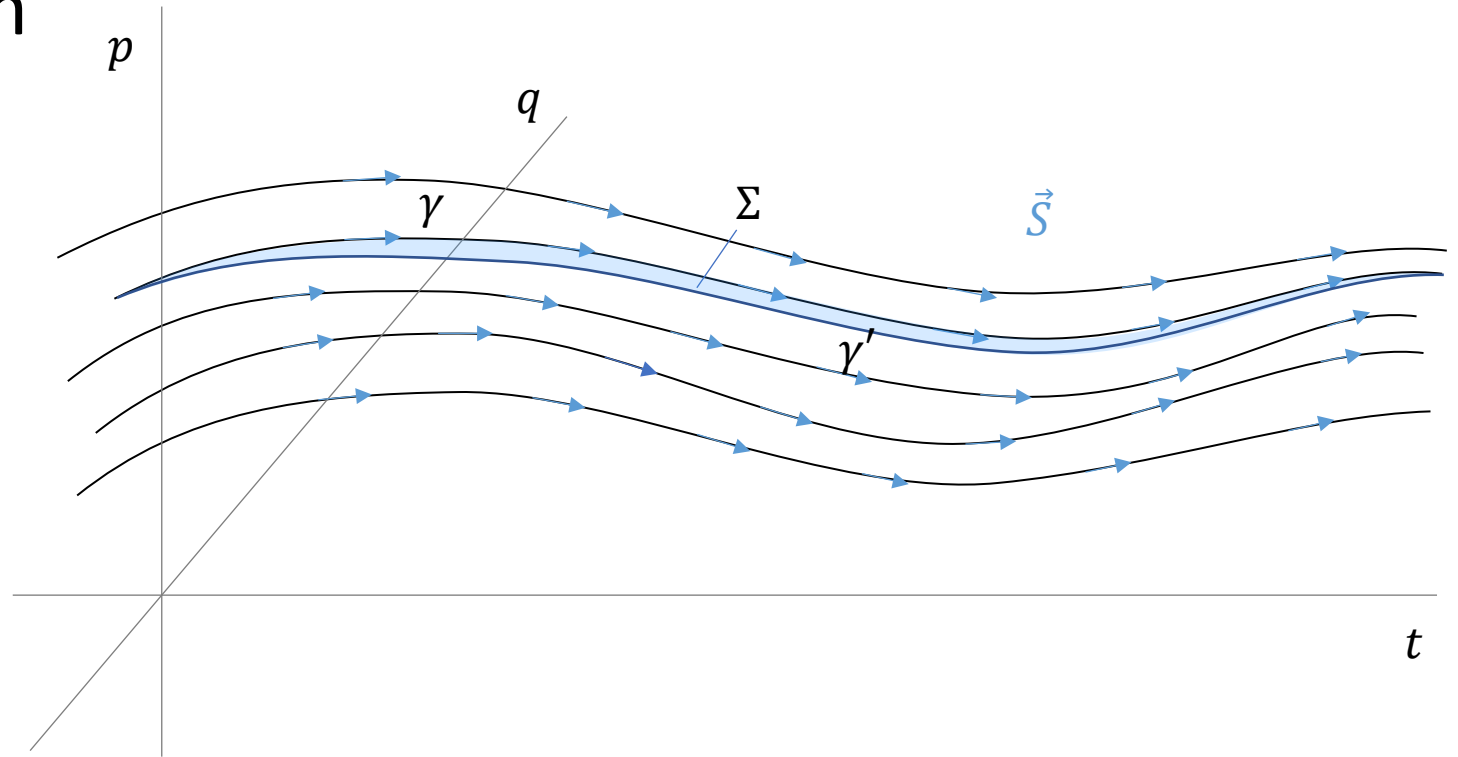
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Assumptions
of
Physics

If the path γ is an evolution of the system

$$\begin{aligned}\Phi[\Sigma] &= - \iint_{\Sigma} \vec{S} \cdot d\vec{\Sigma} \\ &= \delta \int_{\gamma} \vec{\theta} \cdot d\vec{\gamma} = 0\end{aligned}$$

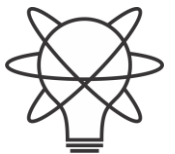
flow is tangent



If the path γ is not an evolution of the system, we can always find a variation γ' such that

$$- \iint_{\Sigma} \vec{S} \cdot d\vec{\Sigma} = \delta \int_{\gamma} \vec{\theta} \cdot d\vec{\gamma} \neq 0$$

Action is stationary over
and only over evolutions



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Assumptions
of
Physics

1 DOF of classical phase space

+

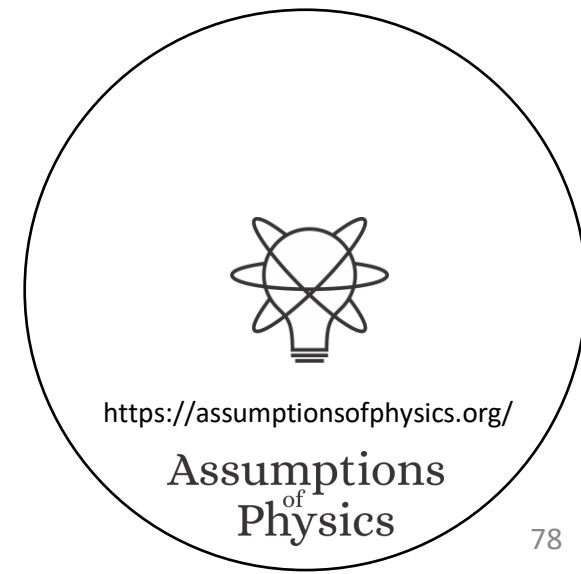
Determinism/Reversibility

+

Kinematic equivalence

↕

Principle of stationary action



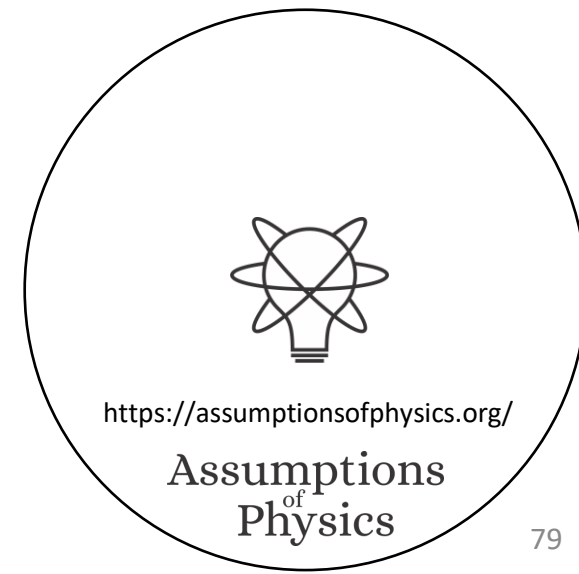
Relativistic mechanics

<https://assumptionsofphysics.org/book/AssumptionsOfPhysicsV3.0.pdf#section.1.1.9>

J. Phys. Commun. **2** 045026 (2018)

<https://arxiv.org/abs/1702.07052>

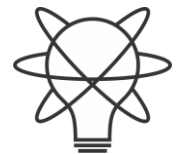
<http://iopscience.iop.org/article/10.1088/2399-6528/aaba25>



Required background

- Special relativity

- Space and time coordinates are grouped together: $q^\alpha = [ct, q^i]$
- Metric tensor $g_{\alpha\beta}$ defines the geometry of space-time (i.e. lengths and angles)
- Using $(-, +, +, +)$ signature: $ds^2 = -dt^2 + dx^2 + dy^2 + dz^2$
- Energy and momentum are grouped together (four-momentum): $p_\alpha = \left[-\frac{E}{c}, p_i\right]$
- Four-velocity is the velocity with respect to proper time (time in the rest frame) and its norm is $-c^2$: $u^\alpha = d_\tau q^\alpha \quad u^\alpha g_{\alpha\beta} u^\beta = -c^2$
- For a free particle, relationship between four-momentum and four-velocity: $p_\beta g^{\beta\alpha} = p^\alpha = mu^\alpha$



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Assumptions
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Physics

Want to extend Hamiltonian mechanics to handle change of coordinates that mix space and time variables

$$q^\alpha = [t, q^i] \quad [q^i(t)] \rightarrow [t(s), q^i(s)]$$

Separate time-the-variable from time-the-parameter

Hamiltonian mechanics
on the extended
phase space

Recall

$$\theta_a = [p_i, 0, -H]$$

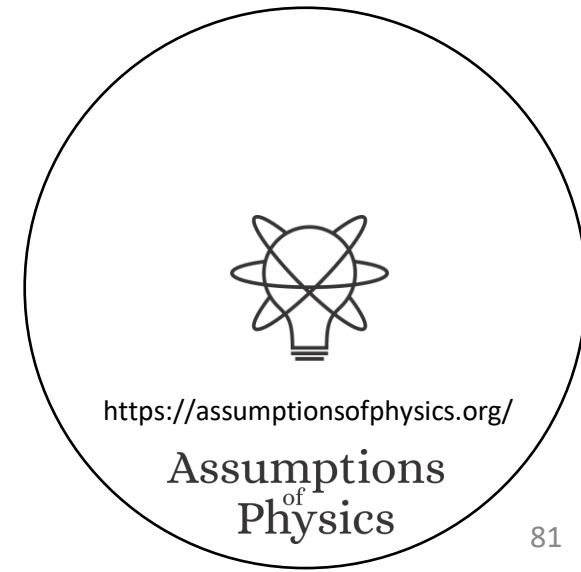
Form a covector

$$p_\alpha = [-E, p_i]$$

$\Rightarrow$ only possible choice for conjugates

$$\frac{dq^\alpha}{ds} = \frac{\partial \mathcal{H}}{\partial p_\alpha} \quad \frac{dp_\alpha}{ds} = -\frac{\partial \mathcal{H}}{\partial q^\alpha}$$

Hamiltonian constraint
 $\mathcal{H}(t, q^i, E, p_i) = 0$
Not just a conserved quantity



$$q^\alpha = [t, q^i] \quad \omega = dq^i dp_i - dt dE$$

$$p_\alpha = [-E, p_i]$$

Energy-momentum covector and negative sign for temporal DOF appear in the extended phase space without knowledge of space-time geometry (or c)

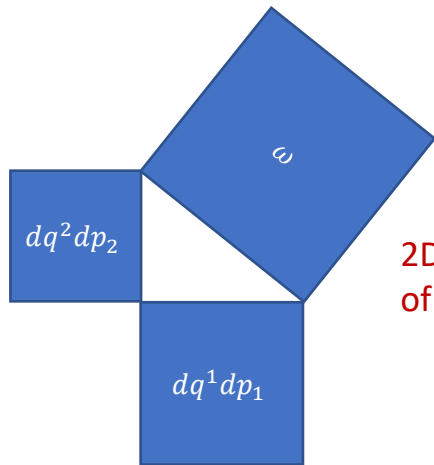
Lorentzian relativity is the only “correct” one

It extends the count of configurations per DOF to the temporal DOF in the correct way

$$\omega = dq^1 dp_1 + dq^2 dp_2$$

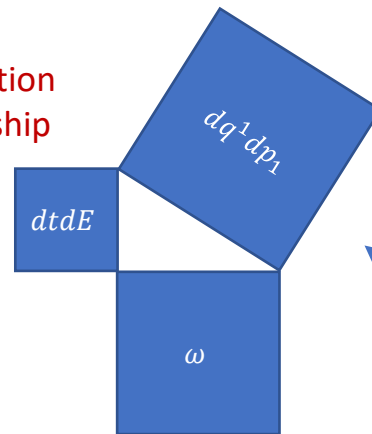
$$\omega = dq^1 dp_1 - dt dE$$

$$dq^1 dp_1 = \omega + dt dE$$



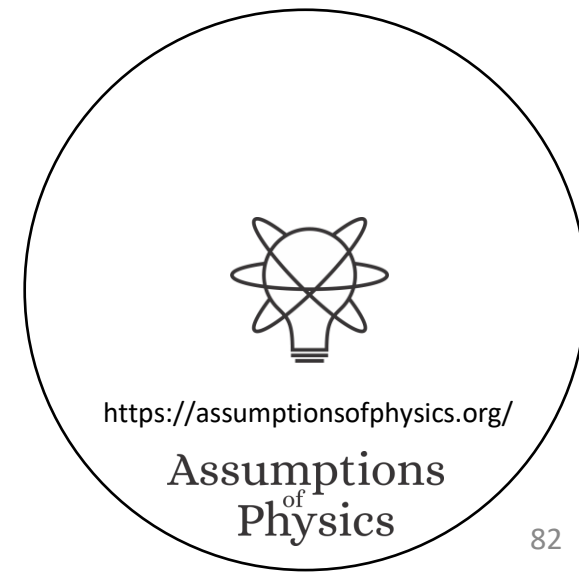
2D representation
of 4D relationship

Independent DOFs are orthogonal



States are counted at
equal time:
temporal DOF
orthogonal to ω

Pythagoras's formula still present



Example: free particle in an inertial frame

$$\mathcal{H} = \frac{1}{2m} \left(p_i \delta^{ij} p_j - \left(\frac{E}{c} \right)^2 + m^2 c^2 \right)$$

$$\frac{dt}{ds} = \frac{\partial \mathcal{H}}{\partial(-E)} = -\frac{1}{2mc^2} \frac{\partial}{\partial(-E)} (-E)^2 = -\frac{1}{2mc^2} (-2E) = \frac{E}{mc^2}$$

$$\frac{dq^i}{ds} = \frac{\partial \mathcal{H}}{\partial p_i} = \frac{1}{2m} \frac{\partial}{\partial p_i} (p_i \delta^{ij} p_j) = \frac{2\delta^{ij} p_j}{2m} = \frac{p_i}{m}$$

$$\frac{dE}{ds} = -\frac{d(-E)}{ds} = \frac{\partial \mathcal{H}}{\partial t} = 0$$

$$\frac{dp_i}{ds} = -\frac{\partial \mathcal{H}}{\partial q^i} = 0$$

This Hamiltonian constraint is the generator for proper time (i.e. affine parameter s corresponds to proper time)

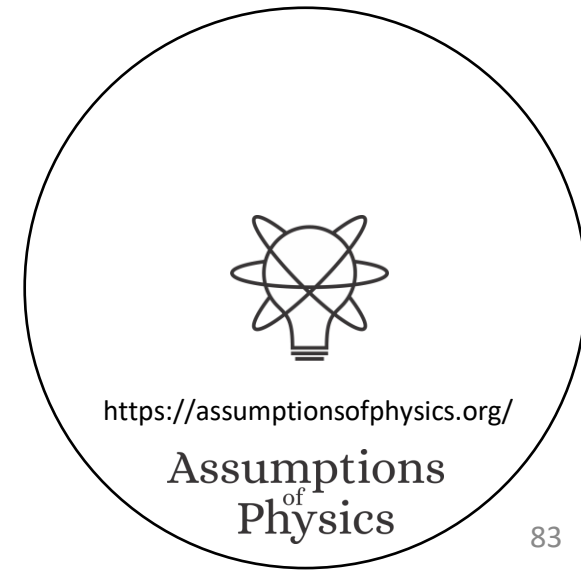
four
momentum

$$\frac{E}{c} = m \frac{d(ct)}{ds}$$

$$p_i = m \frac{dq^i}{ds}$$

four
velocity

proper time!



Compare to “standard” way

Generator for proper time

Nice and quadratic

$$\mathcal{H} = \frac{1}{2m} \left(p_i \delta^{ij} p_j - \left(\frac{E}{c} \right)^2 + m^2 c^2 \right)$$

$$H = c \sqrt{m^2 c^2 + p_i \delta^{ij} p_j}$$

Hamiltonian is time generator

$$\frac{dq^i}{dt} = \frac{\partial H}{\partial p_i} = c \frac{\partial}{\partial p_i} \sqrt{m^2 c^2 + p_i \delta^{ij} p_j}$$

$$= \frac{c}{2} \frac{1}{\sqrt{m^2 c^2 + p_i \delta^{ij} p_j}} \frac{\partial}{\partial p_i} (m^2 c^2 + p_i \delta^{ij} p_j)$$

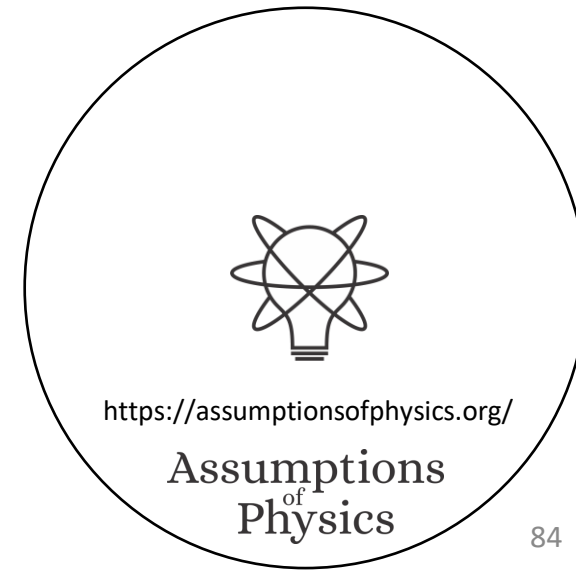
$$= \frac{1}{2} \frac{2 \delta^{ij} p_j}{\sqrt{m^2 + \frac{p_i \delta^{ij} p_j}{c^2}}} = \frac{p_i}{\sqrt{m^2 + \frac{|p|^2}{c^2}}}$$

$$\frac{dp_i}{dt} = - \frac{\partial H}{\partial q^i} = 0$$

$$\frac{dq^i}{dt} = \frac{ds}{dt} \frac{dq^i}{ds} = \frac{mc^2}{E} \frac{p_i}{m}$$

Hamiltonian mechanics on the extended phase-space comes out more naturally and is a nicer formulation

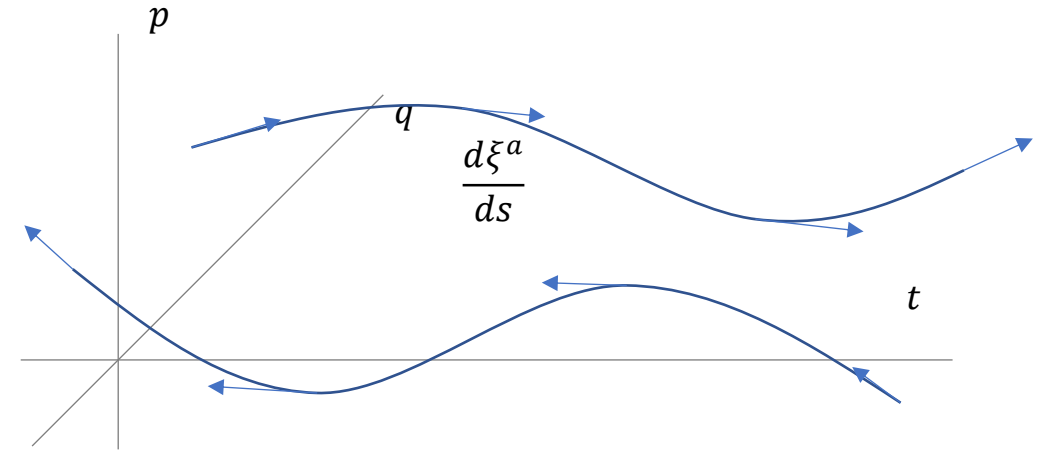
Wait... quadratic?
Can E be negative?



Classical anti-particles

This can be negative

$$\begin{aligned} d_s t &= \partial_{-E} \mathcal{H} & d_s(-E) &= \partial_t \mathcal{H} \\ d_s q^i &= \partial_{p_i} \mathcal{H} & d_s p_i &= \partial_{q^i} \mathcal{H} \end{aligned}$$



Particle: affine parameter aligned with time

Anti-particle: affine parameter anti-aligned with time

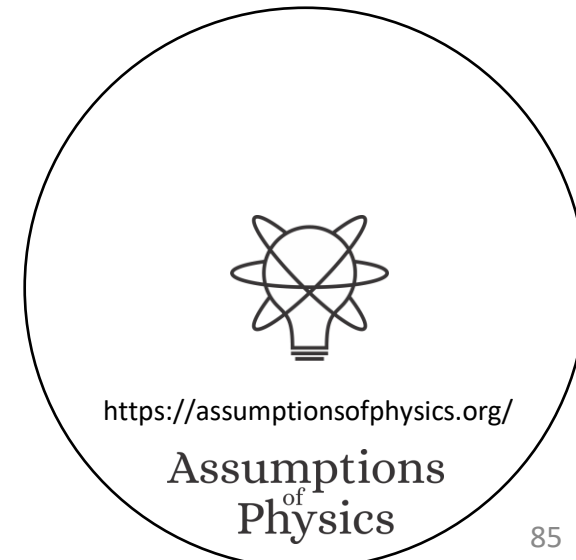
Free particle

$$d_s t = \frac{E}{mc^2}$$

Anti-particle
= negative energy

Parametrization flows
backwards with
respect to time

An evolution cannot
change time alignment



Inconsistent units for quantum observables

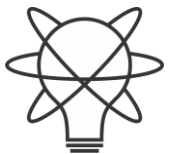
Suppose X is an observable

Expectation $\langle \psi | X | \psi \rangle$ should have units of X

But then $X|\psi\rangle$ cannot have the same units of $|\psi\rangle$

But then $X|\psi\rangle + |\psi\rangle$ makes no physical sense

Therefore, X cannot be a map within the same space



<https://assumptionsofphysics.org/>

Assumptions
of
Physics

Tangent vectors in differential geometry

Defined as derivations

$$v: \mathcal{C}^\infty(X) \rightarrow \mathbb{R}$$

$$v = \sum_i v^i \partial_i$$

component basis

In polar coordinates

$$\partial_r + \partial_\theta = ???$$

[m] [rad]

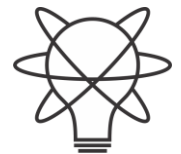
In phase space

$$\partial_q + \partial_p = ???$$

[m] [kg m s⁻¹]

Mathematically precise $\nRightarrow$ physically precise

Doesn't work with units



<https://assumptionsofphysics.org/>

Assumptions
of
Physics

The problem with counting on the continuum

We'd like:

1. Every state is a single case (i.e. $\mu(\{\psi\}) = 1$)
2. Finite continuous range carries finite information (i.e. $\mu(U) < \infty$)
3. Count is additive for disjoint sets (i.e. $\mu(\cup U_i) = \sum \mu(U_i)$)

Incompatible!

Pick two!

Discard 1 $\Rightarrow$ Lebesgue measure

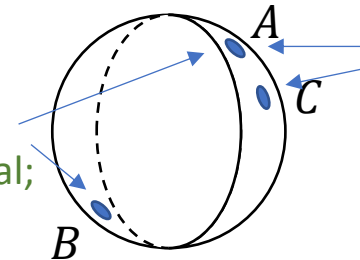
Discard 2 $\Rightarrow$ counting measure

Discard 3 $\Rightarrow$ "Quantum measure"

$$\mu(U) = 2^{\sup(s(\text{hull}(U)))}$$

Exponential of the maximum entropy reachable with convex combinations (statistical mixtures) of U (reduces to counting/Liouville measure)

Orthogonal states: additive
different states all else equal;
mutually exclusive



$$\mu(\{A\}) = 2^0 = 1$$

$$\mu(\{A, B\}) = 2^1 = 2$$

$$\mu(\{A, C\}) < 2 = \mu(\{A\}) + \mu(\{C\})$$

sub-additive
Non-orthogonal states: different states
but in different contexts; not mutually
exclusive

Quantum mechanics $\Rightarrow$ lower bound
on #conf (entropy) on continuous DOF



JOHN
TEMPLETON
FOUNDATION
Inspiring Awe & Wonder



<https://assumptionsofphysics.org/>

Assumptions
of
Physics

Conjecture: quantum gravity $\Rightarrow$ lower bound on DOF count

Same problem!

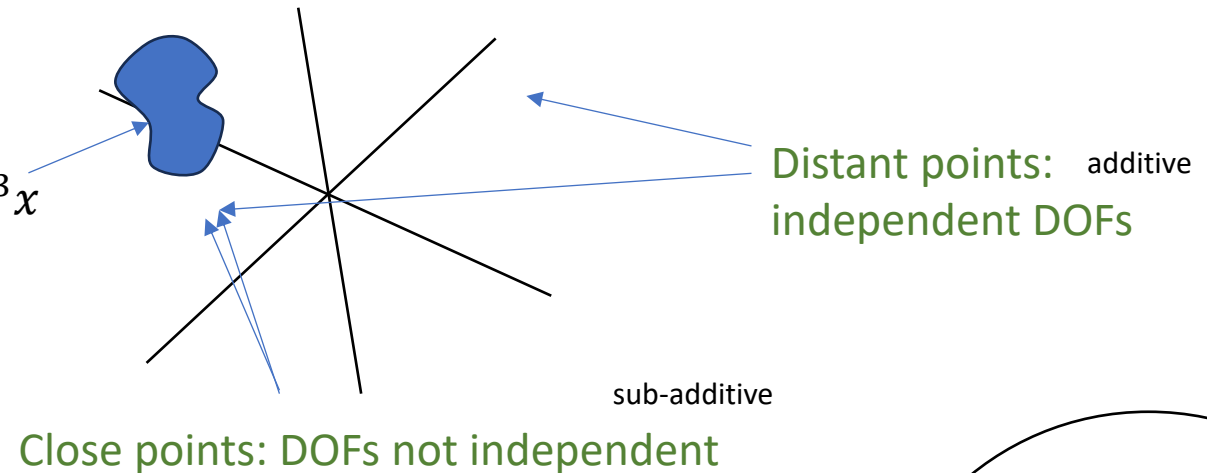
$$\frac{\#conf}{\#DOFs}$$

Lower bound
on this...

...requires a lower
bound on this

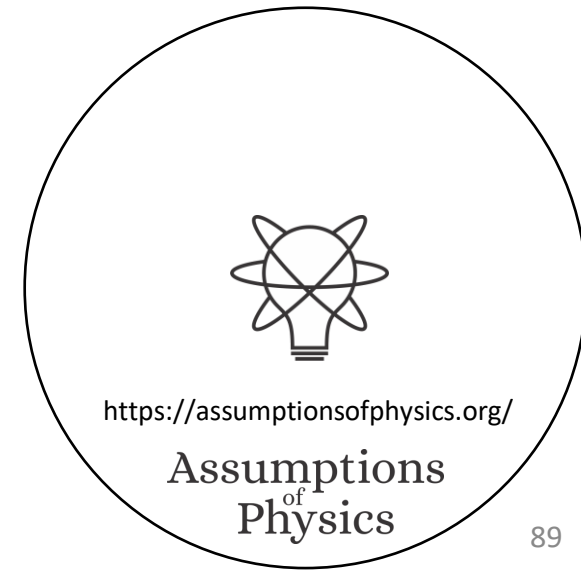
$$\#DOFs \sim \text{Spatial volume } \int_U \sqrt{-g} d^3x$$

1. Every point is a single DOF (i.e. $\mu(\{x\}) = 1$)
2. Finite volume carries finitely many DOFs (i.e. $\mu(U) < \infty$)
3. Count is additive for disjoint regions (i.e. $\mu(\cup U_i) = \sum \mu(U_i)$)



From QM: Lower bound on state count requires a severe revisitation of particle state space

Does lower bound on DOF count require an equally severe revisitation of space-time?



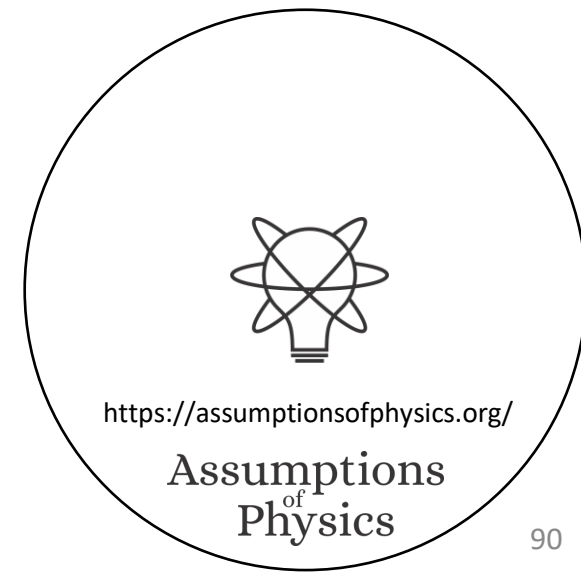
How do we turn these insights into a general theory of physics?

Every formal theory starts with primitive objects

Objects that are NOT defined from other objects

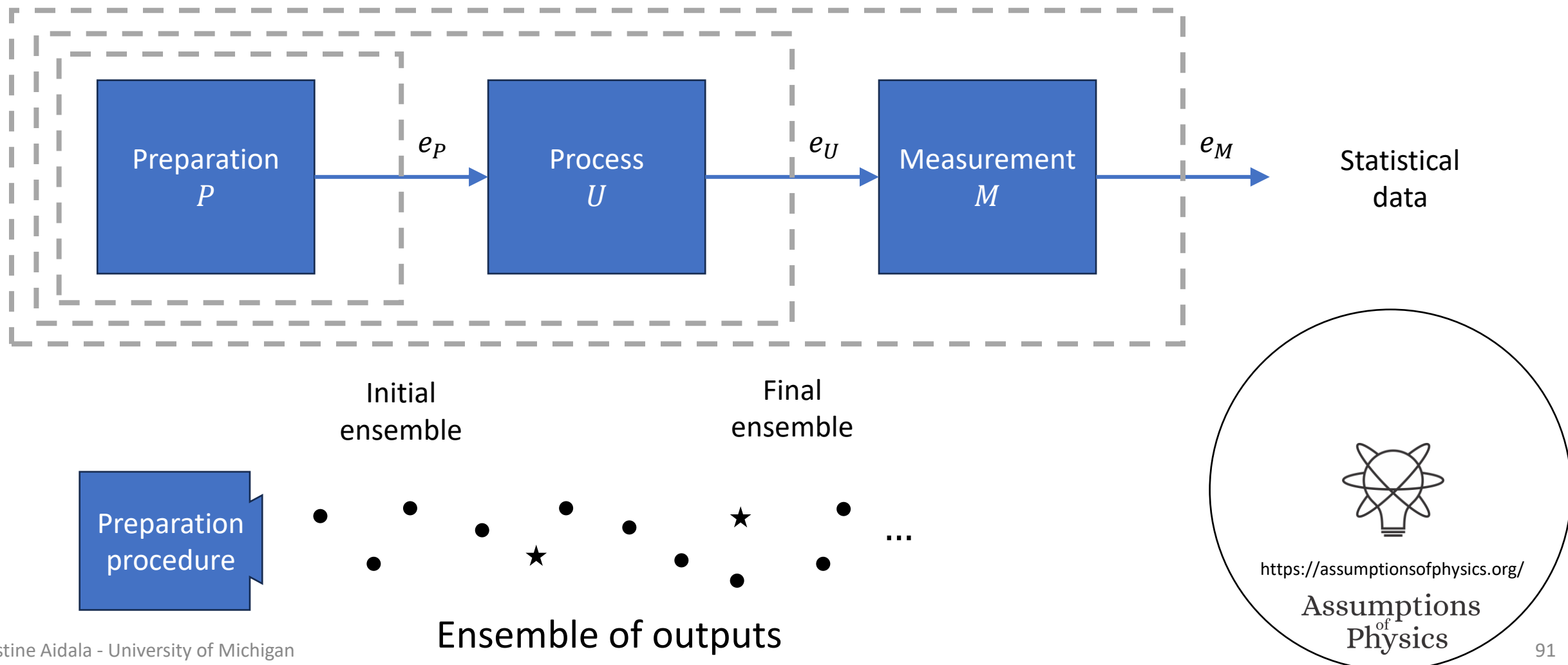
We don't want to kick the can down the road

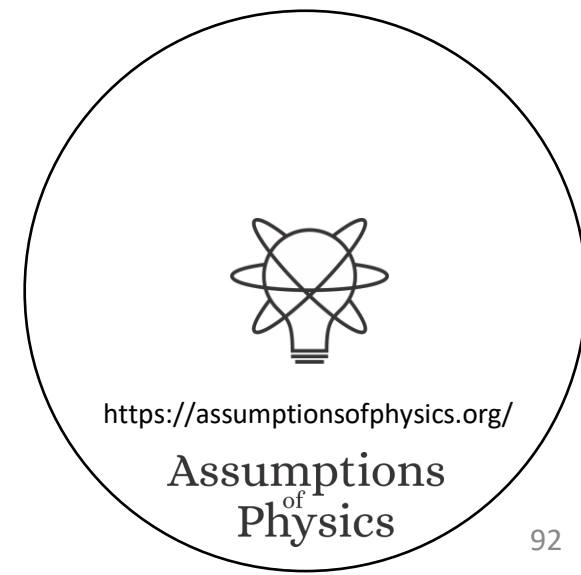
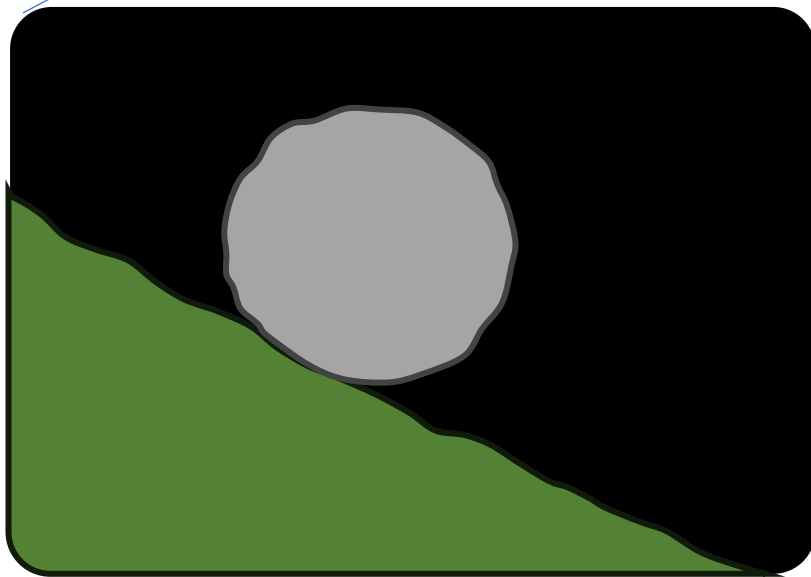
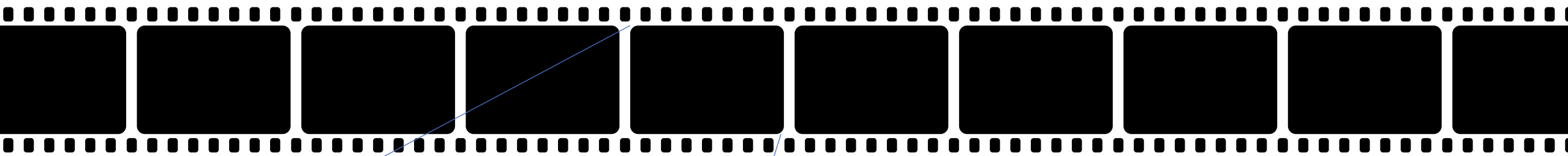
Are ensembles the correct primitive objects for physical theories? How do we know?

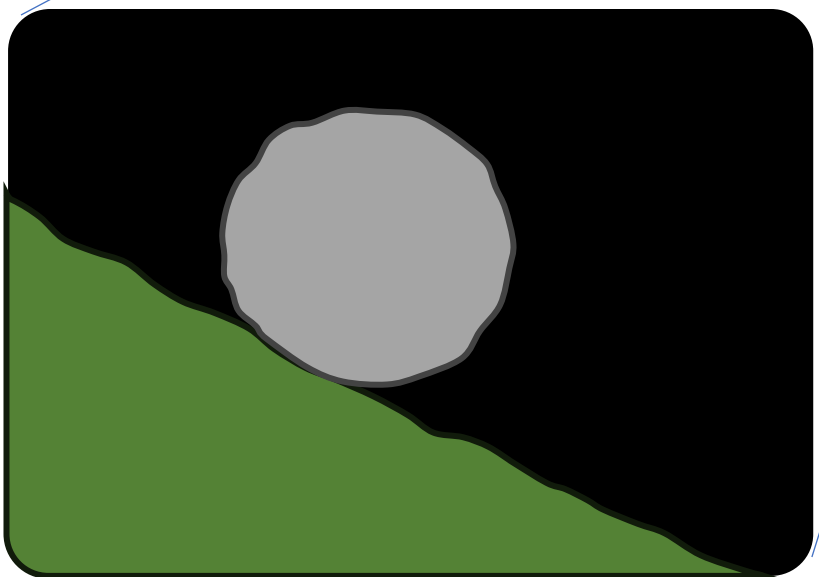
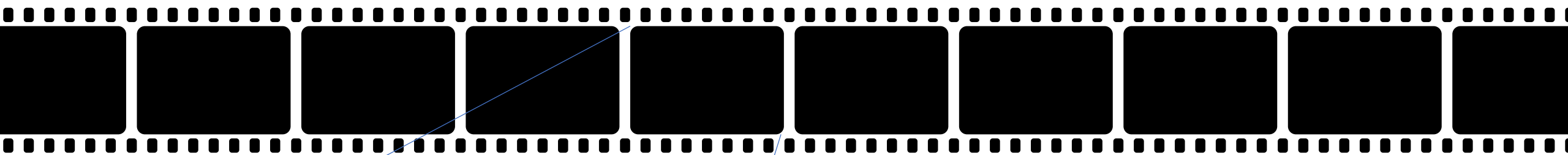


Principle of scientific reproducibility. Scientific laws describe relationships that can always be experimentally reproduced.

⇒ Scientific laws are relationships between ensembles



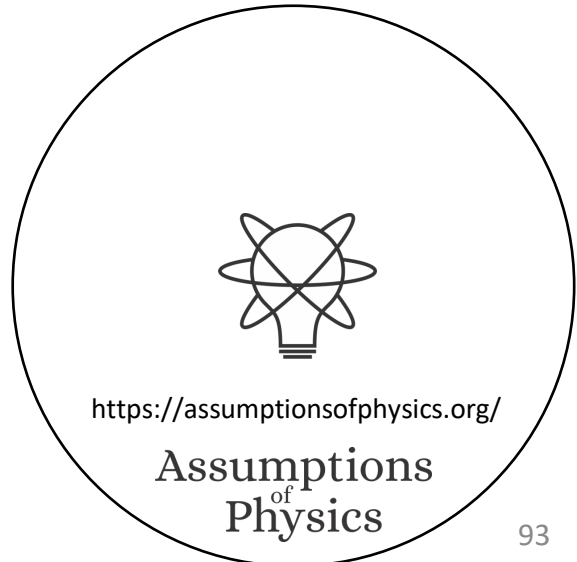


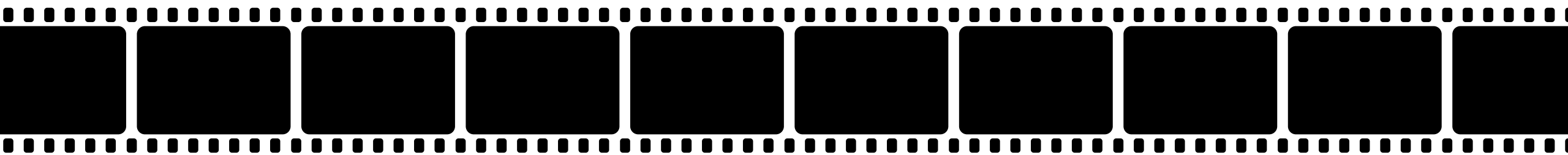


That's not it!!!

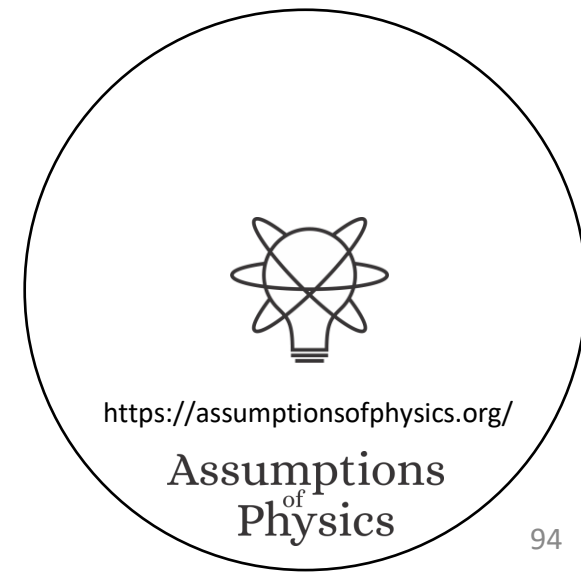
~~Physics describes the evolution
of a particular system like a movie~~

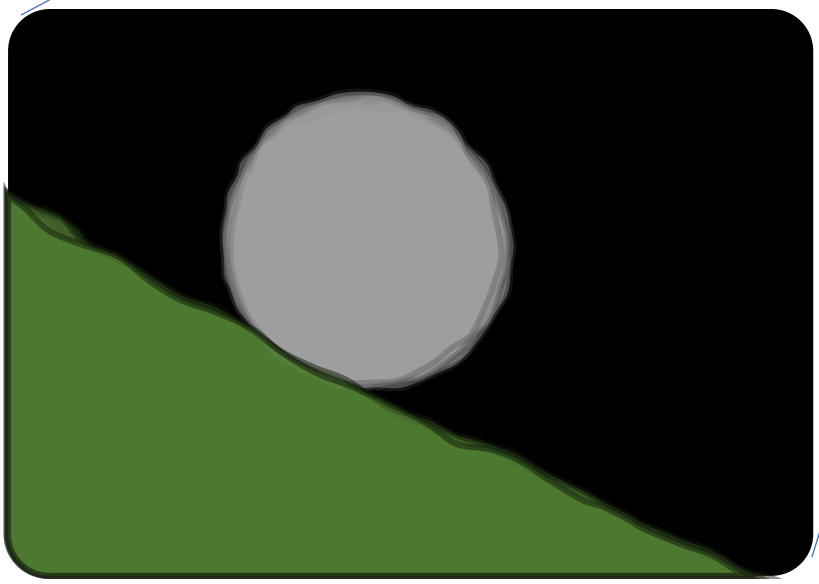
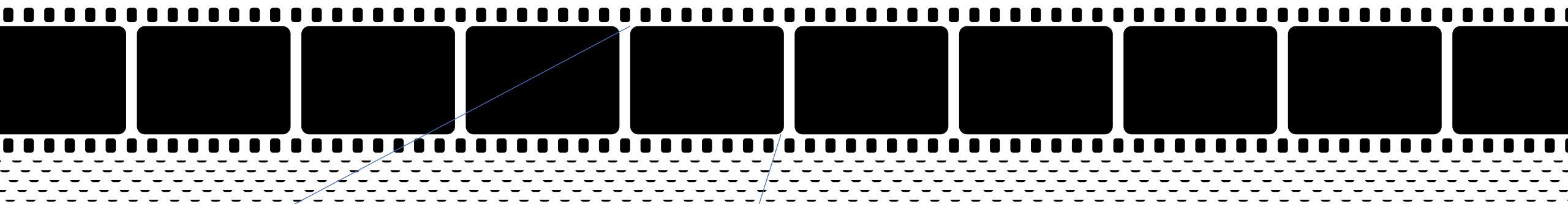
$$x(t)$$



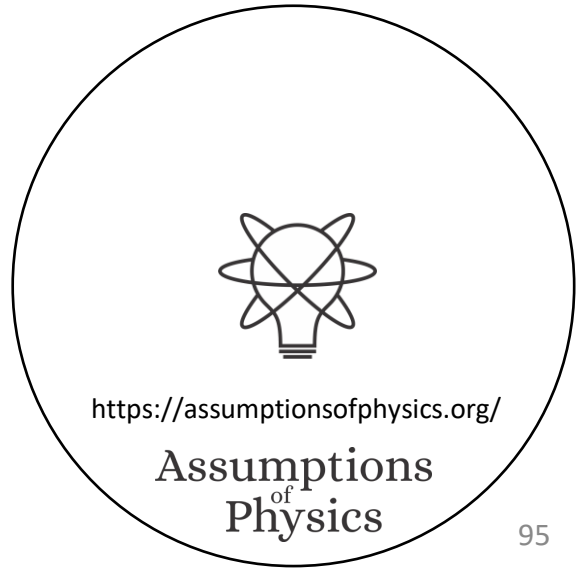


Physics describes the evolution
of similarly prepared systems





Physics describes the evolution
of similarly prepared systems

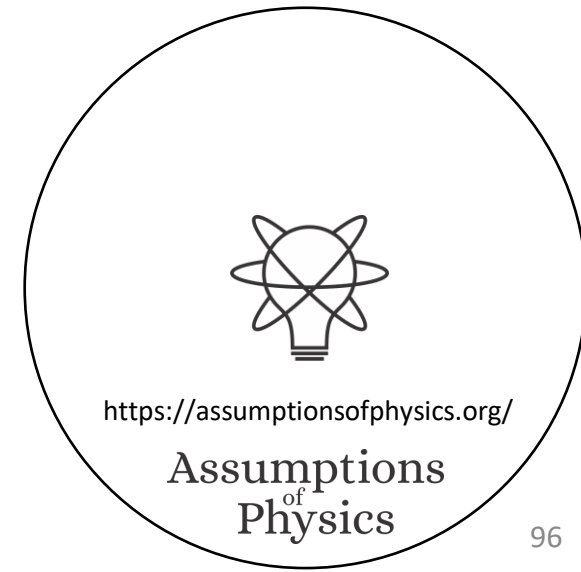


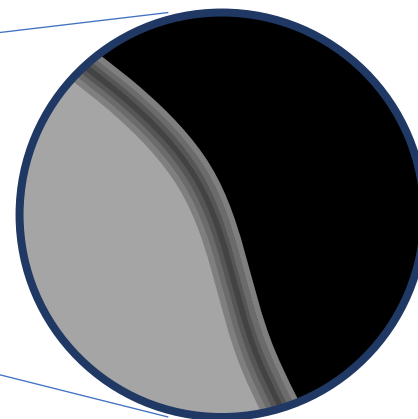
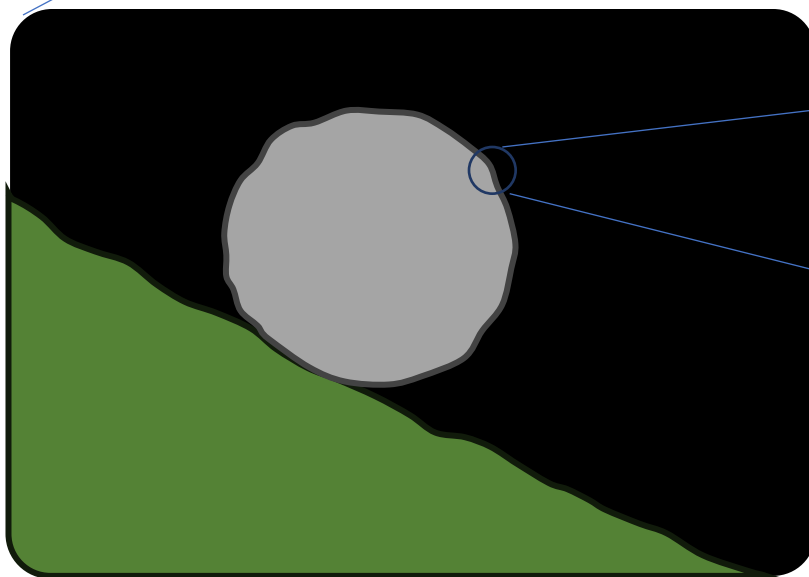
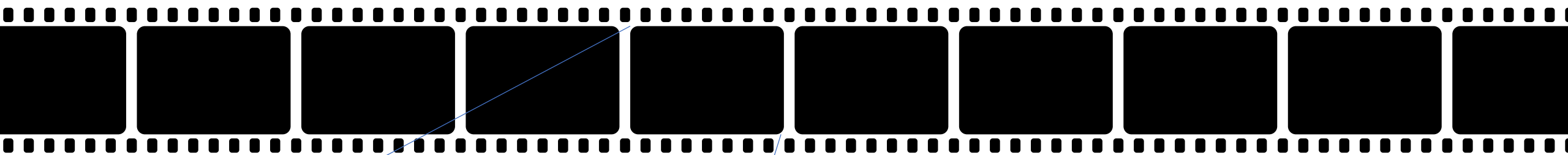
Statistical description of preparations
and measurement outcomes

$x(t)$

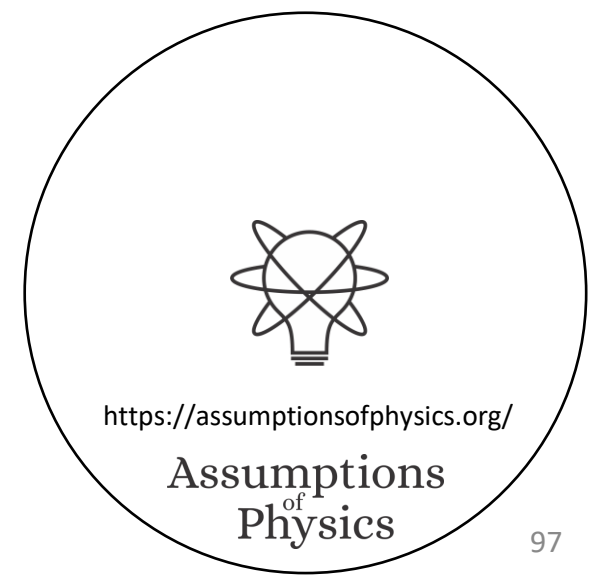
In physics, we can only
study relationships that
can be reproduced:
relationships between
ensembles

Statistical description
of all clocks synchronized
at a particular initial event





Even a single preparation/measurement
is an average over a finite time



To be able to define a probability distribution over heads or tails, you need first to define what heads and tails are

We can do that because we can leave the coin be, and it remains heads or tails

If it kept changing more than 100 times a second, we wouldn't perceive heads or tails: just a blur

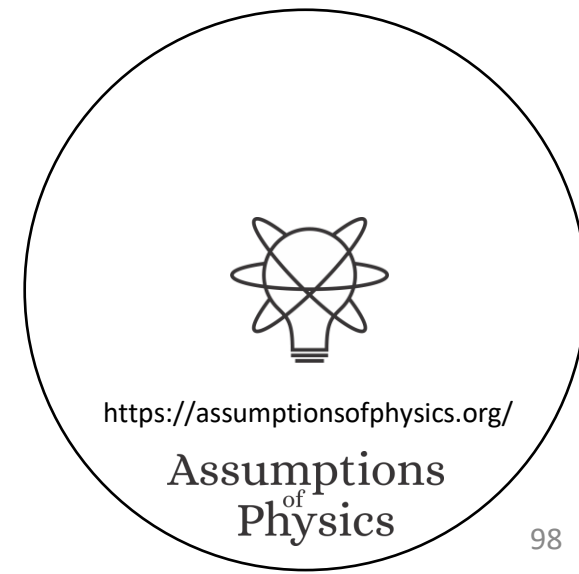


wikipedia

wikipedia

The state of a system is what remains
“stable” for at least a short amount of time

It's ensembles all the way down



Principle of scientific reproducibility. Scientific laws describe relationships that can always be experimentally reproduced.

⇒ Scientific laws are relationships between ensembles

Ensembles MUST be the foundational primitives

Without a generic theory of ensemble spaces there is no well-defined theory of physics, causality, probability, information theory, ...

